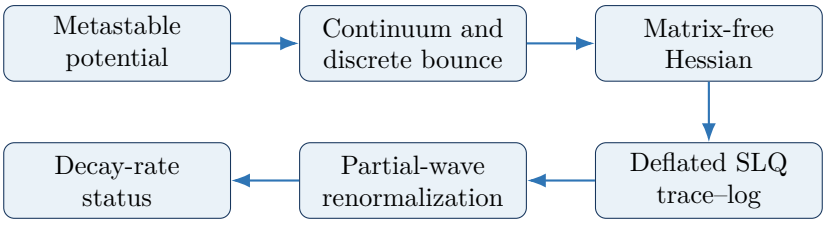
# Higgs Fluctuation Solver in 4D
# – Beyond the Standard Model –

A Technical Guide to the Unified Bounce, Hessian,
Stochastic-Lanczos, and Continuum-Determinant Pipeline




M. S. Cook

In Vero Vacuo

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Source program: `Higgs_fluctuation_solver_stabilized.py`
Production calculation: $L = 26$, $h = 0.8$, global-$O(4)$ scalar model
In Vero Vacuo project website

# Preface and Scope

This document explains the architecture, physics, numerical algorithms, and outputs of the unified source file

`Higgs_fluctuation_solver_stabilized.py`

The program is a unified research solver for a four-component scalar field in four-dimensional Euclidean space, with quadratic $SU(2)$ gauge diagnostics and a nonlinear lattice-vertex generation mode. It contains ten execution modes: analytic validation, $SU(2)$ prerequisite validation, a regulated gauge–Goldstone–ghost determinant scan, an $SU(2)$ counterterm stage, nonlinear SU(2) vertex-table generation, exploratory research spectra, discrete bounce relaxation, stochastic Lanczos quadrature (SLQ), scalar continuum partial-wave renormalization, and a full scalar pipeline. This revision documents the source and production-recovery work through October 2, the single-core validation audits completed on October 5, the matching procedure updates through October 7, 2026, the October 7 implementation and independent audit of nonlinear SU(2) lattice vertices, and the October 8 fixed full-gauged routing/projector deck and independent audit. The October 19 matching update adds a provisional Higgs-slope coefficient, first continuum-minus-lattice six-point and higher-point pilots, and a lattice-only gauge two-point follow-up. The October 14–18 spacing-campaign audit adds the four-translation-mode recovery, paired determinant-shift closure, L36 control-variate checks, verified-subspace spacing comparison, point-source gauge/ghost Green functions, and separate L24/L30 trace-log and gauge-parameter pilots. These results are catalogued in Chapter 14 and Section 9.7.4; the campaigns use distinct profiles, estimators, and probe streams and are not to be merged without matching provenance. Production-precision higher-point integrations, the continuum gauge/ghost two-point tensor, full Ward/Slavnov–Taylor and regulator-cancellation checks, complete finite full-gauged coefficients, determinant completion, and controlled continuum/volume extrapolations remain outstanding.

On October 9, 2026, a provider-neutral process supervisor was added for concurrent $L = 26$ and $L = 30$ scalar relaxed-bounce/SLQ tests. It runs the existing solver stages in separate processes with per-size output trees, restartable checkpoints, per-stage logs, a JSON campaign manifest, and single-threaded numerical-library defaults. This enables repeatable parallel job execution; it does not distribute one lattice operator across nodes or provision cloud resources. The procedure and resource cautions are in Section 15.3.

Chapter 14 records the separate SU(2)-program spacing campaign at fixed physical extent. It audits the scalar longitudinal translation subspace and stochastic determinant bookkeeping; these results are not a full gauge–Goldstone–ghost calculation or a physical decay prefactor.

Chapter 18 incorporates the substantive body of the standalone October 1 modal-bridge implementation guide. Its status statements are retained with their original date; in particular, the October 1 note about the unavailable $L = 26$ checkpoint is superseded by the later recovery and loss history recorded above.

The broader motivation for this research program, together with its companion project on quantum computing, is described on the In Vero Vacuo project website. The site provides project-level context complementary to the technical derivations and numerical results documented here.

The title places the work in a "Beyond the Standard Model" research context, because metastable scalar potentials, additional scalar operators, and vacuum decay are common ingredients in extensions of the Standard Model. The implemented production calculation itself is more specific: it is a *global* $O(4)$ scalar model. The updated source additionally contains the quadratic gauge–Goldstone and ghost operators around a zero-gauge-field radial background, including a link-derived Hessian in positive background $R_\xi$ gauges. The source now also evaluates regulated finite-lattice gauge–Goldstone–ghost determinant ratios, treats the false-vacuum constant modes and their Gaussian measure normalization explicitly, and constructs the local counterterm basis with derived continuum pole residues. Scalar-only radial one-loop finite coefficients have now been calculated at the stated benchmark. The full gauged finite matching constants and controlled continuum and volume limits remain incomplete, so a physical gauged prefactor has not yet been calculated.

Chapter 9 incorporates the October 5 $L = 6$ pilot, increased-probe convergence and complete-multiplet studies, and the fresh $L = 3, 4$ nonconstant-background prerequisite audit. It distinguishes recorded follow-up results from the fresh audit and identifies which raw artifacts remain available. These tests do not determine finite matching coefficients or a physical gauged decay rate.

The October 6 matching review is recorded in `SU2_Matching_Coefficient_Progress_Report_20261006.tex` and summarized in Section 9.7. It confirms that the radial basis, pole residues, and coefficient interface are ready. The October 7 matching update reports finite scalar-only radial one-loop coefficients. The October 8 routing/projector setup was fixed and audited, with a first four-point pilot. The October 18–19 follow-up extracts a provisional finite two-point slope, reports exploratory four- and six-point gauge-linked contacts, and evaluates eight-, ten-, and twelve-leg pilots; the continuum gauge/ghost two-point tensor and production-precision contact integrals remain open. Details are in Section 9.7.4 and the updated source report `SU2_Next_Steps_Finite_Matching_20261006.tex`, which also documents the nonlinear action and its one-loop projection formulas. The scalar-link, Wilson-gauge, and ghost vertex generators for that action have since been implemented in `su2_lattice_vertices.py`, connected to the solver's `su2_vertices` mode, and independently checked against finite transformations and separate action expansions. The update also records one scalar quartic tadpole as a normalization check. A separate October 8 evaluator integrates the gauge-linked contribution to one fixed-route four-radial-Higgs projection and performs a four-point IR scan. Neither this pilot nor the vertex implementation completes full gauged matching.

The foundational progress report, including the adopted EFT matching prescription, is incorporated into the main text as Chapter 17. Its source body remains shared with the standalone report through `SU2_Physical_Prefactor_Progress_Content.tex`, so the two presentations use the same report text, equations, tables, and validation graph.

The numerical figures reproduced here were generated by the solver and its companion reports. The conceptual sequence also incorporates material from the local slide sources `bounce_solution.tex`, `one_loop_fluctuations.tex`, `functional_determinants.tex`, and `Lattice_decay_research.tex`.

**Scope and interpretation**

The finite result quoted in this guide is the continuum-renormalized fluctuation determinant of the global scalar theory. It is not a completed electroweak or gauged-$SU(2)$ vacuum-decay prediction. Three internal orientation collective-coordinate normalizations also remain to be supplied if the symmetry is treated as global.

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# Chapter 1

# The Physical and Computational Problem

## 1.1 Euclidean false-vacuum decay

For a real four-component scalar field $\mathbf{\Phi}(x)$, define its radial amplitude $\rho = (\mathbf{\Phi}\cdot\mathbf{\Phi})^{1/2}$. The Euclidean action is

$$S_E[\mathbf{\Phi}] = \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\mathbf{\Phi}\cdot\partial_\mu\mathbf{\Phi} + V(\rho)\right]. \tag{1.1}$$

The semiclassical decay rate per unit four-volume has the schematic form

$$\frac{\Gamma}{V_4} = \left(\frac{B}{2\pi}\right)^2 \left|\frac{\det{}' H_B}{\det H_F}\right|^{-1/2} e^{-B} \times \mathcal{J}_{\text{internal}} \times [1 + \text{higher loops}], \tag{1.2}$$

where $B = S_E[\rho_B] - S_E[0]$, $H_B$ is the Hessian around the bounce, $H_F$ is the false-vacuum Hessian, and the prime means that collective coordinates are omitted from the determinant [1, 2].

## 1.2 Metastable scalar potential

The code uses the polynomial

$$V(\rho) = -\frac{\mu^2}{2}\rho^2 + \frac{\lambda}{4}\rho^4 + \frac{\gamma}{6}\rho^6 + \frac{\delta}{8}\rho^8. \tag{1.3}$$

The production parameter set is

$$\mu^2 = -0.40, \qquad \lambda = -2.00, \qquad \gamma = 0.80, \qquad \delta = 0.08. \tag{1.4}$$

With the sign convention in Eq. (1.3), the curvature at the false vacuum is $m_F^2 = -\mu^2 = 0.4 > 0$. The negative quartic creates a barrier, while the positive sixth- and eighth-order terms stabilize the potential at large field amplitude.

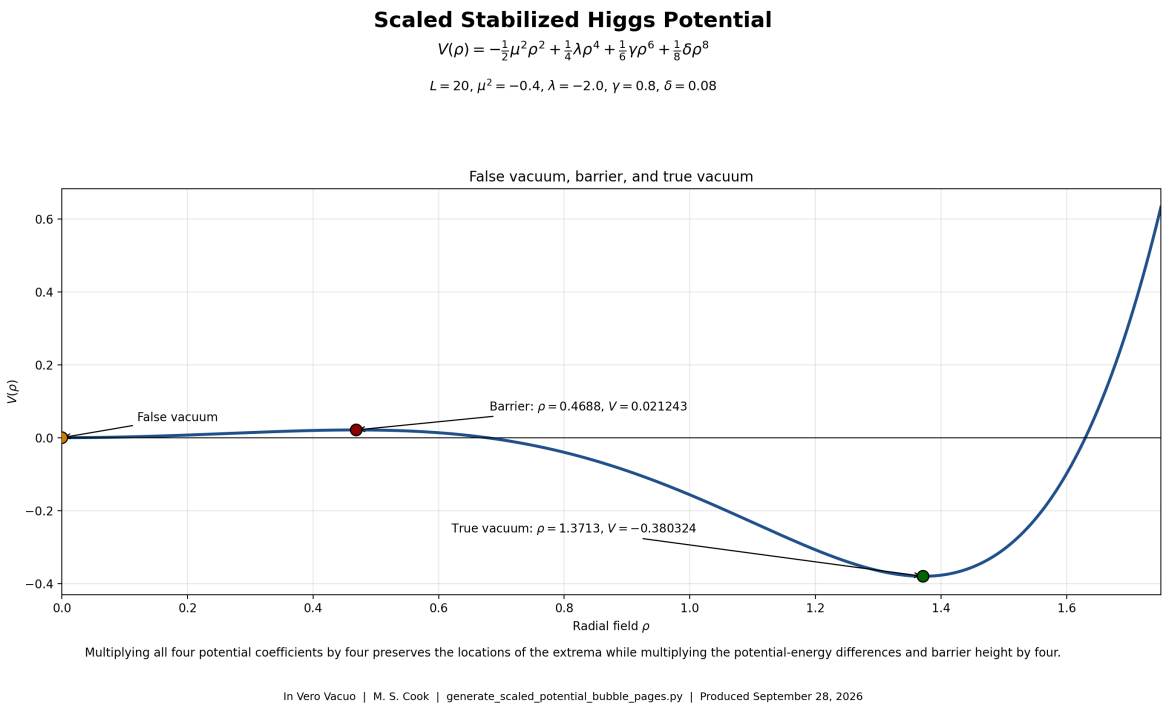



Figure 1.1: Scaled metastable potential used for the production calculation. The false vacuum is at $\rho = 0$, separated by a barrier from the deeper nonzero minimum.

**Implementation note**

Nonzero stationary points are found by solving a cubic in $s = \rho^2$, $\delta s^3 + \gamma s^2 + \lambda s - \mu^2 = 0$. The radial curvature identifies barrier maxima and true-vacuum minima before any bounce shooting is attempted.

```
        ('slq', {}),
        (
            'continuum',
            {
                'HIGGS_MU_SQ': '-0.40',
                'HIGGS_LAMBDA': '-2.00',
                'HIGGS_GAMMA': '0.80',
                'HIGGS_DELTA': '0.08',
                'HIGGS_L': '20',
                'HIGGS_RADIAL_CUTOFF': '20',
                'HIGGS_PARTIAL_WAVE_LMAX': '70',
                'HIGGS_DETERMINANT_OUTPUT':
                    'scaled_potential_continuum_renormalized',
            },
        ),
    ]
    for _stage, _overrides in _stage_environments:
        _environment = _pipeline_os.environ.copy()
        _environment.update(_overrides)
        _environment['HIGGS_RUN_MODE'] = _stage
        print(f'\n=== Running unified pipeline stage: {_stage} ===', flush=True)
        _pipeline_subprocess.run(
            [_pipeline_sys.executable, __file__],
            check=True,
            env=_environment,
```

```
        )
    raise SystemExit(0)

if _PIPELINE_MODE == 'relax':
    """Relax a sampled continuum bounce to a discrete lattice stationary point.

    The nonlinear equation solved is

        -Delta_h rho(x) + V'(rho(x)) = 0

    on a periodic four-dimensional lattice.  Newton corrections are obtained with
    matrix-free MINRES using the longitudinal Hessian.  A residual-norm line search
    keeps the iteration in the basin of the nontrivial saddle rather than blindly
    taking a full Newton step.
    """

    import json
    import os
    import sys
    import time
    from datetime import datetime
    from pathlib import Path

    import matplotlib.pyplot as plt
    import numpy as np
    import scipy.sparse.linalg as spla
    from matplotlib.backends.backend_pdf import PdfPages
    from scipy.integrate import solve_ivp


    L = int(os.getenv('HIGGS_RELAX_L', '26'))
    SPACING = float(os.getenv('HIGGS_RELAX_SPACING', '0.8'))
    MAX_NEWTON = int(os.getenv('HIGGS_RELAX_MAX_NEWTON', '14'))
```

Listing 1.1: Potential, derivatives, and stationary-point classification in the unified source.

# Chapter 2

# Program Architecture and Execution Modes

## 2.1 Unified dispatcher

The merged program begins with a mode dispatcher. Most specialized numerical implementations are embedded directly in the source file; the nonlinear vertex rules are maintained in the imported module `su2_lattice_vertices.py`. The `full` mode launches the relaxation, SLQ, and continuum stages sequentially with a consistent production parameter set. It is a scalar pipeline: setting `HIGGS_SU2_GAUGE=1` in `full`, `relax`, `slq`, or `continuum` now raises an explicit error instead of silently producing scalar-only results. The independent `su2_validation`, `su2_determinant`, `su2_counterterms`, and `su2_vertices` branches run outside the scalar pipeline. This separation prevents a regulated or unmatched gauge result from being silently reported as part of the physical scalar rate.

```
            "basis": "exact periodic Fourier zero-momentum constants",
            "selection": "explicit orthogonal projectors; no eigenvalue threshold"
        },
        "primed_measure_correction_per_generator": "+0.5*log(xi)",
        "raw_regulator_measure_correction_per_generator": "+0.5*log(xi)",
        "gauge_parameter_spreads": gauge_spreads,
        "normalized_gauge_cancellation_status": (
            "PASS" if all(normalized_checks) else "INCONCLUSIVE"
        ),
        "modal_export_enabled": modal_export,
        "modal_datasets": modal_datasets,
        "complete_rate": None,
        "scope": (
            "Finite-lattice gauge--Goldstone--ghost determinant only. A physical "
            "rate still requires counterterm matching, continuum and volume limits, "
            "and a fixed gauge-volume/false-vacuum zero-mode prescription."
        ),
        "probe_csv": str(probe_file), "summary_csv": str(aggregate_file),
        "low_modes_csv": str(low_file)
    }
    with open(output / "determinant_summary.json", "w", encoding="utf-8") as stream:
        json.dump(summary, stream, indent=2)

    with PdfPages(output / "SU2_Determinant_Gauge_Scan.pdf") as report:
        figure, axes = plt.subplots(1, 2, figsize=(12, 7))
        for shift in shifts:
            selected = [row for row in aggregates if row["steps"] == final_steps
                        and row["ir_shift"] == shift
                        and row["zero_mode_policy"] == "primed_normalized"]
```

```
            axes[0].errorbar(
                [row["xi"] for row in selected],
                [row["mean_log_ratio"] for row in selected],
                yerr=[row["standard_error"] for row in selected], marker="o",
                label=f"shift={shift:g}"
            )
```

Listing 2.1: Top-level execution-mode dispatcher.

Table 2.1: Execution modes in the unified solver.

| Mode | Purpose | Principal output |
|---|---|---|
| `validation` | Exactly solvable periodic-lattice test of matrix-free operators and low spectra. | PASS/FAIL diagnostics |
| `su2_validation` | Link-Hessian Ward identities, zero-coupling limit, homogeneous Fourier spectra, and optional stationary-bounce checks. | JSON and validation PDF |
| `su2_determinant` | Regulated coupled gauge–Goldstone and ghost determinant scan over positive $R_\xi$ gauges, with exact and deflated-SLQ backends. | CSV, JSON, low modes, and gauge-scan PDF |
| `su2_counterterms` | Evaluate the local radial EFT basis, derive the continuum pole residues, and apply supplied finite matching coefficients. | Basis CSV, JSON summary, and matching PDF |
| `su2_vertices` | Generate nonlinear scalar-link, Wilson-gauge, and ghost momentum/color tables for configured momentum routings. | Compressed NPZ tables and JSON metadata |
| `research` | Sample a stationary or Gaussian profile and calculate a low-mode Hessian spectrum; optionally add $SU(2)$ diagnostics. | CSV tables and two/three-page PDF |
| `relax` | Solve the nonlinear discrete lattice equation with Newton–MINRES. | Relaxed field checkpoint |
| `slq` | Estimate the full bare lattice trace–log with deflated stochastic Lanczos quadrature. | Checkpointed probe statistics |
| `continuum` | Evaluate the renormalized primed determinant with radial partial waves and Gel'fand–Yaglom ratios. | Finite determinant summary |
| `full` | Run relaxation, SLQ, and continuum renormalization in order. | Complete scalar calculation set |

## 2.2 Nonlinear lattice SU(2) vertex generation

The nonlinear action remains a proposed matching convention rather than the action used by the scalar production pipeline. Its perturbative vertices are now available as reusable functions in `su2_lattice_vertices.py` and through `HIGGS_RUN_MODE=su2_vertices`. Its principal API is:

- `scalar_link_vertex` and `wilson_gauge_vertex`;
- `ghost_gauge_vertex` and `ghost_scalar_vertices`;
- `faddeev_popov_matrix` and `generate_one_loop_vertex_tables`.

The implementation keeps the specified logarithmic link coordinate $U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a]$, midpoint Fourier phases, and all-incoming momentum convention. It provides scalar-link tensors with two Higgs legs and $r = 0, 1, 2$ gauge legs; Wilson-plaquette tensors with $n = 2, 3, 4$ gauge legs; gauge–ghost tensors with $r = 0, 1, 2$ gauge legs; local scalar-dependent ghost tensors; and the nonzero-background sparse Faddeev–Popov matrix. The scalar potential derivative rules remain as documented in the matching report.

The table-generation mode writes compressed arrays and a JSON sidecar. The default outputs are `su2_nonlinear_vertex_tables_20261007_conserving.npz` and `su2_nonlinear_vertex_tables_20261007_conserving.json`; existing files are not overwritten. These tables include every color and polarization component for their recorded sample momenta, not a scan over every possible external momentum. Reusable vertex functions accept other momentum routings. For example:

```
HIGGS_RUN_MODE=su2_vertices python Higgs_fluctuation_solver_stabilized.py
```

Listing 2.2: Generate the configured nonlinear SU(2) vertex tables.

The independent audit is run separately with `su2_nonlinear_vertex_audit.py`. It compares scalar-link coefficients with finite differences of the exact link exponentials, checks Wilson coefficients against an independent ordered-word expansion and the quadratic transverse kernel, checks ghost vertices through finite transformations in the logarithmic link coordinate, and compares the nonzero-background FP matrix with finite gauge transformations. At $a = 0.8$, $g_0 = 0.65$, and $\xi = 1$, the maximum discrepancies were $6.204 \times 10^{-9}$ for the complete scalar-link table, $7.216 \times 10^{-16}$ for the Wilson quadratic kernel, zero for the complete Wilson tensor comparison at the tested routings, $2.398 \times 10^{-6}$ for the complete gauge–ghost table, and $3.234 \times 10^{-10}$ absolute ($1.679 \times 10^{-11}$ relative) for the nonzero-background FP matrix. The finite-transformation audit also checks local scalar-ghost tensors and gauge-invariance identities. These are implementation tolerances, not matching uncertainties.

The vertex audit validates rule construction, not the complete gauged matching calculation. Fixed nonexceptional projection kinematics are now available, and a separate evaluator has integrated one gauge-linked four-Higgs sector with a common IR mass regulator. The generators are still not connected to a complete multi-point matching integrator or physical determinant calculation. The remaining integrated sectors, regulator cancellation, full Ward/Slavnov–Taylor checks, and gauge/parameter matching are outstanding. Accordingly, the coefficient template and full-matching status must remain unset/false.

## 2.3 Data flow

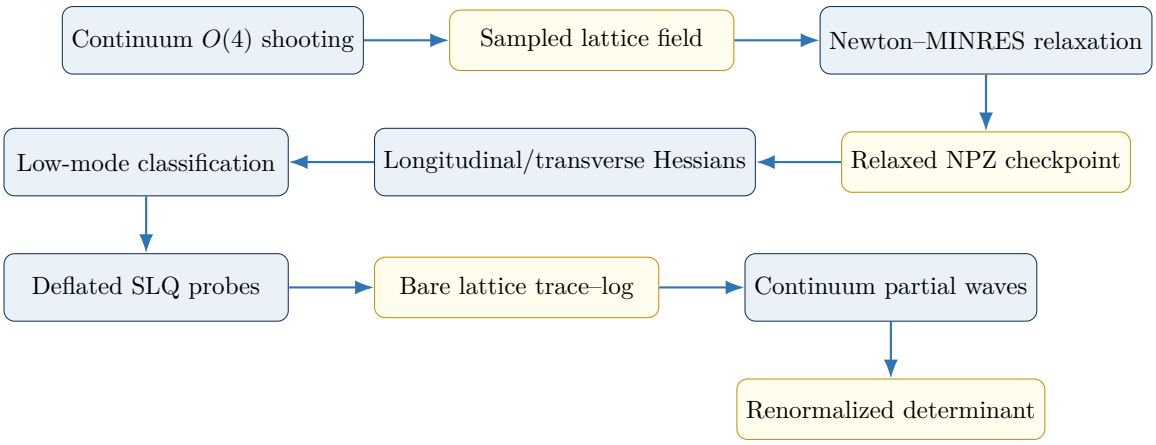



Figure 2.1: Computational data flow. The bare lattice and continuum calculations are complementary: SLQ diagnoses the complete finite-lattice trace, while the partial-wave calculation supplies the regulator-subtracted finite result.

# Chapter 3

# Lattice Geometry, Units, and Matrix-Free Operators

## 3.1 Four-dimensional lattice

For a periodic hypercubic lattice with $L$ sites per direction,

$$N_{\text{sites}} = L^4, \qquad N_{\text{dof}} = 4L^4. \tag{3.1}$$

At $L = 26$, each scalar channel has 456,976 sites and the full four-component fluctuation space has 1,827,904 degrees of freedom. An explicit dense Hessian is therefore impossible, and even an assembled sparse matrix is unnecessary.

The periodic nearest-neighbor negative Laplacian is

$$(-\Delta_h f)_x = \frac{1}{h^2}\left[8f_x - \sum_{\mu=1}^{4}\left(f_{x+\hat{\mu}} + f_{x-\hat{\mu}}\right)\right]. \tag{3.2}$$

It is applied by reshaping a vector into an $L^4$ array and using `numpy.roll`. This reduces permanent operator storage to local potential diagonals and working vectors.

```
        )


    def negative_laplacian(field):
        result = 8.0 * field
        for axis in range(4):
            result -= np.roll(field, 1, axis=axis)
            result -= np.roll(field, -1, axis=axis)
        return result / SPACING**2


    def residual(field):
        return negative_laplacian(field) + potential_derivative(field)


    def hessian_operator(field, channel):
        diagonal = (
            longitudinal_curvature(field)
            if channel == 'longitudinal' else transverse_curvature(field)
        )

        def matvec(vector):
```

```
            grid = np.asarray(vector).reshape(SHAPE)
            return (negative_laplacian(grid) + diagonal * grid).ravel()

        return spla.LinearOperator(
            shape=(N, N), matvec=matvec, rmatvec=matvec, dtype=np.float64
        )


    def continuum_initial_guess():
        roots = np.roots([DELTA, GAMMA, LAMBDA, -MU_SQ])
        points = []
        for root in roots:
            if abs(root.imag) < 1.0e-10 and root.real > 0.0:
                rho_sq = float(root.real)
                rho = np.sqrt(rho_sq)
                curvature = 2.0 * rho_sq * (
                    LAMBDA + 2.0 * GAMMA * rho_sq + 3.0 * DELTA * rho_sq**2
                )
                points.append((rho, potential(rho), curvature))
        rho_barrier = min(point[0] for point in points if point[2] < 0.0)
        rho_true = min(
            (point for point in points if point[2] > 0.0),
            key=lambda point: point[1],
        )[0]

        def rhs(radius, state):
            rho, rho_prime = state
            return [rho_prime, potential_derivative(rho) - 3.0 * rho_prime / radius]

        def overshoots(rho_center):
            epsilon = 1.0e-6
            force = potential_derivative(rho_center)
```

Listing 3.1: Matrix-free periodic Laplacians and scalar Hessian construction.

## 3.2 Lattice-to-physical scaling

The exploratory research mode includes an explicit lattice spacing $a$ and uses

$$\Phi_{\mathrm{phys}} = \frac{\Phi_{\mathrm{lat}}}{a}, \quad m^2_{\mathrm{phys}} = \frac{m^2_{\mathrm{lat}}}{a^2}, \quad V_{4,\mathrm{phys}} = (La)^4, \quad \Lambda_{\mathrm{UV}} \sim \frac{\pi}{a}. \tag{3.3}$$

The production relaxation and SLQ stages instead work in dimensionless lattice units with a refinement spacing $h = 0.8$. The continuum matching stage uses $m_F$ as the reference scale. These two spacing conventions must not be silently identified. The production gauged benchmark is stated in scaled units; an explicit physical energy conversion is still needed for a rate in $\mathrm{GeV}^4$.

# Chapter 4

# Analytic Validation Before Production

## 4.1 Why validation is separate

The default mode replaces the bounce by a homogeneous periodic problem whose Fourier spectrum is known exactly. This tests indexing, periodic boundaries, component ordering, operator symmetry, eigensolver sorting, and degenerate multiplet counting without relying on the difficult bounce calculation.

For momentum components $p_\mu = 2\pi n_\mu / L$, the lattice kinetic eigenvalue is

$$\widehat{p}^2 = \frac{1}{a^2} \sum_{\mu=1}^{4} (2 - 2\cos p_\mu) \,. \tag{4.1}$$

The homogeneous Hessian eigenvalues are $\widehat{p}^2 + U_L$ and $\widehat{p}^2 + U_T$, with one longitudinal and three transverse copies.

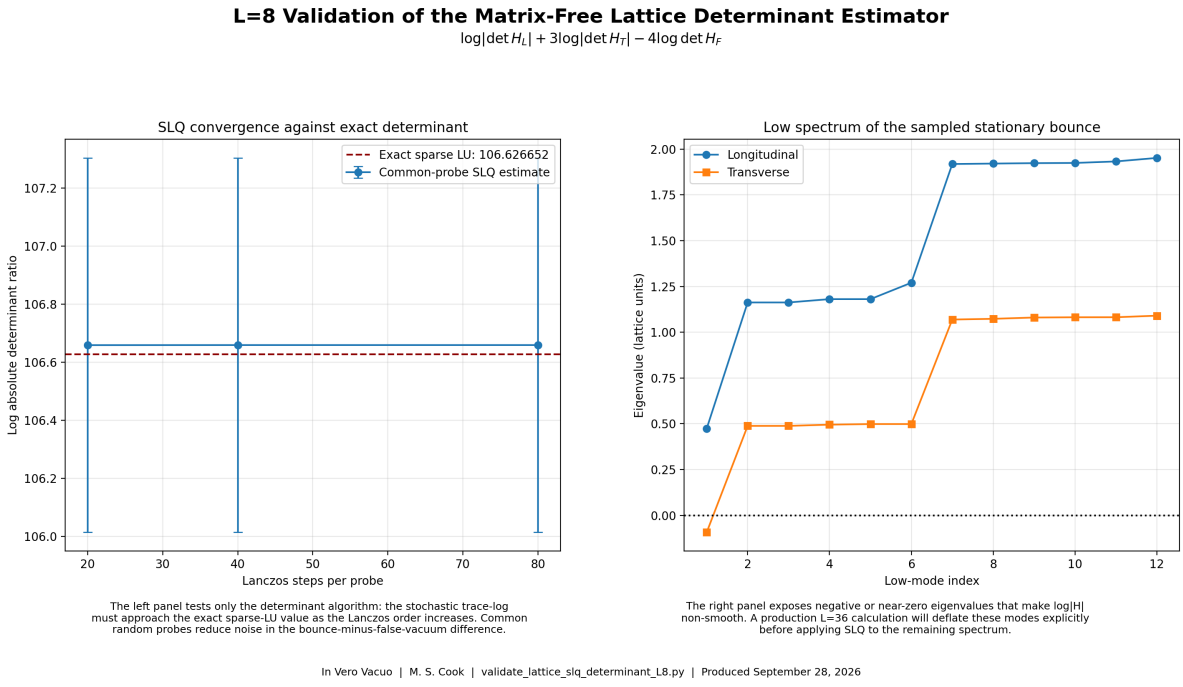



Figure 4.1: Small-lattice validation of the stochastic-Lanczos determinant machinery against a direct reference. Production calculations are accepted only after the matrix-free construction passes analytic checks.

Table 4.1: Core validation diagnostics used by the program.

| Diagnostic | Failure detected |
|---|---|
| Matrix-free/reference product error | Incorrect stencil or component layout |
| Symmetry residual | Inconsistent `matvec`/`rmatvec` or indexing |
| Laplacian row-sum residual | Broken periodic constant mode |
| Low-spectrum error | Eigensolver or operator error |
| Multiplet multiplicities | Lost degeneracies or incomplete spectral window |

## 4.2 New SU(2) prerequisite validation

The `su2_validation` mode tests the quadratic link-action operator defined in Chapter 7. On a $3^4$ grid it checks the exact off-shell lattice Ward identity at $\xi = 0.5, 1, 2$, the $g = 0$ decoupling limit, self-adjointness, and complete homogeneous Fourier spectra. Its optional `HIGGS_SU2_BOUNCE_FILE` input repeats the identity tests on the production $L = 26$, $h = 0.8$ field using the scaled potential.

Table 4.2: Measured SU(2) prerequisite checks, September 30, 2026.

| Check | Result |
|---|---|
| Small-grid off-shell Ward identity | Relative errors below $3.3 \times 10^{-16}$ |
| Stationary-bounce off-shell identity | Relative errors below $2.8 \times 10^{-16}$ |
| Stationary-bounce on-shell identity | Relative errors below $5.6 \times 10^{-14}$ |
| Homogeneous Fourier spectra | Relative errors below $9.3 \times 10^{-16}$ |
| Zero-coupling decoupling | Zero measured relative error |
| Homogeneous normalized one-loop expression | Gauge-parameter spread $5.12 \times 10^{-13}$ |

The homogeneous determinant test includes the gauge-fixing normalization and uses a positive constant broken background. It is not a bounce/false-vacuum determinant ratio. These prerequisites therefore do not establish gauge independence of a completed decay prefactor.

# Chapter 5

# The Continuum $O(4)$ Bounce

## 5.1 Radial equation and shooting method

The least-action bounce may be chosen $O(4)$ symmetric under the standard assumptions for scalar false-vacuum decay [1, 3]. It depends only on $r = (x_1^2 + x_2^2 + x_3^2 + x_4^2)^{1/2}$ and satisfies

$$\rho''(r) + \frac{3}{r}\rho'(r) = V'(\rho), \qquad \rho'(0) = 0, \qquad \rho(\infty) = 0. \tag{5.1}$$

The $3/r$ term is Euclidean radial friction. The code integrates from a small $\epsilon > 0$ using the regular series

$$\rho(\epsilon) = \rho_0 + \frac{V'(\rho_0)}{8}\epsilon^2, \qquad \rho'(\epsilon) = \frac{V'(\rho_0)}{4}\epsilon. \tag{5.2}$$

A bisection search adjusts $\rho_0$ between undershooting and overshooting solutions until the separatrix approaching the false vacuum is isolated.

```
    TRANSLATION_MODES = 4
    ORIENTATION_MODES = 3

    if not BOUNCE_FILE.exists():
        raise FileNotFoundError(f'Relaxed bounce file not found: {BOUNCE_FILE}')
    if DEFLATED_MODES < 6:
        raise ValueError('At least six deflated modes are required.')

    data = np.load(BOUNCE_FILE)
    rho = np.asarray(data['field'], dtype=float)
    if rho.shape != SHAPE:
        raise ValueError(f'Expected relaxed field shape {SHAPE}, found {rho.shape}.')
    rho_sq = rho**2
    u_longitudinal = (
        -MU_SQ + 3.0 * LAMBDA * rho_sq
        + 5.0 * GAMMA * rho_sq**2 + 7.0 * DELTA * rho_sq**3
    )
    u_transverse = (
        -MU_SQ + LAMBDA * rho_sq
        + GAMMA * rho_sq**2 + DELTA * rho_sq**3
    )
    u_false = np.full(SHAPE, FALSE_MASS_SQ)


    def negative_laplacian(field):
        result = 8.0 * field
        for axis in range(4):
            result -= np.roll(field, 1, axis=axis)
```

```
            result -= np.roll(field, -1, axis=axis)
        return result / SPACING**2


    def make_operator(diagonal):
        diagonal = np.asarray(diagonal, dtype=float)

        def matvec(vector):
            field = np.asarray(vector).reshape(SHAPE)
            return (negative_laplacian(field) + diagonal * field).ravel()

        return spla.LinearOperator(
            shape=(N, N), matvec=matvec, rmatvec=matvec, dtype=np.float64
        )


    h_longitudinal = make_operator(u_longitudinal)
    h_transverse = make_operator(u_transverse)
    h_false = make_operator(u_false)


    def load_low_modes(operator, channel):
        cache = OUTPUT_DIRECTORY / f'low_modes_{channel}_k{DEFLATED_MODES}.npz'
        if cache.exists():
            cached = np.load(cache)
            print(f'Loaded {channel} low-mode cache.', flush=True)
            return cached['eigenvalues'], cached['eigenvectors']
        print(f'Computing {DEFLATED_MODES} lowest {channel} modes...', flush=True)
        values, vectors = spla.eigsh(
            operator, k=DEFLATED_MODES, which='SA',
            ncv=max(80, 2 * DEFLATED_MODES + 1), tol=2.0e-9,
            maxiter=30000,
        )
        order = np.argsort(values)
        values = values[order]
        vectors = vectors[:, order]
        np.savez_compressed(cache, eigenvalues=values, eigenvectors=vectors)
        return values, vectors


    long_values, long_vectors = load_low_modes(h_longitudinal, 'longitudinal')
    trans_values, trans_vectors = load_low_modes(h_transverse, 'transverse')

    # Exact mode classification established by the relaxed Hessian calculation:
    # longitudinal index 0 is the physical negative mode; indices 1:5 are the
    # four lattice-lifted translation collective coordinates.  The transverse
    # mode of smallest absolute eigenvalue is one orientation direction, repeated
    # over the three internal transverse components.
    translation_indices = np.arange(1, 1 + TRANSLATION_MODES)
    orientation_index = int(np.argmin(np.abs(trans_values)))
    long_keep = np.ones(DEFLATED_MODES, dtype=bool)
    long_keep[translation_indices] = False
    trans_keep = np.ones(DEFLATED_MODES, dtype=bool)
```

Listing 5.1: Continuum bounce equation, shooting bracket, and physical-tail event.

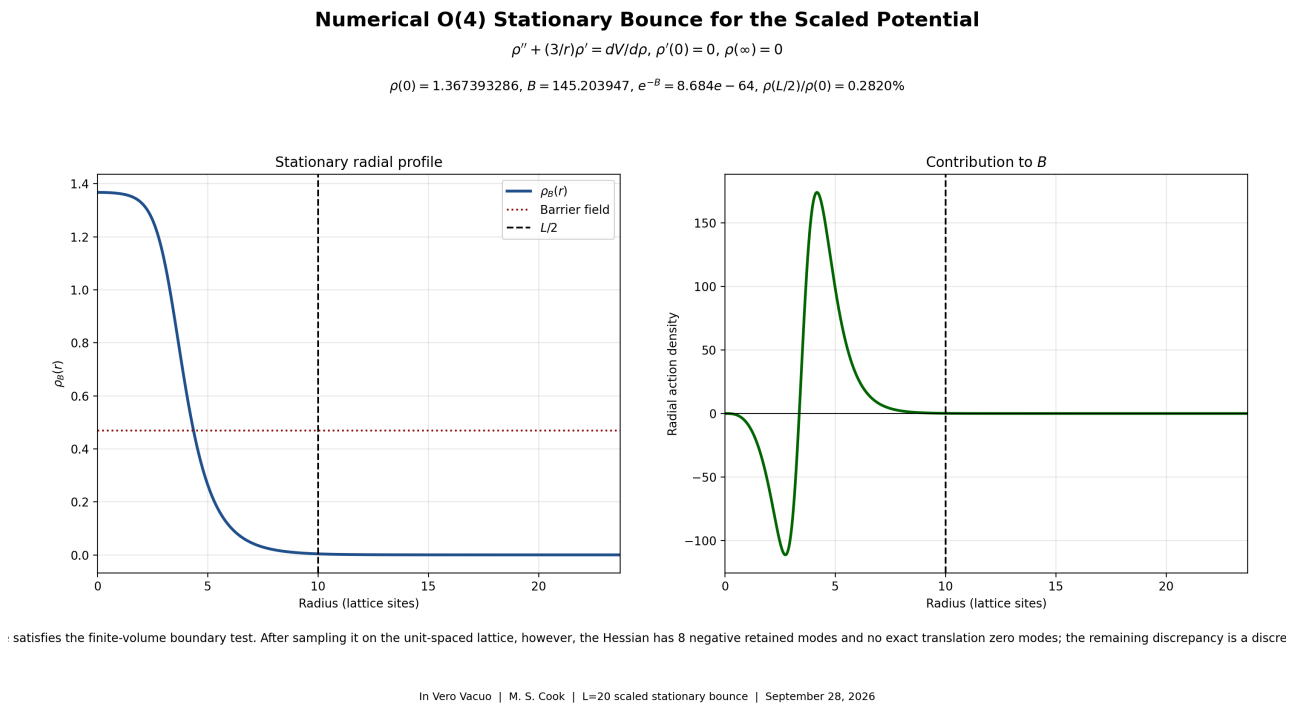



Figure 5.1: Stationary radial bounce and its sampling on the finite periodic lattice. The boundary residual is approximately 0.282%, below the adopted one-percent finite-volume criterion.

## 5.2 Bounce action

For the radial solution,

$$B = 2\pi^2 \int_0^\infty r^3 \left[\frac{1}{2}\rho_B'(r)^2 + V(\rho_B(r)) - V(0)\right] \mathrm{d}r. \tag{5.3}$$

The production calculation gives

$$B = 145.2039474492, \qquad e^{-B} = 8.6841 \times 10^{-64}. \tag{5.4}$$

The integration terminates when the field reaches its physical false-vacuum tail; extending an inaccurate dense-output interpolant beyond this point would contaminate the action.

# Chapter 6

# Discrete-Action Bounce Relaxation

## 6.1 Why continuum sampling is insufficient

A continuum stationary solution sampled on a coarse hypercubic lattice is not automatically stationary under the discrete action. The mismatch lifts the translation modes and can introduce artificial negative deformation modes. The production pipeline therefore solves

$$F[\rho] \equiv -\Delta_h \rho + V'(\rho) = 0 \tag{6.1}$$

directly on the periodic grid.

## 6.2 Newton–MINRES iteration

At iteration $k$, the Newton correction obeys

$$H_L[\rho_k], \delta\rho_k = -F[\rho_k], \qquad H_L = -\Delta_h + V''(\rho_k). \tag{6.2}$$

The linear system is solved by matrix-free MINRES because the longitudinal Hessian is symmetric but indefinite near the saddle. A backtracking line search selects $\alpha$ so that

$$\|F[\rho_k + \alpha\delta\rho_k]\|_2 < \|F[\rho_k]\|_2. \tag{6.3}$$

Every accepted step is checkpointed in compressed NPZ form.

```
            xi_label = str(gauge_xi).replace(".", "p")
            count = min(modal_modes, coupled_values.size)
            metadata_path, _, _ = export_su2_modal_dataset(
                modal_output, f"homogeneous_L3_xi_{xi_label}_coupled_A_chi",
                coupled, coupled_values[:count], coupled_vectors[:, :count],
                sector="coupled_A_chi",
                background={
                    "type": "homogeneous_true_vacuum", "field_array": homogeneous_rho,
                    "rho": float(np.sqrt(true_rho_sq)), "goldstone_curvature": 0.0
                },
                lattice={"shape": list(shape), "spacing": spacing,
                         "boundary_conditions": "periodic"},
                parameters={"g": gauge_coupling, "xi": gauge_xi,
                            "ir_shift": 0.0, "generator_multiplicity": 3},
                explicit_operator=(coupled_matrix if coupled.shape[0]
                                   <= modal_max_operator_dim else None)
            )
            modal_datasets.append(str(metadata_path))
```

Listing 6.1: Discrete residual and channel Hessian used by Newton–MINRES.

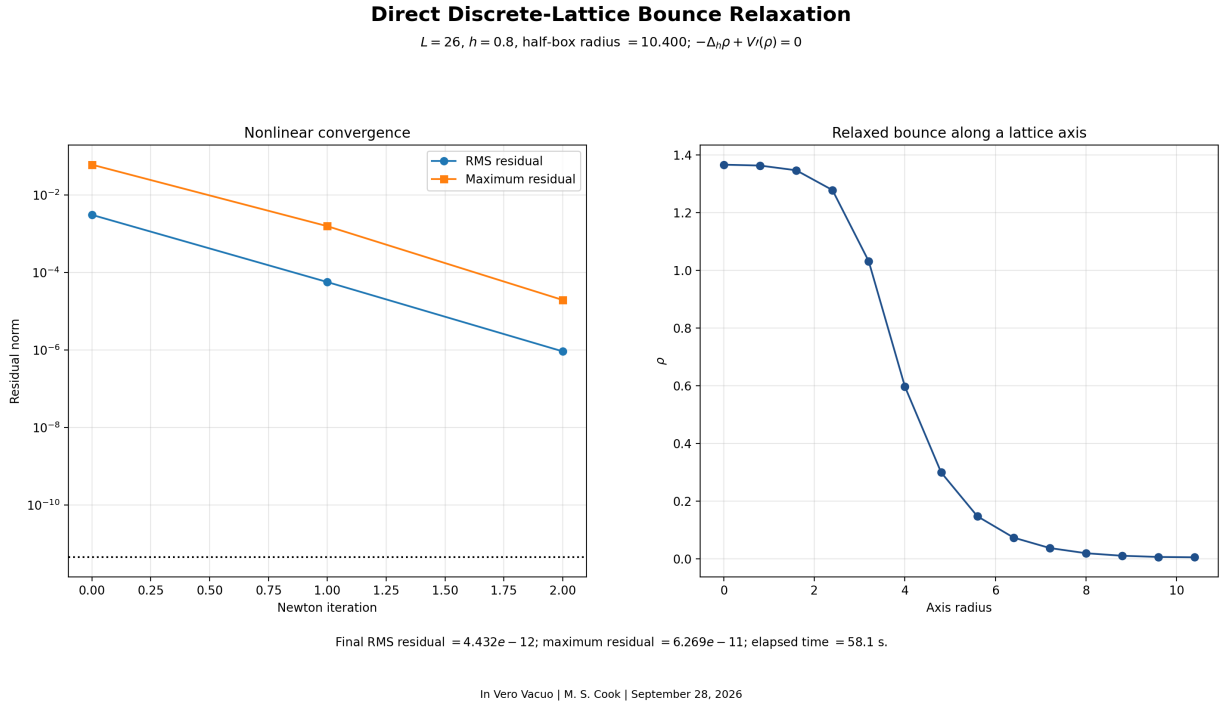



Figure 6.1: Nonlinear convergence and the final relaxed bounce along a lattice axis. The final RMS equation-of-motion residual is $4.43 \times 10^{-12}$.

**Key result**

For $L = 26$ and $h = 0.8$, the relaxed field has RMS residual $4.43266 \times 10^{-12}$ and maximum residual $6.30906 \times 10^{-11}$ in the regenerated September 30 field. The identity $H_T \rho = F[\rho]$ is satisfied to the same accuracy, independently confirming the internal-orientation zero mode.

# Chapter 7

# Hessian Construction and Fluctuation Channels

## 7.1 Longitudinal and transverse curvatures

For an $O(4)$-invariant potential, fluctuations separate into one radial longitudinal channel and three degenerate transverse channels. On the bounce,

$$U_L(\rho) = -\mu^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \tag{7.1}$$

$$U_T(\rho) = -\mu^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6. \tag{7.2}$$

The scalar channel operators are

$$H_L = -\Delta_h + U_L(x), \qquad H_T = -\Delta_h + U_T(x), \qquad H_F = -\Delta_h + m_F^2. \tag{7.3}$$

Only one transverse operator must be diagonalized; its determinant contribution is multiplied by three.

## 7.2 Spectral meaning of the special modes

The physical bounce spectrum and its treatment by collective coordinates follow the standard semiclassical construction [2]. It should contain:

1. one negative radial mode, producing the imaginary part associated with decay;
2. four translation collective modes, one for each Euclidean coordinate;
3. three internal-orientation collective modes for global $O(4) \to O(3)$;
4. a positive continuum above these special modes.

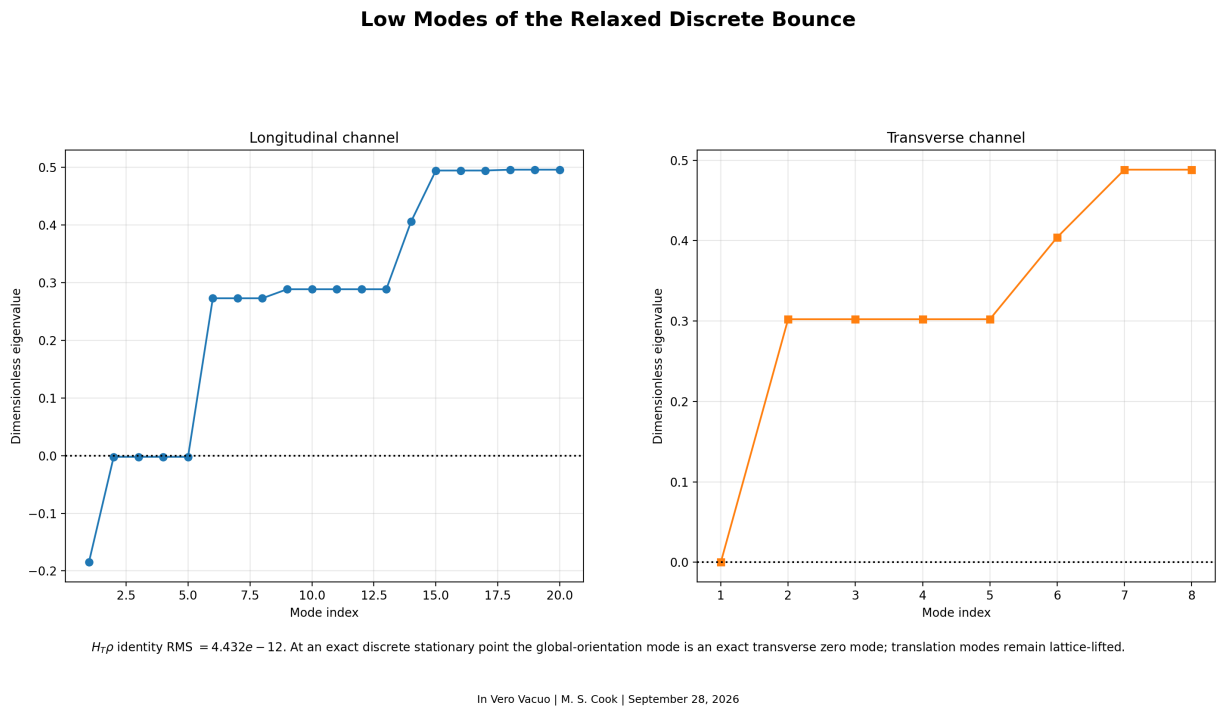



Figure 7.1: Low longitudinal and transverse spectra around the relaxed discrete bounce. The principal longitudinal instability is separated from the four translation modes; the transverse orientation mode is numerically zero.

Table 7.1: Low-mode structure of the relaxed $L = 26$, $h = 0.8$ configuration.

| Sector | Multiplicity | Dimensionless eigenvalue |
|---|---|---|
| Radial instability | 1 | $-1.8504151 \times 10^{-1}$ |
| Translations | 4 | $-2.1265306 \times 10^{-3}$ |
| Internal orientation | 3 physical copies | $-5.33 \times 10^{-12}$ |
| First positive longitudinal level | 3-fold | $2.7292064 \times 10^{-1}$ |
| First positive transverse level | 4-fold per channel | $3.0226725 \times 10^{-1}$ |

**Scope and interpretation**

The translation modes are not exactly zero on a fixed lattice because continuous translations are broken to discrete shifts. They are identified by their fourfold degeneracy and collective-coordinate character, then removed from the primed determinant. The radial negative mode is retained through $\log|\lambda_-|$.

## 7.3 New link-derived SU(2) gauge–Goldstone Hessian

The scalar production pipeline is unchanged, but research mode now provides a separate quadratic gauge extension when `HIGGS_SU2_GAUGE=1`. The background is a radial Higgs doublet with vanishing gauge field. Each of the three generators has the same five-field block $(A_1, A_2, A_3, A_4, \chi)$; the radial Higgs remains a separate scalar block. Thus the bosonic dimension is $16L^4$, or 7,311,616 at $L = 26$, while the three complex ghost operators are handled separately.

### 7.3.1 Why the link stencil replaces the centered stencil

The initial gauge diagnostic used a centered continuum gradient and a site-local gauge mass. It was self-adjoint, but self-adjointness alone does not imply the discrete gauge identity. The current implementation instead expands the covariant scalar-link and zero-background gauge-link actions. Gauge fields occupy outgoing links, scalar fields occupy sites, and the positive background $R_\xi$ parameter is no longer restricted to $\xi = 1$. Define the forward gradient $G$, backward divergence $D$, $G^\dagger = -D$, and $K = -DG$. With $q = g/2$ and $F = DA - \xi q \rho \chi$, the implemented one-generator operators are

$$H_{A\chi} = \begin{pmatrix} K\mathbf{1} + (1-\xi^{-1})GD + M^2_{\mathrm{link}} & C \\ C^\dagger & K + U_T + \xi q^2 \rho^2 \end{pmatrix}, \qquad H_c = K + \xi q^2 \rho^2, \tag{7.4}$$

where

$$(M^2_{\mathrm{link}})_{x\nu} = q^2 \rho_x \rho_{x+\hat{\nu}}, \qquad (C\chi)_{x\nu} = \frac{q}{h}(\rho_{x+\hat{\nu}} - \rho_x)(\chi_x + \chi_{x+\hat{\nu}}). \tag{7.5}$$

The continuum limit of the latter is $2q(\partial_\nu \rho)\chi$. The stored research field is converted to $\rho_{\mathrm{phys}} = \rho/a$ before calling the shared operator constructor. Production validation instead supplies the field and spacing directly in scaled units.

```
for reproducibility, but they are no longer required by this program.
"""

import os as _pipeline_os
import subprocess as _pipeline_subprocess
import sys as _pipeline_sys

from su2_lattice_vertices import (
    faddeev_popov_matrix as make_nonlinear_su2_ghost_matrix,
    generate_one_loop_vertex_tables,
    ghost_gauge_vertex as nonlinear_su2_ghost_gauge_vertex,
    ghost_scalar_vertices as make_nonlinear_su2_scalar_ghost_vertices,
    scalar_link_vertex as nonlinear_su2_scalar_link_vertex,
    wilson_gauge_vertex as nonlinear_su2_wilson_gauge_vertex,
)


def _pipeline_env_flag(name, default=False):
    """Read a conservative Boolean environment flag."""
    value = _pipeline_os.getenv(name)
    if value is None:
        return bool(default)
    normalized = value.strip().lower()
    if normalized in {"1", "true", "yes", "on"}:
        return True
    if normalized in {"0", "false", "no", "off"}:
        return False
    raise ValueError(f"{name} must be one of 0/1, false/true, no/yes, or off/on.")


def make_periodic_scalar_operator(curvature, spacing):
    """Return -Delta plus a site curvature on a periodic four-dimensional grid."""
    import numpy as np
    import scipy.sparse.linalg as spla

    curvature_grid = np.asarray(curvature, dtype=float)
    if curvature_grid.ndim != 4 or not np.all(np.isfinite(curvature_grid)):
        raise ValueError("The scalar curvature must be a finite four-dimensional grid."
            )
    if not np.isfinite(spacing) or spacing <= 0.0:
```

```
        raise ValueError("The scalar-operator spacing must be finite and positive.")

    def matvec(vector):
        scalar = np.asarray(vector, dtype=float).reshape(curvature_grid.shape)
        laplacian = sum(
            (2.0 * scalar - np.roll(scalar, 1, axis=axis)
             - np.roll(scalar, -1, axis=axis)) / spacing**2
            for axis in range(4)
        )
        return (laplacian + curvature_grid * scalar).ravel()

    site_count = curvature_grid.size
    return spla.LinearOperator(
        (site_count, site_count), matvec=matvec, rmatvec=matvec,
        dtype=np.float64
    )


def export_su2_modal_dataset(output_directory, dataset_name, operator,
                             eigenvalues, eigenvectors, *, sector,
                             background, lattice, parameters,
                             explicit_operator=None, degeneracy_tolerance=1.0e-9):
    """Export versioned eigenpairs and metadata for quantum-algorithm benchmarks.

    Eigenvectors are stored as columns with ordinary discrete l2 norm one.  A
    deterministic sign is chosen by making each vector's largest-magnitude
    entry positive.  Degenerate subspaces remain basis-dependent and are
    identified in the JSON sidecar rather than assigned physical vector labels.
    """
    import hashlib
    import json
    from datetime import datetime, timezone
    from pathlib import Path

    import numpy as np
    import scipy.sparse as sp

    output = Path(output_directory)
    output.mkdir(parents=True, exist_ok=True)
```

Listing 7.1: Shared link-action gauge–Goldstone and ghost constructors.

### 7.3.2 Ward identities, normalization, and determinant scope

For $R\alpha = (G\alpha, q\rho\alpha)$, the off-shell identity contains the scalar EOM residual:

$$H_{A\chi} R\alpha = \begin{pmatrix} \xi^{-1} G(H_c \alpha) \\ q\rho H_c \alpha \end{pmatrix} + \begin{pmatrix} 0 \\ q\alpha[K\rho + V\prime(\rho)] \end{pmatrix}. \tag{7.6}$$

The new validation mode retains this term on arbitrary profiles and checks its smallness on the regenerated stationary bounce. A Gaussian cannot be substituted for the stationary field in a physical prefactor calculation.

The homogeneous test combines $\frac{1}{2}\log\det H_{A\chi}$, $-\log\det H_c$, and the gauge-fixing normalization $\frac{N}{2}\log\xi$. Its measured agreement across three gauges is a finite-volume prerequisite, not a completed bounce-rate calculation. In the research report, coupled and ghost spectra are written to a separate CSV and a third diagnostic PDF page. They are explicitly excluded from the scalar IR determinant and from scalar SLQ/continuum results.

**Scope and interpretation**

The previous centered-stencil SU(2) Gaussian report is historical diagnostic output. Its numbers must not be reused as results of the revised link operator. Chapter 17 records the foundational report and defines the adopted finite EFT matching conditions, including the one-loop-order $\rho^{10}$ and $\rho^{12}$ coefficients.

# Chapter 8

# Regulated SU(2) Determinants and Counterterms

## 8.1 Gauge–Goldstone–ghost determinant stage

For one $SU(2)$ generator, the new `su2_determinant` mode evaluates the regulated finite-lattice contribution

$$\Delta\Gamma^{(a)}_{1,g} = \frac{1}{2}\log\frac{\det H_{A\chi,B}}{\det H_{A\chi,F}} - \log\frac{\det H_{c,B}}{\det H_{c,F}}. \tag{8.1}$$

The radial background makes the three generator blocks identical, so the reported gauge-sector logarithm is three times Eq. (8.1). The mode provides an exact eigensystem backend for coupled dimensions no larger than 3000 and a production-oriented backend using low-mode deflation and stochastic Lanczos quadrature. Common Rademacher probes are reused at every value of $\xi$, reducing the stochastic uncertainty in gauge-parameter differences.

At the symmetric false vacuum, $\rho_F = 0$, each generator contains four constant gauge modes and one constant ghost mode on the periodic box. The output records both raw IR-regulated ratios and primed ratios in which these constant modes are omitted. Removing the constant longitudinal gauge mode also removes a factor from the gauge-fixed Gaussian measure; the implemented relative correction is

$$\Delta\Gamma^{(a)}_{\text{measure}} = \frac{1}{2}\log\xi \tag{8.2}$$

per generator. Raw, raw-normalized, primed, and primed-normalized results are therefore retained separately so that apparent gauge dependence can be audited rather than hidden.

Table 8.1: Exact $3^4$ homogeneous determinant cancellation test.

| IR mass-squared shift | Spread over $\xi = 0.5, 1, 2$ | Status |
|---|---|---|
| $10^{-8}$ | $1.079 \times 10^{-7}$ | PASS |
| $10^{-6}$ | $1.079 \times 10^{-5}$ | PASS |
| $10^{-4}$ | $1.078 \times 10^{-3}$ | PASS |

The decreasing spread is consistent with restoration of the normalized gauge-independent result as the auxiliary IR shift is removed. This is an exact homogeneous validation, not the renormalized production-bounce determinant. The production checkpoint was regenerated and validated on October 2 as described below, but it must be regenerated or transferred into the execution environment before the production scan can begin. Counterterm matching and spacing and volume limits remain necessary afterward.

## 8.2 October 2 production-checkpoint recovery

The missing $L = 26$, $h = 0.8$ stationary-bounce checkpoint was regenerated from the continuum initial profile using the Newton–MINRES relaxation mode. Three accepted Newton corrections reduced the residual from $3.06017 \times 10^{-3}$ to the previously archived stationary solution. The recovered field had center value 1.365689301 and was written to

`su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz`.

Before the workspace interruption, the regenerated checkpoint passed the complete prerequisite suite. The measured production-background quantities were

$$B_h = 139.9422784742, \tag{8.3}$$

$$\|\mathcal{E}_h\|_{\mathrm{RMS}} = 4.4316323 \times 10^{-12}, \tag{8.4}$$

$$\|\mathcal{E}_h\|_{\infty} = 6.2693073 \times 10^{-11}. \tag{8.5}$$

The stationary-background on-shell Ward residuals at $\xi = 0.5, 1, 2$ were respectively $1.38 \times 10^{-14}$, $2.76 \times 10^{-14}$, and $5.52 \times 10^{-14}$. The independent homogeneous normalized gauge spread was $1.14 \times 10^{-13}$. The resulting validation status was **PASS**, and the surviving machine-readable record is `su2_prefactor_validation_20261002/validation_summary.json`.

**Workspace-persistence event**

A subsequent cloud-workspace reconnection restored the project without the untracked NPZ directory. Consequently, the production determinant scan did not start and no determinant probes were lost. The checkpoint must be regenerated once more, or copied from persistent external storage, before the scan is launched. The numerical agreement with the archived solution shows that regeneration is deterministic at the reported precision.

## 8.3 Local EFT basis and pole closure

The `su2_counterterms` mode evaluates the radial restriction of the adopted gauge-invariant local basis,

$$\Delta S_{\mathrm{ct}} = \int \mathrm{d}^4x \Big[ \frac{1}{2}\delta Z_{\mathcal{H}} (\partial_\mu \rho)^2 + \frac{1}{2}\delta m^2 \rho^2 + \frac{1}{4}\delta\lambda\rho^4 + \frac{1}{6}\delta\gamma\rho^6 + \frac{1}{8}\delta\delta\rho^8 + \frac{1}{10}\delta c_{10}\rho^{10} + \frac{1}{12}\delta c_{12}\rho^{12} \Big]. \tag{8.6}$$

Each basis integral is stored independently, allowing finite matching coefficients to be updated without recomputing the background invariants.

With $m_F^2 = -\mu^2$, the longitudinal and transverse scalar curvatures are

$$U_L = m_F^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \tag{8.7}$$

$$U_T = m_F^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6. \tag{8.8}$$

The scalar pole polynomial obtained from $U_L^2 + 3U_T^2 - 4m_F^4$ closes through $\rho^{12}$. In background Feynman gauge, with $q = g/2$, the radial-background pole density including the three gauge generators is

$$P_{\mathrm{total}} = U_L^2 + 3U_T^2 + 6q^2\rho^2 U_T + 9q^4\rho^4 + 24q^2(\partial_\mu\rho)^2. \tag{8.9}$$

Writing $C = (64\pi^2\bar{\epsilon})^{-1}$, the residues that cancel the positive $CP_{\mathrm{total}}$ pole for the production parameters and $g = 0.65$ are listed in Table 8.2.

Table 8.2: Derived continuum Feynman-gauge pole residues.

| Coefficient | Residue divided by $C$ |
|---|---|
| $\delta Z_{\mathcal{H}}$ | $-5.070000$ |
| $\delta m^2$ | $18.693000$ |
| $\delta\lambda$ | $-207.8116390625$ |
| $\delta\gamma$ | $338.718000$ |
| $\delta\delta$ | $-82.325600$ |
| $\delta c_{10}$ | $-48.640000$ |
| $\delta c_{12}$ | $-3.993600$ |

The direct matrix-invariant expression and Eq. (8.9) agree to $3.11 \times 10^{-15}$, while the normalized residues cancel the integrated pole density to $1.70 \times 10^{-15}$. These are continuum pole residues, not the finite constants converting the implemented lattice regulator to $\overline{\mathrm{MS}}$. The finite entries in `su2_counterterm_coefficients_template.json` therefore remain explicitly unset until the lattice momentum and derivative matching calculations are completed.

**Scope and interpretation**

The October 1 implementation supplies a regulated determinant architecture, zero-mode and measure policies, the retained local operator basis, and continuum pole closure. It does not yet supply a counterterm-matched, continuum- and volume-extrapolated $SU(2)$ decay prefactor.

# Chapter 9

# October 5 Single-Core SU(2) Validation and Variance Audits

## 9.1 Scope, execution policy, and evidence status

The October 5, 2026 work advanced the quadratic $SU(2)$ branch through four stages: a small four-dimensional operator pilot, a homogeneous increased-probe/quadrature audit, a complete-low-mode-multiplet variance comparison, and a nonconstant-background prerequisite audit. This chapter incorporates their test results into the technical guide rather than leaving them only in separate reports. All backgrounds in these October 5 studies were prescribed off-shell diagnostics, not stationary bounces. No production gauged decay prefactor was calculated.

Execution was sequential on one CPU, with CPU affinity {0} and `OPENBLAS_NUM_THREADS`, `OMP_NUM_THREADS`, `MKL_NUM_THREADS`, and `NUMEXPR_NUM_THREADS` set to one. The recorded environment was Python 3.13.15, NumPy 2.4.1, and SciPy 1.17.0. The separate harnesses used definitions from the original source; they did not execute its top-level scalar pipeline or edit the production solver. Its recorded unchanged SHA-256 was

`c592079bdb8c3632f7d0dc413c98c012bff43fbc393931439af4be1b3f9e0b1f`

**Result provenance for this revision**

The initial pilot summary is present under `su2_L6_single_core_pilot_20261005/`. The increased-probe and multiplet values below are recorded results transcribed from their standalone October 5 reports; the earlier follow-up raw directories and saved bases are absent from the current snapshot. The nonconstant study was a fresh run, and its harness, aggregate audit summary, symbolic closure, and ten counterterm summaries are present. However, its complete raw matrices, spectra, probes, and comparison CSVs are no longer present in this snapshot. This document update checks the available summaries and incorporates the reports; it does not rerun the calculations or claim a new independent verification of missing raw data.

## 9.2 Four-dimensional $L = 6$ operator pilot

The pilot used a periodic $6^4$ lattice with $h = 0.8$, $g = 0.65$, $\xi = 1$, and auxiliary IR mass-squared shift $\epsilon = 0.003$. One generator has 6,480 coupled gauge–Goldstone and 1,296 ghost degrees of freedom. The two profiles were

$$\rho_{\mathrm{hom}}(n) = 0.1, \qquad \rho_{\mathrm{var}}(n) = 0.1 + 0.04 \prod_{\mu=1}^{4} \cos(2\pi n_\mu/6).$$

The varying profile spans 0.06–0.14 and activates gradient mixing. Eight smallest-algebraic eigenpairs were requested in each sector; this fixed count did not certify complete low-mode degeneracies.

Table 9.1: October 5 $L = 6$ direct operator checks. H and V denote homogeneous and varying profiles. Residuals use $\|Hv - \lambda v\|/\max(1, |\lambda|)$ for normalized $v$.

| Case | Sector | Symmetry | Linearity | Minimum $\lambda$ | Max. residual |
|---|---|---|---|---|---|
| H | Coupled | $1.17 \times 10^{-17}$ | $1.10 \times 10^{-16}$ | 0.00105625 | $3.62 \times 10^{-9}$ |
| H | Ghost | 0 | $1.07 \times 10^{-16}$ | 0.00105625 | $9.06 \times 10^{-12}$ |
| V | Coupled | $1.05 \times 10^{-17}$ | $1.22 \times 10^{-16}$ | 0.00105406 | $6.58 \times 10^{-10}$ |
| V | Ghost | $2.55 \times 10^{-17}$ | $1.04 \times 10^{-16}$ | 0.00106681 | $1.33 \times 10^{-10}$ |

The reported observable throughout these audits is the regulated three-generator gauge-sector combination

$$\begin{aligned}\mathcal{D}_\epsilon = 3\big[&\tfrac{1}{2}(\log\det H_{A\chi,B,\epsilon} - \log\det{}' H_{A\chi,F,\epsilon}) \\ &- (\log\det H_{c,B,\epsilon} - \log\det{}' H_{c,F,\epsilon}) + \tfrac{1}{2}\log\xi\big],\end{aligned} \tag{9.1}$$

where $H_\epsilon = H + \epsilon I$ and $B$ denotes the diagnostic profile. The false-vacuum prime removes four constant gauge modes and one constant ghost mode per generator before shifting. The radial Higgs determinant is not included. Deflated diagnostic modes remain in the determinant through their explicit log eigenvalues; deflation is not this prime policy.

Table 9.2: Initial four-probe SLQ means and one estimated standard error, using integer seed 20261005 and the same probes at each order.

| Order | Homogeneous | Varying |
|---|---|---|
| 10 | $-29.6992 \pm 12.0430$ | $-9.8218 \pm 12.7586$ |
| 20 | $-45.2381 \pm 11.5188$ | $-18.1826 \pm 9.9116$ |
| 30 | $-45.2406 \pm 11.5187$ | $-18.1851 \pm 9.9117$ |

The paired mean order-30 minus order-20 changes were $-0.002522$ and $-0.002580$; maximum absolute per-probe changes were 0.003087 and 0.003108. Ten steps were insufficient. Stable higher-order estimates at fixed probes did not establish a precise trace. The homogeneous exact Fourier target was $\mathcal{D}_{\epsilon,\text{exact}} = -19.56733155726579$. The initial order-30 mean differed by $-25.6733154$, or 2.23 estimated standard errors; this discrepancy is retained rather than relabelled as precision validation. The tested operator gate required symmetry and linearity below $10^{-12}$, low-pair residuals below $10^{-7}$, positive computed minimum eigenvalue plus shift, and finite estimates. Both profiles passed that gate, not a determinant-accuracy gate. The complete pilot took 7.6775 seconds; peak memory was not recorded. A single-$\xi$ automatic gauge-spread PASS is vacuous and is not evidence of gauge independence.

## 9.3 Increased probes and independent quadrature references

The homogeneous follow-up kept the same lattice, parameters, prime policy, and eight deflation vectors per sector. It used 4,096 independent Rademacher probe pairs with seed 20261005 and orders $10, 20, 30, 40, 60, 80$. Its four-probe prefix is not the initial pilot: a different deflation basis was generated. The 1,024- and 4,096-probe follow-ups reused one saved basis.

The final discrepancy was 0.727 estimated standard errors. The uncertainty approximately halved from 1,024 to 4,096 probes. Only one of the four gauge constant modes was covered by the eight coupled deflation vectors: they were not the complete eight lowest modes counted with multiplicity. The recorded

Table 9.3: Recorded nested prefixes of the homogeneous follow-up at 80 steps. Prefixes are correlated, not independent ensembles.

| Probes | Mean | SEM | Mean minus exact |
|---|---|---|---|
| 4 | −27.734582 | 18.319092 | −8.167251 |
| 16 | −40.690843 | 25.294243 | −21.123511 |
| 64 | −26.002996 | 13.888968 | −6.435665 |
| 256 | −21.893954 | 6.720888 | −2.326622 |
| 1024 | −20.992059 | 3.247177 | −1.424727 |
| 4096 | −20.750938 | 1.628882 | −1.183606 |

Fourier expected-trace bias check was approximately $1.21 \times 10^{-12}$, negligible here but not a coverage certificate.

For every probe, an orthonormal Fourier transform evaluated the same projected quadratic form directly, with $P = I - VV^T$:

$$q_{\mathrm{ref}}(z) = \sum_{j \in \mathrm{low}} \log(\lambda_j + \epsilon) + \sum_k |\mathcal{F}(Pz)_k|^2 \log s_k.$$

SLQ minus this individual reference isolates quadrature error rather than sampling error relative to the trace.

Table 9.4: Recorded quadrature error over 4,096 homogeneous probes.

| Order | Mean error | Maximum absolute probe error |
|---|---|---|
| 10 | 13.6303 | 66.2927 |
| 20 | $2.5171 \times 10^{-3}$ | $4.4397 \times 10^{-3}$ |
| 30 | $-9.18 \times 10^{-12}$ | $6.00 \times 10^{-11}$ |
| 40 | $-1.15 \times 10^{-11}$ | $7.09 \times 10^{-11}$ |
| 60 | $-1.19 \times 10^{-11}$ | $6.55 \times 10^{-11}$ |
| 80 | $-1.14 \times 10^{-11}$ | $1.04 \times 10^{-10}$ |

An independent exact-arithmetic Fourier-measure Gauss/left-Radau bracket used 34 coupled and 17 ghost distinct spectral values. For $\log x$, the sector bounds are $R_m(a) \le \int \log x \, d\mu \le G_m$, with $a = 0.99 \min s_k > 0$. The combined lower and upper bounds use $3(\frac{1}{2}L_c - U_g) + C$ and $3(\frac{1}{2}U_c - L_g) + C$, respectively, where $C$ is the exact false-vacuum contribution. Maximum widths at 10, 20, and 30 steps were 68.5729, 0.26145, and $7.0215 \times 10^{-8}$. Finite-support quadrature terminates by 34/17 sector steps in exact arithmetic; the 40–80-step theoretical widths therefore vanish. Numerical reference-bracket violations reached $8.19 \times 10^{-11}$. These are not directed-rounding certificates for the production recurrence, and theoretical termination does not imply machine-exact results. This finite-support result does not transfer to a nonconstant background. The recorded 4,096-probe run took 256.1 seconds, including Fourier and bracket audits, and produced 24,576 probe/order rows.

## 9.4 Complete low-mode multiplets and variance reduction

For the homogeneous control, the audit built real sine/cosine bases for conjugation-closed Fourier masks. The four gauge constant modes, one Goldstone constant mode, and one ghost constant mode can then be covered explicitly. Completing an eight-mode target requires 37/9 coupled/ghost vectors because the boundary intersects degenerate shells. Two independent seeds, 20261005 and 20261006, each used 4,096

probe pairs; the same pairs were reused across six schemes within each seed. These integers are seeds, not execution dates.

For an orthonormal deflation basis, the Rademacher residual estimator has

$$M = P\log(H+\epsilon I)P, \qquad \mathrm{Var}(z^T M z) = 2(\|M\|_F^2 - \sum_i M_{ii}^2).$$

Independent coupled and ghost probes give combined single-probe variance $9(\frac{1}{4}\,\mathrm{Var}_c + \mathrm{Var}_g)$. The homogeneous Fourier structure allowed direct double-precision variance calculation rather than cutoff selection from sampled fluctuations.

Table 9.5: Recorded complete-multiplet comparison. Positive reduction is a variance decrease relative to the freshly generated eight-vector baseline.

| Scheme | Coupled/ghost | Probe variance | SEM at 4096 | Reduction |
|---|---|---|---|---|
| No deflation | 0/0 | 11840.37 | 1.70021 | −6.21% |
| Fresh eight-vector | 8/8 | 11148.31 | 1.64977 | Reference |
| Complete constants | 5/1 | 9753.25 | 1.54310 | 12.51% |
| Complete eight-mode target | 37/9 | 10327.65 | 1.58789 | 7.36% |
| First shell, all fields | 45/9 | 10430.68 | 1.59579 | 6.44% |
| Second shell, all fields | 165/33 | 14603.28 | 1.88819 | −30.99% |

Complete constants were the best tested cutoff: 12.51% less variance, or about 6.47% less standard error at fixed count. This is a modest gain, not a universal optimum. More deflation need not reduce this uncentered estimator variance; the second shell increased the residual off-diagonal log-operator norm. The fresh eight-vector basis again covered only one gauge constant mode and must not be equated with the earlier saved 4,096-probe basis. Complete schemes recorded maximum orthogonality error $4.76\times10^{-15}$ and eigenpair residual $4.59\times10^{-16}$; the constant scheme expected combined trace-bias check was $8.31\times10^{-13}$.

Full ensembles used Fourier quadratic forms, not 8,192 full SLQ runs. For complete constants the seed means were $-21.266991\pm1.530133$ and $-17.952936\pm1.551980$. Pooling only these two independent ensembles yielded

$$\widehat{\mathcal{D}}_\epsilon = -19.609964 \pm 1.089806 \quad \text{(8192 probes; one estimated standard error)}.$$

The difference from exact was $-0.042632$; uncertainty near 1.09 does not establish four-decimal accuracy. Shared-probe schemes cannot be pooled as independent estimates. Original SLQ was checked on 64 matched pairs per scheme at orders 20, 30, and 40. For complete constants, maximum quadrature errors were $7.55\times10^{-9}$, $7.64\times10^{-11}$, and $9.09\times10^{-11}$; for no deflation they were 1.74, $3.17\times10^{-8}$, and $7.82\times10^{-11}$. These subset measurements are not certified full-ensemble bounds. The recorded finalized harness run took approximately 20.3 seconds, dominated by Fourier and subset checks.

## 9.5 Fresh nonconstant-background prerequisites on $L = 3, 4$

The fresh audit retained $h = 0.8$ and $g = 0.65$ and used periodic $3^4$ and $4^4$ grids, with coupled/ghost dimensions 405/81 and 1280/256. It swept $\xi = 0.5, 1, 2$ and $\epsilon = 0.001, 0.003, 0.01$ on profiles

$$\rho_A(n) = 0.1 + A\prod_{\mu=1}^{4}\cos(2\pi n_\mu/L).$$

Determinant and SLQ checks used $A = 0, 0.04$; local-basis tests also used $A = 0.005, 0.01, 0.02$. Each determinant case used 64 independent pairs per seed for two seeds, orders 20, 40, and 80, and a target-eight

low window expanded through degeneracy boundaries. Both grids have the same spacing and different physical volumes: this is not a continuum extrapolation.

Explicit matrices were assembled by applying the original matrix-free operators to coordinate unit vectors. Tests used the unsymmetrized matrices; full dense eigensystems supplied deterministic traces and individual projected-probe references. Matrix assembly and operator action share an implementation, so agreement is a consistency check complemented by analytic homogeneous spectra and the off-shell Ward identity.

Table 9.6: Fresh nonconstant-audit maxima, checked against $10^{-10}$ gates.

| Diagnostic | Maximum error |
|---|---|
| Matrix self-adjointness | 0 |
| Linearity | $2.04 \times 10^{-16}$ |
| Matrix-free versus assembled action | $2.59 \times 10^{-16}$ |
| Low-eigenpair relative residual | $6.28 \times 10^{-14}$ |
| Off-shell lattice Ward identity | $2.99 \times 10^{-16}$ |
| Zero-coupling decoupling at $\xi = 1$ | $1.48 \times 10^{-16}$ |
| Homogeneous versus analytic spectrum | $4.97 \times 10^{-14}$ |
| Independent local pole-basis closure | $1.53 \times 10^{-15}$ |

With $q = g/2$ and $R_\rho = -\Delta\rho + U_T\rho$, the Ward test retained the equation-of-motion term:

$$H_{A\chi}\begin{pmatrix}\nabla\alpha\\ q\rho\alpha\end{pmatrix} = \begin{pmatrix}\nabla(H_c\alpha)/\xi\\ q\rho H_c\alpha + q\alpha R_\rho\end{pmatrix}.$$

Three random $\alpha$ fields were used per gauge parameter. The varying profiles had RMS equation-of-motion residuals 0.194615 ($L = 3$) and 0.133865 ($L = 4$), not approximately zero. Passing this off-shell identity does not establish physical gauge independence.

The low window closes when adjacent levels differ by more than $10^{-9}\max(1, |\lambda|)$. Varying coupled/ghost ranks were 9/8 for all three gauges at $L = 3$, and 17/9, 9/9, 9/9 at $L = 4$ in increasing $\xi$ order. Homogeneous coupled ranks were $29, 37, 13$, with ghost rank nine on both grids. No homogeneous constant-mode shortcut was imposed on nonconstant fields.

Table 9.7: Fresh varying-profile exact determinants and 80-step estimates at $\epsilon = 0.003$. Each seed mean uses 64 pairs; uncertainties are one SEM.

| $L$ | $\xi$ | Exact | Seed 20261005 | Seed 20261006 |
|---|---|---|---|---|
| 3 | 0.5 | $-17.4139$ | $-23.4876 \pm 5.3701$ | $-15.0475 \pm 4.7876$ |
| 3 | 1 | $-16.7945$ | $-24.6894 \pm 4.7614$ | $-14.8426 \pm 4.4228$ |
| 3 | 2 | $-16.4506$ | $-26.0028 \pm 4.7508$ | $-14.3035 \pm 4.3497$ |
| 4 | 0.5 | $-17.8121$ | $-9.2855 \pm 7.1145$ | $-14.8401 \pm 7.4645$ |
| 4 | 1 | $-17.1935$ | $-12.0467 \pm 6.6016$ | $-19.8538 \pm 6.9532$ |
| 4 | 2 | $-16.8513$ | $-17.4302 \pm 7.3510$ | $-21.9011 \pm 7.1914$ |

Maximum individual combined SLQ-minus-dense-reference errors across cases and shifts were $1.00 \times 10^{-11}$ at 20 and 40 steps and $1.05 \times 10^{-11}$ at 80 steps. Sampling error remained dominant. The largest discrepancy from exact among 72 correlated 80-step ensemble comparisons was 2.01 estimated standard errors; no arbitrary two-SEM pass/fail gate was imposed. Probe reuse across parameters and profiles means those comparisons are correlated.

At $\xi = 1$, the exact varying-profile values were:

| $L$ | $\epsilon = 0.001$ | $\epsilon = 0.003$ | $\epsilon = 0.01$ |
|---|---|---|---|
| 3 | −18.851936 | −16.794478 | −13.772544 |
| 4 | −19.247069 | −17.193485 | −14.173802 |

The minimum shifted eigenvalue was 0.001528125, so all real logs were defined. The convention check gave $\mathcal{D}_{\text{primed}} - \mathcal{D}_{\text{raw}} = 3\log\epsilon$ at the same gauge-measure normalization. The observed off-shell gauge and IR dependence is retained, not interpreted as a physical cancellation test.

## 9.6 Local UV-basis checks and finite-matching boundary

The unchanged counterterm stage evaluated ten profiles with no external finite coefficients. Unlike Eq. (9.1), it includes the radial scalar as well as three generator sectors. At $\xi = 1$, an independent sitewise minimal continuum $E$ matrix with $E_{\mu\nu} = m_W^2\delta_{\mu\nu}$, $E_{\chi\chi} = U_T + m_W^2$, and $E_{\mu\chi} = 2q\partial_\mu\rho$ gave the pole density

$$P_{\text{pole}} = U_L^2 + 3\left[2m_W^4 + (U_T + m_W^2)^2 + 8q^2(\partial\rho)^2\right] - 4m_F^4, \qquad m_F^2 = 0.4.$$

A separate symbolic expansion retained potential terms through $\rho^{12}$. The independent matrix-versus-polynomial error was $1.53 \times 10^{-15}$; scalar/total closure and residue-cancellation identities also passed the $10^{-10}$ diagnostic gate. The gradient coefficient was $24q^2 = 6g^2 = 2.535$ and $\delta Z_H/C = -12g^2 = -5.07$, using the kinetic basis $\frac{1}{2}\int(\partial\rho)^2$ and source pole convention $C = 1/(64\pi^2\epsilon_{\text{bar}})$. The dimensional pole variable $\epsilon_{\text{bar}}$ is distinct from the auxiliary IR mass-squared shift $\epsilon$.

Forward-difference evaluation of these continuum local expressions obeyed

$$h^4\sum|\nabla\rho_A|^2 = h^2L^4A^2\sin^2(\pi/L).$$

For $A = 0.005, 0.01, 0.02, 0.04$, the gradient integrals were $0.000972, 0.003888, 0.015552, 0.062208$ at $L = 3$ and $0.002048, 0.008192, 0.032768, 0.131072$ at $L = 4$; they vanish at $A = 0$. These tests verify gradient implementation and local pole algebra. They do not measure lattice UV asymptotics, independently derive continuum heat-kernel residues, or determine a finite wave-function constant. Small amplitude on these coarse grids is not evidence of a resolved continuum derivative expansion.

**October 5 status and next calculation**

The fresh deterministic operator, off-shell Ward, convention, and local pole-algebra checks passed their stated tolerances. The run took 72.23 seconds and recorded 36 determinant references, 13,824 probe/order/shift rows, 24 operator-check rows, and ten local-basis profiles. Quadrature error was negligible relative to sampling uncertainty. All ten counterterm summaries retained `BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED`; full matching remained false and no finite counterterm action was supplied.

Finite matching is the next distinct calculation. It requires an explicit EFT truncation/operator basis, matching scale, gauge convention, and consistent lattice/continuum IR prescription. The finite gradient coefficient must be matched independently, not inferred from its pole residue or fitted blindly to the tiny-grid determinant values. Nonzero gauge backgrounds are additionally required for full gauge-field and coupling matching. A stationary-bounce/collective-mode audit and controlled volume, spacing, and regulator limits remain necessary before interpreting a physical rate.

## 9.7 October 6 matching-coefficient progress audit

The October 6 morning review audited the coefficient template, the solver's counterterm interface, the symbolic pole closure, the aggregate prerequisite summary, and all ten retained profile summaries. It

also reran Python syntax compilation and the lightweight analytic validation; the latter completed with `VALIDATION RESULT: PASS`. No production-source edit or finite matching calculation was performed during that morning audit.

The completed infrastructure consists of the kinetic and potential basis through $\rho^{12}$, continuum background-Feynman-gauge pole residues, local closure checks at approximately $10^{-15}$, and a JSON interface that validates supplied coefficients and evaluates their action against measured basis integrals. All ten retained nonconstant-profile summaries report `BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED`.

The finite status remains unambiguous:

```
"coefficients": null,    finite_matching_coefficients_derived=false,
                         full_su2_matching_confirmed=false.
```

After the morning audit, a first finite integral was evaluated as a narrow normalization check: the scalar $O(4)$ quartic tadpole at $a = 0.8$, $m^2 = \mu^2 = 0.4$, and $\lambda = -2$. The implemented forward-difference kinetic term and corresponding free kernel are

$$S_{\rm kin}^{\rm lat} = \frac{a^4}{2}\sum_{x,\nu}\left(\frac{\rho_{x+a\hat{\nu}} - \rho_x}{a}\right)^2, \qquad \hat{k}^{\,2} = \frac{4}{a^2}\sum_{\nu}\sin^2\!\left(\frac{ak_\nu}{2}\right).$$

For $V_4 = \lambda(\Phi^A\Phi^A)^2/4$ with four real scalar components, the single diagram has $\Sigma_{\rm tad} = 6\lambda I$. The numerical result, using $m^2_{\rm lat} = m^2_{\overline{\rm MS}} + \delta m^2$ as the additive matching convention, is

$$\begin{aligned} I_{\rm lat} &= 0.2268778187404072, \\ I_{\overline{\rm MS}} &= -0.002533029591058444, \\ \delta m^2_{\rm tad} &= 6\lambda(I_{\overline{\rm MS}} - I_{\rm lat}) = 2.752930179977589. \end{aligned}$$

This is only one cutoff-dependent scalar mass contribution. It is not the complete finite mass coefficient, must not be supplied as `delta_m2`, and does not determine the momentum-dependent kinetic coefficient. No complete potential or wave-function matching coefficient was computed. The continuum pole residues listed earlier in this guide are not finite matching constants and must not be substituted for them. The template still has `"coefficients": null`, the full-matching flag remains false, and no finite counterterm action or gauged decay rate follows from this partial example.

The default stationary-bounce checkpoint expected by `su2_counterterms` is absent from the current workspace. An explicitly selected retained diagnostic background can exercise the basis evaluator, but neither route derives the coefficients. The next calculation is an infinite-volume radial matching calculation at $\xi = 1$ and $\mu = \sqrt{0.4}$: derive lattice momentum-space vertices, evaluate regulated Brillouin-zone potential and small-external-momentum amplitudes, subtract the corresponding $\overline{\rm MS}$ amplitudes with a common IR prescription, and verify regulator cancellation and numerical convergence. Bare mass shifts can have power-law cutoff dependence, so test matched amplitudes or observables under spacing refinement rather than requiring each bare coefficient to be $a$-independent. Keep `includes_full_su2_lattice_to_msbar_matching=false` for any radial-only result. Full gauge-field and gauge-coupling matching additionally requires nonzero background gauge fields.

**October 6 matching status**

The achieved milestone is basis and pole-residue validation plus a working coefficient-input interface, with one scalar tadpole contribution evaluated as a first matching check. The complete finite lattice-to-$\overline{\rm MS}$ coefficient calculation remains incomplete. Do not present the tadpole alone as a derived mass coefficient or full matching result.

### 9.7.1 October 7–8 nonlinear action, routing, and matching projections

The follow-up procedure report makes the target theory explicit while preserving the implementation boundary: the production fluctuation and determinant operators still use the scalar forward-difference action and quadratic gauge–Goldstone–ghost operators. The nonlinear vertex generators are implemented in `su2_lattice_vertices.py`, imported by the solver, and available through its table-generation mode; they are not yet consumed by a full-gauged matching integrator. For full-theory matching, the proposed convention is a fundamental Higgs doublet with links $U_{x\mu} = \exp[-iag_0 A^a_{x\mu} T^a]$, $T^a = \sigma^a/2$, and the Wilson plaquette action

$$S_g = \frac{4}{g_0^2} \sum_{x,\mu<\nu} \left(1 - \frac{1}{2}\,\mathrm{ReTr}\, U_{x,\mu\nu}\right), \qquad U_{x,\mu\nu} = U_{x\mu} U_{x+a\hat{\mu},\nu} U^\dagger_{x+a\hat{\nu},\mu} U^\dagger_{x\nu}.$$

With this generator and link normalization its continuum limit is $\frac{1}{4}\int d^4x\, F^a_{\mu\nu} F^a_{\mu\nu}$, the Wilson plaquette formulation [18]. The gauge coupling remains the bare $g_0$; the benchmark $g(\mu) = 0.65$ alone does not specify $g_0(a)$.

The matter action is the covariant nearest-neighbor term

$$S_{H,\mathrm{link}} = a^4 \sum_{x,\mu} \frac{(U_{x\mu} H_{x+a\hat{\mu}} - H_x)^\dagger (U_{x\mu} H_{x+a\hat{\mu}} - H_x)}{a^2}, \qquad \rho_x = \sqrt{2 H_x^\dagger H_x}.$$

At $U = 1$ it reduces to the canonical four-real-component forward difference already used by the solver. The tree potential remains $V_0 = -\mu_{\mathrm{sq}}\rho^2/2 + \lambda\rho^4/4 + \gamma\rho^6/6 + \delta\rho^8/8$; the $\rho^{10}$ and $\rho^{12}$ terms belong to the perturbative matching-counterterm basis, not the leading tree potential. The gauge-covariant kinetic counterterm is $\delta Z_H$ times $S_{H,\mathrm{link}}$.

For a radial background and zero background gauge field, the proposed background gauge is

$$F^a_x = (DA^a)_x - \xi q \bar{\rho}_x \chi^a_x, \qquad S_{\mathrm{gf}} = \frac{a^4}{2\xi} \sum_{x,a} (F^a_x)^2, \qquad q = g_0/2,$$

with the Faddeev–Popov operator defined as the exact variation of $F$ under the stated lattice gauge transformations. This defines the nonlinear ghost rules consistently with the link-field coordinate; it is not part of the gauge-invariant bare action. The proposed perturbative action is $S_g + S_{H,\mathrm{link}} + a^4 \sum_x (V_0 + V_{\mathrm{ct}}) + S_{\mathrm{gf}} + S_{\mathrm{gh}}$. Its quadratic matter-link expansion agrees with the recorded link Hessian, but that check does not validate the missing nonlinear vertices.

The procedure report specifies exact vertex generators rather than a continuum derivative approximation: expand the two exponentials in each scalar link to generate scalar–scalar–gauge vertices (including the terms linear and quadratic in $A$), differentiate the ordered plaquette product for pure-gauge vertices, and differentiate $M = \delta F/\delta\omega$ for ghost vertices. Scalar-potential vertices follow by differentiating $f(X)$ with $X = \phi^A\phi^A$; the compact derivative rule and explicit link expansions are given in the procedure report. This is the standard perturbative lattice-vertex construction [19].

At one loop, for constant $\rho$, $\xi = 1$, and $M_A^2 = q^2\rho^2$, define

$$U_L = m_0^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \qquad U_T = m_0^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6.$$

The lattice determinant contribution is

$$V_{1,\mathrm{lat}}(\rho) = \frac{1}{2}\int_{\mathrm{BZ}} \frac{d^4k}{(2\pi)^4} \left[\log(\hat{k}^2 + U_L) + 3\log(\hat{k}^2 + U_T + M_A^2) + 6\log(\hat{k}^2 + M_A^2)\right],$$

up to a background-independent constant. For the continuum comparison in $d = 4 - 2\epsilon$, the vector/ghost logarithm carries multiplicity $3(d-2)$ before $\overline{\mathrm{MS}}$ subtraction; replacing it by its four-dimensional value

too early can lose a finite term. This determinant is a background-field check, not by itself the full set of local matching coefficients [20, 21].

The kinetic coefficient is defined by the difference of off-shell self-energy slopes at the subtraction point, with lattice $\widehat{p}^2$ and continuum $p^2$:

$$\delta Z_H(\mu) = \left.\frac{\partial \Sigma^{(2)}_{\overline{\mathrm{MS}},\mathrm{ren}}}{\partial p^2}\right|_{p^2=\mu^2} - \left.\frac{\partial \Sigma^{(2)}_{\mathrm{lat}}}{\partial \widehat{p}^2}\right|_{\widehat{p}^2=\mu^2}.$$

Potential coefficients require matching corresponding multi-point 1PI vertices with an operator projector that separates local potential terms from momentum-dependent operators. Use a common infrared prescription and identical external kinematics on both sides. In particular, the massless gauge contribution contains nonanalytic $\rho^4 \log \rho^2$ behavior, so a Taylor expansion of the constant-background determinant at $\rho = 0$ is not a valid full-gauge-theory potential projector. Nonexceptional Euclidean momenta at a declared subtraction scale, or a common explicit regulator with demonstrated cancellation, are needed before extracting local coefficients.

**October 8 full-gauged Step 1 status: no full matching result yet**

The October 8 routing/projector deck and its independent tree-level audit complete the selected kinematic setup for full-gauged Step 1. The deck fixes the nonexceptional routings, stores complete generated color/polarization tensors at those selected momenta, and records projectors and a common infrared-regulator sequence. A separate pilot now evaluates the gauge-linked four-radial-Higgs insertion sum at the fixed local-contact routes. It is one projected sector only: it neither supplies the complete four-point amplitude nor establishes regulator cancellation for the full gauge/ghost system. The coefficient JSON must remain null and `includes_full_su2_lattice_to_msbar_matching` must remain false. The scalar quartic tadpole reported above is only one scalar mass-diagram check; it cannot be promoted to the complete `delta_m2` or used to infer `delta_ZH`. Likewise, the gauge-linked four-point diagnostic is not the full `delta_lambda`.

### 9.7.2 October 8 fixed full-gauged kinematics and projectors

The fixed deck is generated by `su2_full_gauged_projection_setup.py`; its full-precision arrays and metadata are `su2_full_gauged_projection_tables_20261008.npz` and `su2_full_gauged_projection_tables_20261008.json`. The setup uses $a = 0.8$, $g_0 = 0.65$, $\xi = 1$, $\mu_R^2 = 0.4$, and $\bar{\rho} = 0.37$ for local vertex tests. All legs are incoming; site fields use $e^{+ip\cdot x}$ and gauge fields are assigned to link midpoints.

For the Higgs two-point slope, choose $p = \sqrt{s}(1,2,3,4)/\sqrt{30}$ at $s = p^2 = 0.2, 0.4, 0.6$, with legs $(p, -p)$. Fit the three samples to $c_0 + c_1 x + c_2 x^2$, where $x = p^2$ in the continuum and $x = \widehat{p}^2$ on the lattice. The derivative row $(0, 1, 2x_\mu)$, applied to the inverse design matrix, defines the slope at $p^2 = \mu_R^2$ or its corresponding lattice $\widehat{p}^2$; at the subtraction point the latter is 0.3966567139724254. Both continuum and lattice tree kernels return unit slope within $1.11 \times 10^{-15}$. This is only a tree-projector test; applying it to $\Sigma^{(2)}$ remains Step 2.

The deck also includes an asymmetric nonexceptional $H^\dagger H A$ triangle, a symmetric three-leg triangle and four-leg tetrahedron at the reference scale, and fixed-shape nonexceptional four-, six-, eight-, ten-, and twelve-scalar contact routings at average squared-momentum scales $0.2, 0.4, 0.6$. For the gauge two-point function define $K = p$ in the continuum and $K_\mu = 2\sin(ap_\mu/2)/a$ on the lattice; use

$$T_{\mu\nu} = K^2\delta_{\mu\nu} - K_\mu K_\nu, \qquad L_{\mu\nu} = K_\mu K_\nu, \qquad P_T^{\mu\nu} = \frac{T^{\mu\nu}}{3(K^2)^2}, \quad P_L^{\mu\nu} = \frac{L^{\mu\nu}}{(K^2)^2}.$$

For each local scalar $2n$-point contact, contract the full tensor and fit the three routing samples to $d_0 + d_1\bar{s} + d_2\bar{s}^2$; retain the intercept $d_0$. The lattice and continuum fits use their respective $\widehat{p}^2$ and $p^2$ coordinates.

The common Step 2 infrared prescription is background Feynman gauge and the same additive regulator on every massless gauge and ghost denominator, $k^2 \to k^2 + \lambda_{\rm IR}^2$ and $\widehat{k}^{\,2} \to \widehat{k}^{\,2} + \lambda_{\rm IR}^2$, at the deck's base points $\lambda_{\rm IR}/\mu_R = 1/8, 1/16, 1/32$. Massive scalar lines remain unshifted. The gauge-linked four-point pilot applies this prescription at those points and adds $1/64$ as a check on its three-point extrapolation. This is not yet a cancellation test for the complete gauge/ghost system: regulator cancellation and vanishing of regulator-induced Ward-identity breaking must still be demonstrated in the full $\lambda_{\rm IR} \to 0$ extrapolation.

The independent audit `su2_full_gauged_projection_audit.py` records PASS in `su2_full_gauged_projection_audit_20261008.json`. Its selected checks include momentum conservation and nonexceptionality, scalar-link tensors against finite-action differences, the asymmetric scalar-link Ward identity, Wilson tensors against an independent expansion and two-point transversality, ghost tensors against finite transformations, the nonzero-background Faddeev–Popov Jacobian, and tree/local projector algebra. Maximum discrepancies range from $4.65 \times 10^{-16}$ for momentum conservation to $2.72 \times 10^{-6}$ for the finite-transformation ghost-tensor comparison; each passes its declared tolerance. These are selected tree-level implementation tests, not loop Ward identities or matching results.

### 9.7.3 October 8 gauge-linked four-point pilot

The first integrated gauge-linked multi-point sector is a fixed-route four-radial-Higgs projection. It is evaluated by the standalone `su2_four_point_gauge_sector_20261008.py`; its full-precision record is `su2_four_point_gauge_sector_20261008_ircheck.json`. The evaluator expands the one-loop bosonic trace-log through four source derivatives and keeps only the gauge-linked insertion classes. The scalar-only $O(4)$ loops are not included in this sector result; the direct Higgs–ghost contribution vanishes at the symmetric background in the stated gauge.

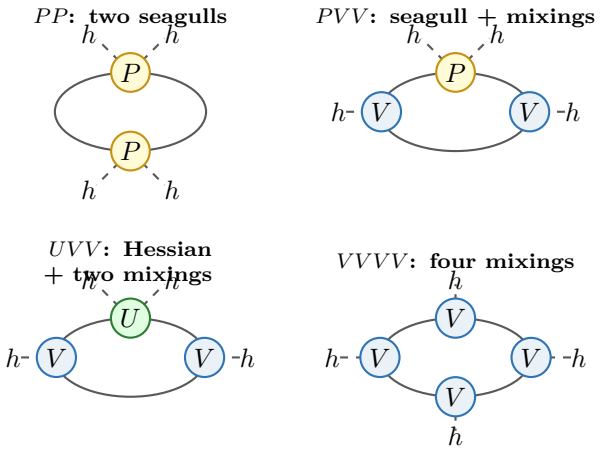



Figure 9.1: Closed-loop insertion classes retained in the gauge-linked four-Higgs pilot. $P$ is the $hhAA$ seagull and carries two external $h$ legs; $V = hA\chi$ is the scalar-link mixing and carries one; $U$ is the $\lambda h^2$ Goldstone-Hessian insertion and carries two. The loop lines represent the coupled gauge/Goldstone propagators; at each $V$ insertion the internal species changes between gauge and Goldstone. Scalar-only loops are excluded from this pilot.

The continuum integrand is renormalized by subtracting

$$\frac{C}{(k^2+\mu_R^2)^2}, \qquad C = -54q^4 + 36q^2\lambda = -8.20745859375, \qquad q = \frac{g_0}{2}.$$

At the chosen scale this subtraction has zero renormalized $\overline{\rm MS}$ integral. Lattice and continuum amplitudes use the same three fixed external momentum shapes and the same additive regulator on every internal

gauge line. Figure 9.2 shows the projected continuum-minus-lattice result as the regulator is reduced.

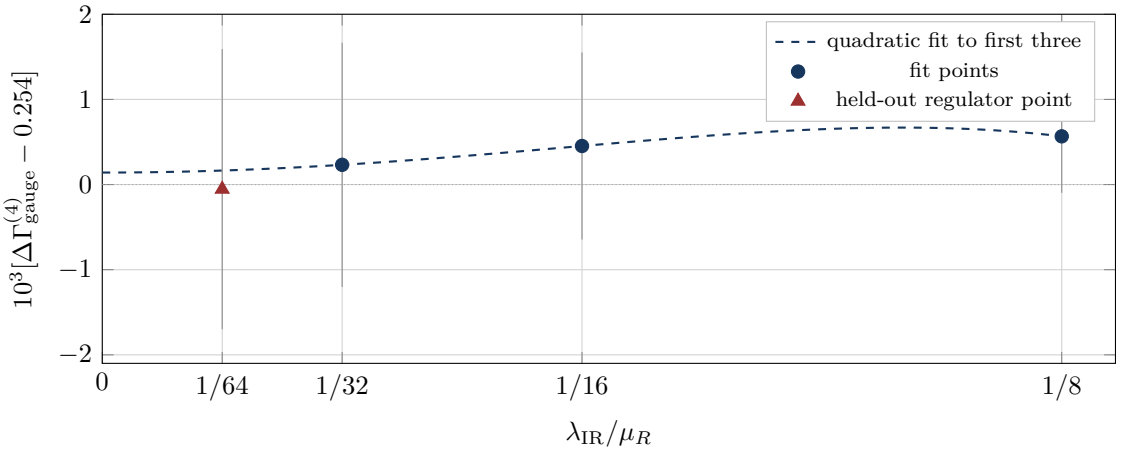



Figure 9.2: Four-point gauge-linked continuum-minus-lattice result versus IR regulator. Error bars are replicate QMC standard errors only. The dashed curve is quadratic in $(\lambda_{\text{IR}}/\mu_R)^2$, fitted to the three blue points; the red triangle is a held-out check, not part of the fit. The zero horizontal line marks the reference value $\Delta\Gamma^{(4)} = 0.254$.

For $\lambda_{\text{IR}}/\mu_R = 1/8, 1/16, 1/32, 1/64$, respectively, $\Delta\Gamma^{(4)}_{\text{gauge}}$ is $0.2545663, 0.2544528, 0.2542318, 0.2539455$; the corresponding QMC standard errors are $6.64 \times 10^{-4}, 1.10 \times 10^{-3}, 1.43 \times 10^{-3}, 1.65 \times 10^{-3}$. The fit to the first three points gives the intercept $0.2541409 \pm 0.0015671$ (QMC error only). The fourth point's residual from the fit is $-2.19 \times 10^{-4} \pm 1.56 \times 10^{-4}$, about 1.4 QMC standard errors. This is a consistency check, not a robust bound on the extrapolation-model error. Dividing by 3! gives the diagnostic $+0.04236 \pm 0.00026$ for this gauge-linked local-contact projection; it is not the full $\delta\lambda$ and is not written to the coefficient template.

The insertion evaluator reproduces the constant-source gauge/Goldstone determinant to below $7 \times 10^{-13}$, agrees with the exact link-generated $hA\chi$ and $hhAA$ vertices to below $6 \times 10^{-17}$, and reproduces the large-momentum $C/k^4$ tail to relative error $8.1 \times 10^{-7}$. Its projected integrand is Bose symmetric to numerical precision. These checks validate the selected insertion sum and numerical implementation; they do not constitute a full gauge/ghost Ward or Slavnov–Taylor test.

### 9.7.4 October 18–19 coefficient and higher-point follow-up

The updated matching report fixes the sign convention $\Delta\Gamma = \Gamma_{\overline{\text{MS}},\text{ren}} - \Gamma_{\text{lat}}$ and retains the benchmark $a = 0.8$, $\mu_R^2 = m_0^2 = 0.4$, $g_0 = 0.65$, $\xi = 1$. Its principal new numerical results are summarized here; values below are sector diagnostics, not a completed full-$SU(2)$ coefficient set.

| Result | Estimate | Interpretation and boundary |
|---|---|---|
| Higgs kinetic slope | $\delta Z_H = 0.0329117974 \pm 3.60 \times 10^{-7}$ | Three-regulator quadratic intercept; QMC error only and finite at the stated $a, \mu_R, \xi$. |
| Four-Higgs contact | Scalar $-1.45103929 \pm 0.0000516$; gauge pilot $+0.04233687 \pm 0.000123$; sum $-1.40870242 \pm 0.000133$ | Conditional sum only. Scalar zero-momentum/contact closure is unresolved and gauge fit-model error is not bounded. |
| Six-Higgs gauge insertion | $\delta\gamma_{\text{gauge}} = -0.01887194 \pm 0.00058491$ | Four scrambles, 64 points per route and replicate; QMC-only, separate from scalar-only result. |
| Gauge-linked 8/10/12 legs | $-23.0962 \pm 14.4191$; $-1.23780 \times 10^4 \pm 1.31017 \times 10^3$; $-2.16743 \times 10^6 \pm 3.61715 \times 10^5$ | Three-regulator intercepts; relative QMC errors 62.4%, 10.6%, 16.7%. Ten-/twelve-leg runs use only four points per scramble. |
| Gauge two-point tensor | Largest normalized transverse residual 0.4123; one Ward component $-0.01198 \pm 0.00052$ | Lattice-only projected follow-up; no continuum pure-gauge/ghost tensor or MS-bar match. |
| Higher-point UV poles | $C_6 = 91.26$, $C_8 = 511.056$, $C_{10} = C_{12} = 0$ | Pole subtraction/power counting only; vanishing higher-leg poles do not imply vanishing finite projections. |

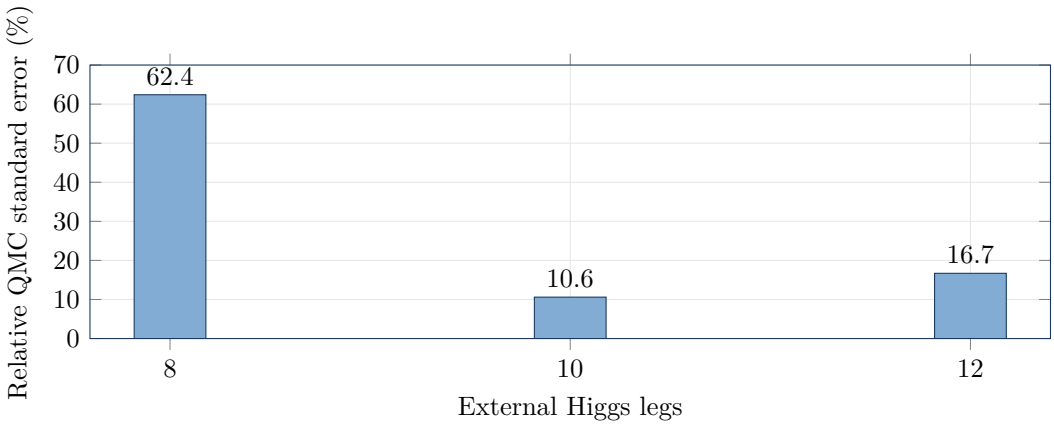



Figure 9.3: Relative replicate-QMC errors of the October 19 three-regulator continuum-minus-lattice intercepts. The $L = 8$ design uses $8 \times 32$ points per route and regulator; the $L = 10, 12$ designs use $8 \times 4$. These bars omit IR-fit, continuum-tail, routing, and other systematic errors.

The two-point slope is the only complete one-loop two-point-sector extraction at this benchmark; its quoted uncertainty excludes IR-fit-model and perturbative errors. The gauge-linked four-point contact is assembled with the scalar nonexceptional projection for a compatible projector, but it differs from the independent zero-momentum scalar result by about three quoted contact standard errors. The six-point continuum-minus-lattice pilot uses the common six-regulator sequence; its fit-window shifts are unresolved at QMC precision, which does not bound extrapolation-model uncertainty. The 8-, 10-, and 12-leg values likewise remain exploratory finite projections: their reported errors are replicate-QMC errors and do not cover continuum-tail, routing, regulator-fit, or other model uncertainties. The gauge two-point Ward residual is neither a pass nor a failure because the auxiliary IR mass breaks the identity and the continuum tensor is absent. Keep the full coefficient JSON null and the full-matching flag false.

The pole-subtraction and finite lattice–continuum comparison follow standard lattice perturbation methods [14, 19, 15]. The Ward/Slavnov–Taylor checks use the background-field and gauge-identity framework [16, 20]. Replicate scrambled-net uncertainty is kept distinct from extrapolation-model error, following the distinction in randomized-net variance analysis [17].

## 9.8 Companion reports and retained artifacts

The detailed standalone presentations are:

- `SU2_L6_Single_Core_Pilot_Narrative_Report_20261005.tex`;

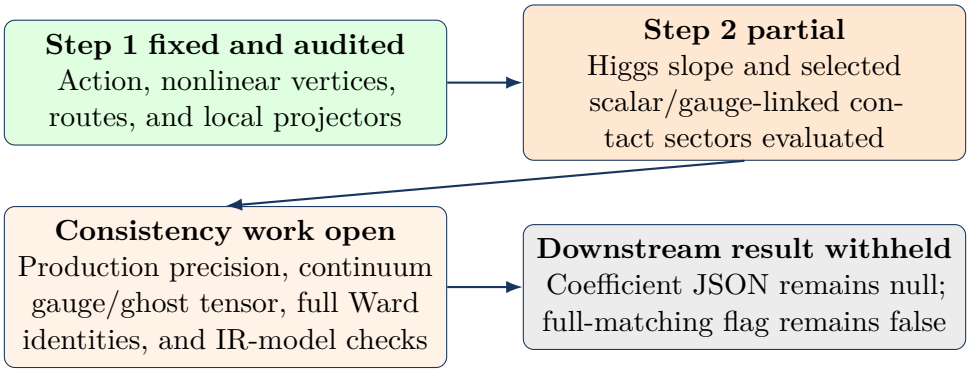



Figure 9.4: Status of the October 19 finite-matching update. "Evaluated" denotes selected sector calculations only; it does not imply complete amplitudes or accepted full-$SU(2)$ coefficients.

- `SU2_L6_Complete_Multiplet_Report_20261005.tex`;
- `SU2_Nonconstant_Prerequisite_Report_20261005.tex`;
- `SU2_Matching_Coefficient_Progress_Report_20261006.tex`;
- `SU2_Next_Steps_Finite_Matching_20261006.tex` (updated October 19; nonlinear action, audited routes/projectors, scalar-only matching, and October 18–19 finite-contact and gauge two-point pilots).
- `SU2_Spacing_Campaign_Mode_Audit_Report_20261014.tex` (14 October audit with follow-ups through October 18; translation subspace, paired determinant shifts, controls, Green functions, and trace-log stability).

The October 18–19 matching values are supported by the run records

- `su2_continuum_uv_asymptotic_paired_20261018.json`
- `su2_higher_point_gauge_insertions_20261018_L6_msbar_qmc.json`
- `su2_higher_point_gauge_insertions_20261019_L8_msbar_qmc_power5.json`
- `su2_higher_point_gauge_insertions_20261019_L10_msbar_qmc_power2_8rep_followup.json`
- `su2_higher_point_gauge_insertions_20261019_L12_msbar_qmc_power2_8rep_followup.json`
- `su2_gauge_two_point_followup_20261018_qmc_power7_haar.json`

The campaign audit and its machine-readable provenance are listed at the end of Chapter 14. The original pilot summary is `su2_L6_single_core_pilot_20261005/pilot_summary.json`. Historical follow-up directories `su2_L6_convergence_20261005/`, `su2_L6_convergence_4096_20261005/`, and `su2_L6_complete_multiplets_20261005/` are not present here; their saved bases and raw probes must be restored to reproduce those basis-dependent comparisons.

The fresh harness `su2_nonconstant_prerequisite_audit.py` is retained.

Within `su2_nonconstant_prerequisites_20261005/`, the retained files include `audit_summary.json`, `symbolic_pole_closure.json`, the archive README, and ten per-profile counterterm summaries. The recorded full archive additionally contained background arrays, sparse operators, complete spectra, low eigenpairs, saved probes, CSVs, and logs. Those raw artifacts and the previously generated bundle are absent from the current snapshot; aggregate summaries are not a substitute. No production bounce or determinant rerun was performed for this guide revision; the separate October 8 vertex/projector audit is recorded below.

# Chapter 10

# Low-Mode Extraction and the IR Determinant

## 10.1 ARPACK and degeneracy-safe channel solves

The code applies `scipy.sparse.linalg.eigsh` separately to $H_L$ and $H_T$. Separate scalar-channel solves reduce memory and preserve the exact threefold transverse degeneracy analytically. The false-vacuum spectrum is constructed from periodic Fourier levels rather than requesting an arbitrary cut through a highly degenerate ARPACK multiplet.

Mode vectors are normalized and diagnosed using their peak amplitude, inverse-participation ratio, and longitudinal/transverse component weights. These quantities separate localized bounce fluctuations from extended periodic plane-wave states.

## 10.2 Finite spectral-window determinant

For the new homogeneous SU(2) reference operators, the program also avoids ARPACK and retains complete momentum multiplicities analytically. With $k = \widehat{p}^2$ and constant $m_W^2 = q^2\rho^2$, one generator has three levels $k + m_W^2$, one $k/\xi + m_W^2$, and one $k + U_T + \xi m_W^2$ per momentum; its ghost level is $k + \xi m_W^2$. These are sorted without dropping degenerate members. The symmetric false vacuum includes exactly zero gauge and ghost levels. Their gauge-volume prescription remains necessary before forming a physical determinant ratio.

For a retained set of modes,

$$\log\left|\frac{\det{}' H_B}{\det H_F}\right|_{\mathrm{IR}} = \sum_{i\in B'} \log|\lambda_i^{(B)}| - \sum_{j\in F} \log \lambda_j^{(F)}. \tag{10.1}$$

This IR window is a useful diagnostic, but it is not the full determinant because the ultraviolet spectrum is absent. The source therefore reports it explicitly as an IR quantity rather than silently interpreting it as the physical prefactor.

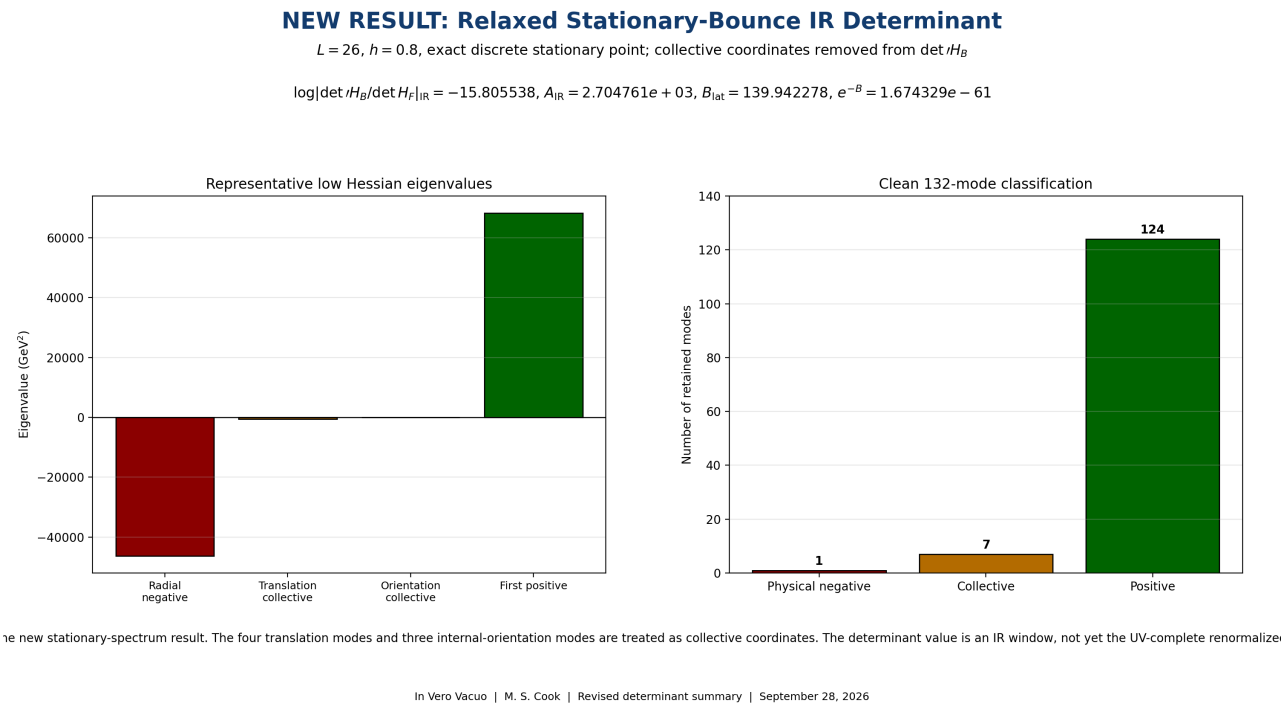



Figure 10.1: Summary of the relaxed stationary spectrum and its primed IR determinant window. This intermediate result motivated the subsequent full SLQ trace–log calculation.

# Chapter 11

# Deflated Stochastic Lanczos Quadrature

## 11.1 Trace–log formulation

The determinant is converted into a trace,

$$\log\det H = \operatorname{Tr}\log H. \tag{11.1}$$

For independent Rademacher vectors $z_k$ with entries $\pm 1$,

$$\operatorname{Tr} f(H) \approx \frac{1}{N_p}\sum_{k=1}^{N_p} z_k^T f(H) z_k. \tag{11.2}$$

Lanczos iteration replaces each large quadratic form by Gaussian quadrature on a small tridiagonal matrix $T_m$,

$$z^T f(H) z \approx \|z\|^2 e_1^T f(T_m) e_1. \tag{11.3}$$

The Rademacher trace estimator and its combination with Lanczos quadrature are described in Refs. [5, 9]. The same random probe is used in bounce and false-vacuum channels, which correlates the estimates and reduces noise in their difference.

## 11.2 Low-mode deflation

The logarithm varies rapidly near zero. The lowest eigenpairs are therefore calculated exactly, projected out of every random probe, and restored analytically. This both protects collective-coordinate classification and improves Krylov convergence.

```
                    (-1.0, 1.0), size=coupled.shape[0]
                )
                ghost_vector = probe_random.choice(
                    (-1.0, 1.0), size=ghost.shape[0]
                )
                probe_data.append((
                    lanczos_data(coupled, coupled_vector, coupled_basis, max(steps_list
                        )),
                    lanczos_data(ghost, ghost_vector, ghost_basis, max(steps_list))
                ))

        if modal_export:
            xi_label = str(gauge_xi).replace(".", "p")
            if method == "exact":
                coupled_export_values = coupled_values[:min(modal_modes, coupled_values
                    .size)]
```

```
                coupled_export_vectors = coupled_vectors[:, :coupled_export_values.size
                    ]
                ghost_export_values = ghost_values[:min(modal_modes, ghost_values.size)
                    ]
                ghost_export_vectors = ghost_vectors[:, :ghost_export_values.size]
                coupled_explicit = (coupled_matrix if coupled.shape[0]
                                    <= modal_max_operator_dim else None)
                ghost_explicit = (ghost_matrix if ghost.shape[0]
                                  <= modal_max_operator_dim else None)
            else:
                coupled_export_values = coupled_computed[:modal_modes]
                coupled_export_vectors = coupled_vectors[:, :coupled_export_values.size
                    ]
                ghost_export_values = ghost_computed[:modal_modes]
                ghost_export_vectors = ghost_vectors[:, :ghost_export_values.size]
                coupled_explicit = None
                ghost_explicit = None
            common_background = {
                "type": "relaxed_bounce", "field_array": bounce,
                "checkpoint": str(bounce_file)
            }
            common_lattice = {
                "shape": list(bounce.shape), "spacing": spacing,
                "boundary_conditions": "periodic"
            }
            metadata_path, _, _ = export_su2_modal_dataset(
                modal_output,
                f"bounce_L{bounce.shape[0]}_xi_{xi_label}_coupled_A_chi",
                coupled, coupled_export_values, coupled_export_vectors,
                sector="coupled_A_chi", background=common_background,
                lattice=common_lattice,
                parameters={"g": gauge_coupling, "xi": gauge_xi,
                            "ir_shift": 0.0, "generator_multiplicity": 3},
                explicit_operator=coupled_explicit
            )
```

Listing 11.1: Matrix-free SLQ operator, low-mode cache, Lanczos recurrence, and quadrature.

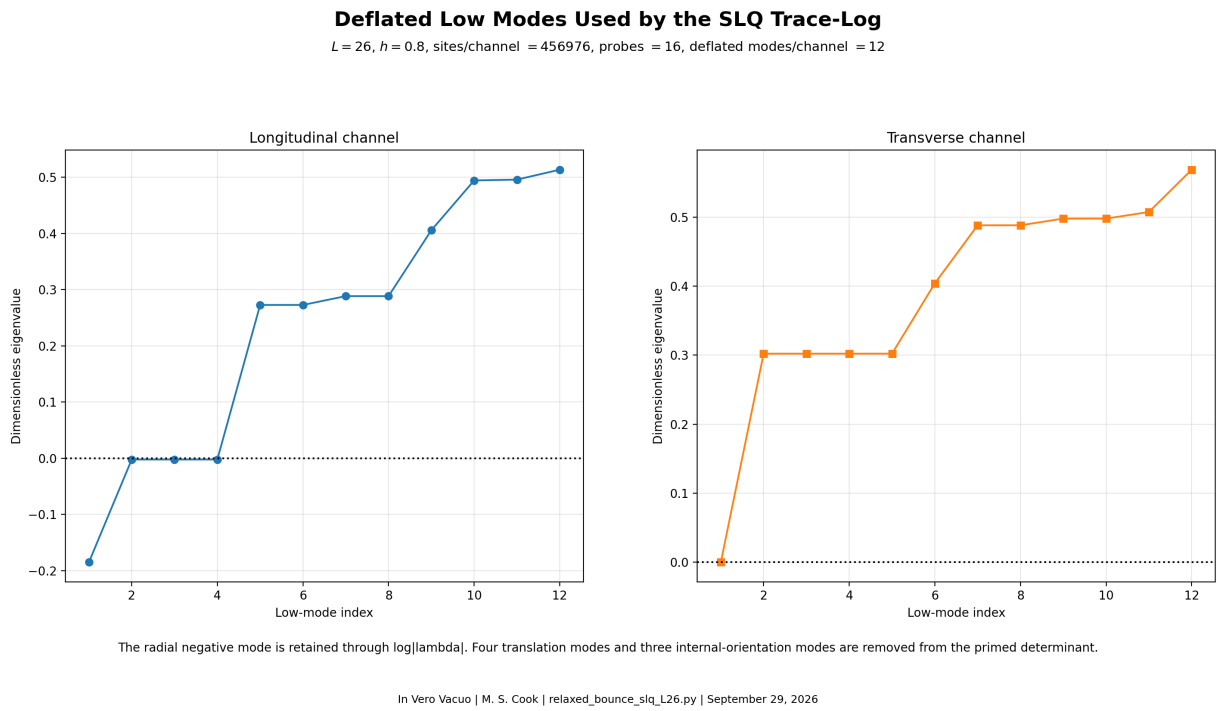
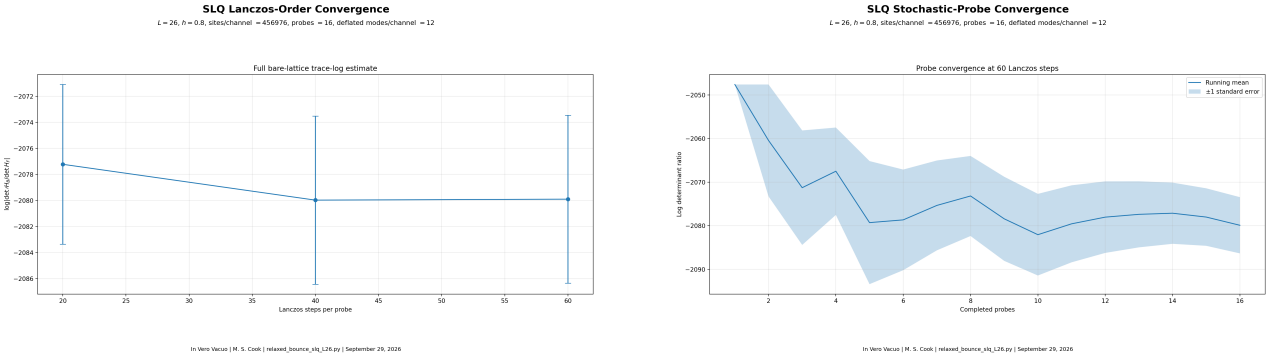



Figure 11.1: Deflated longitudinal and transverse low modes. Four translation modes and three orientation modes are removed; the physical negative mode is retained through its absolute value.




(a) Lanczos-order convergence.

(b) Running stochastic-probe convergence.

Figure 11.2: Two convergence tests required for the stochastic trace–log: Krylov order and probe count.

Table 11.1: Production SLQ settings and result.

| Quantity | Value |
|---|---|
| Lattice | $26^4$, $h = 0.8$ |
| Sites per channel | 456,976 |
| Rademacher probes | 16 |
| Deflated eigenpairs per bounce channel | 12 |
| Lanczos orders tested | 20, 40, 60 |
| Bare finite-lattice trace–log | $-2079.9042 \pm 6.4504$ |
| $\log_{10}$ of bare determinant factor | $451.6454 \pm 1.4007$ |

**Scope and interpretation**

The enormous bare factor is not a physical enhancement. It is dominated by the ultraviolet regulator and diverges as the lattice spacing is removed. The result is valuable as a complete finite-lattice diagnostic, but it must be matched to a renormalized continuum prescription.

# Chapter 12

# Continuum Partial-Wave Renormalization

## 12.1 Radial decomposition

For an $O(4)$-symmetric background, each scalar operator decomposes into radial partial waves labeled by $\ell = 0, 1, 2, \ldots$ with degeneracy

$$g_\ell = (\ell + 1)^2. \tag{12.1}$$

The radial operator has the form

$$\mathcal{H}_{\ell,c} = -\frac{\mathrm{d}^2}{\mathrm{d}r^2} - \frac{3}{r}\frac{\mathrm{d}}{\mathrm{d}r} + \frac{\ell(\ell+2)}{r^2} + U_c(r), \qquad c \in \{L, T, F\}. \tag{12.2}$$

Gel'fand–Yaglom ratios replace functional determinants by ratios of radial initial-value solutions [4]. Their radial partial-wave extension and renormalized angular-momentum summation are discussed in Ref. [8]. The code uses a stable logarithmic formulation and subtracts the large-$\ell$ asymptotic terms before summation.

```
    total_closure_polynomial = {
        name: closure_polynomial[name] + gauge_closure_polynomial[name]
        for name in closure_polynomial
    }
    gauge_kinetic_coefficient = 24.0 * charge_sq
    total_residues_over_C = {
        "delta_ZH": -2.0 * gauge_kinetic_coefficient,
        "delta_m2": -2.0 * total_closure_polynomial["rho2"],
        "delta_lambda": -4.0 * total_closure_polynomial["rho4"],
        "delta_gamma": -6.0 * total_closure_polynomial["rho6"],
        "delta_delta": -8.0 * total_closure_polynomial["rho8"],
        "delta_c10": -10.0 * total_closure_polynomial["rho10"],
        "delta_c12": -12.0 * total_closure_polynomial["rho12"],
    }
    gauge_mass_sq = charge_sq * bounce**2
    direct_total_density = (
        longitudinal_curvature**2
        + 3.0 * (
            2.0 * gauge_mass_sq**2
            + (transverse_curvature + gauge_mass_sq)**2
            + 8.0 * charge_sq * np.sum(gradient**2, axis=0)
        )
        - 4.0 * positive_mass_sq**2
    )
    direct_total_integral = float(cell_volume * np.sum(direct_total_density))
```

```
    expanded_total_integral = float(
        sum(
            coefficient * cell_volume * np.sum(bounce**powers[name])
            for name, coefficient in total_closure_polynomial.items()
        )
        + gauge_kinetic_coefficient * cell_volume * np.sum(gradient**2)
    )
    total_closure_relative_error = abs(
        direct_total_integral - expanded_total_integral
    ) / max(1.0, abs(direct_total_integral))
    total_residue_action_over_C = float(sum(
        total_residues_over_C[name] * basis_integrals[name]
        for name in total_residues_over_C
    ))
    total_residue_cancellation_error = abs(
        total_residue_action_over_C + direct_total_integral
    ) / max(1.0, abs(direct_total_integral))
    if max(total_closure_relative_error, total_residue_cancellation_error) > 1e-11:
        raise RuntimeError("The Feynman-gauge SU(2) pole-residue identity failed.")

    supplied_coefficients = None
    coefficient_metadata = {}
    counterterm_action = None
    full_su2_matching = False
    if coefficient_file_name:
        coefficient_file = Path(coefficient_file_name)
        with open(coefficient_file, encoding="utf-8") as stream:
            coefficient_document = json.load(stream)
        if "coefficients" in coefficient_document:
            supplied_coefficients = coefficient_document["coefficients"]
            coefficient_metadata = coefficient_document.get("metadata", {})
        else:
            supplied_coefficients = coefficient_document
        if supplied_coefficients is not None:
            allowed = set(basis_integrals)
            unknown = set(supplied_coefficients) - allowed
            if unknown:
                raise ValueError(f"Unknown counterterm coefficients: {sorted(unknown)}"
                    )
            supplied_coefficients = {
                name: float(supplied_coefficients.get(name, 0.0)) for name in allowed
            }
            if not all(np.isfinite(list(supplied_coefficients.values()))):
                raise ValueError("All supplied counterterm coefficients must be finite.
                    ")
            counterterm_action = float(sum(
                supplied_coefficients[name] * integral
                for name, integral in basis_integrals.items()
            ))
        full_su2_matching = bool(coefficient_metadata.get(
            "includes_full_su2_lattice_to_msbar_matching", False
        ))

    status = (
        "FULL_SU2_MATCHING_EVALUATED" if full_su2_matching
        else "COEFFICIENTS_EVALUATED_NOT_FULL_SU2" if supplied_coefficients is not None
        else "BASIS_READY_MATCHING_COEFFICIENTS_REQUIRED"
    )
    basis_rows = []
    for name, integral in basis_integrals.items():
        basis_rows.append({
```

```
            "coefficient": name, "basis_integral": integral,
            "supplied_value": (supplied_coefficients or {}).get(name),
            "action_contribution": (
                (supplied_coefficients or {}).get(name, 0.0) * integral
                if supplied_coefficients is not None else None
            ),
            "scalar_residue_over_C": scalar_residues_over_C.get(name),
            "feynman_gauge_total_residue_over_C": total_residues_over_C.get(name),
        })
    basis_file = output / "counterterm_basis.csv"
    with open(basis_file, "w", newline="", encoding="utf-8") as stream:
        writer = csv.DictWriter(stream, fieldnames=list(basis_rows[0]))
        writer.writeheader()
        writer.writerows(basis_rows)
```

Listing 12.1: Channel potentials, radial Gel'fand–Yaglom ratio, and asymptotic partial-wave subtraction.

## 12.2 Primed zero-mode limit

The translation modes occur in the longitudinal $\ell = 1$ sector, with multiplicity four. The global-orientation modes occur in the transverse $\ell = 0$ sector, with three physical copies. A small positive spectral shift $q$ regulates these channels. After explicitly dividing by the appropriate powers of $q$, the calculation is extrapolated to $q \to 0$. This regulated primed-determinant construction is closely related to standard zero-mode determinant treatments [6, 7].

## 12.3 Renormalization result

This section retains the previously reported global-scalar calculation. The partial-wave, zero-mode, and renormalization strategy follows the established numerical one-loop bounce literature [7]. The quoted $\pm 0.0300$ is hardcoded as an estimated uncertainty in the source summary, rather than derived by an automated convergence analysis. It must not be treated as a measured uncertainty or applied to the gauged calculation. No lattice-to-continuum gauge matching has been implemented.

At the scale $\mu = m_F = \sqrt{0.4}$, with radial cutoff 20 and $\ell_{\max} = 70$, the calculation gives

**Key result**

$$\log \left| \frac{\det' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}} = 40.9324 \pm 0.0300,$$
$$\left| \frac{\det' H_B}{\det H_F} \right|_{\overline{\mathrm{MS}}}^{-1/2} = 1.2931 \times 10^{-9}.$$

The bare value $-2079.9$ and the finite value 40.93 are not competing answers: they refer to different regulator treatments. The second is the renormalized global-scalar determinant.

Table 12.1: Bare and renormalized determinant quantities.

| Quantity | Value | Interpretation |
|---|---|---|
| Bare lattice trace–log | $-2079.904 \pm 6.450$ | Regulator dependent |
| Renormalized trace–log | $40.9324 \pm 0.0300$ | Finite global-scalar result |
| Renormalized factor | $1.2931 \times 10^{-9}$ | One-loop suppression |
| Renormalization scale | $\mu = m_F = \sqrt{0.4}$ | Stated scheme choice |
| Bounce action | 145.203947 | Controls $e^{-B}$ |

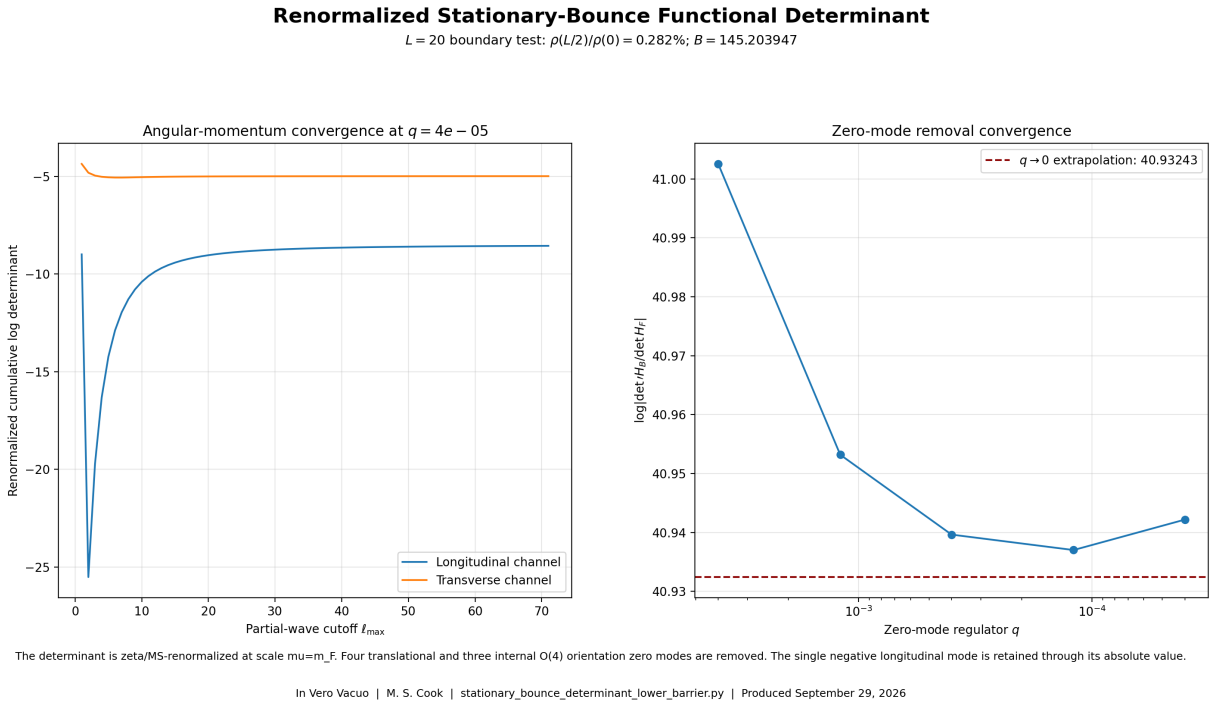



Figure 12.1: Continuum determinant convergence. The panels show the regulated zero-mode limit, partial-wave behavior, and the resulting finite primed determinant.

# Chapter 13

# Spacing Refinement and Regulator Diagnostics

The continuum bounce sampled on a coarse lattice produced lifted translation and orientation modes. A spacing-refinement study separated finite-volume effects from discretization effects and motivated the direct discrete-action relaxation. The sequence was:

1. verify that the bounce tail fits comfortably inside the box;
2. reduce the spacing while holding the physical box sufficiently large;
3. observe the collective-mode shifts move toward zero;
4. solve the discrete equation directly at the selected refinement;
5. use the relaxed configuration for the Hessian and SLQ calculations.

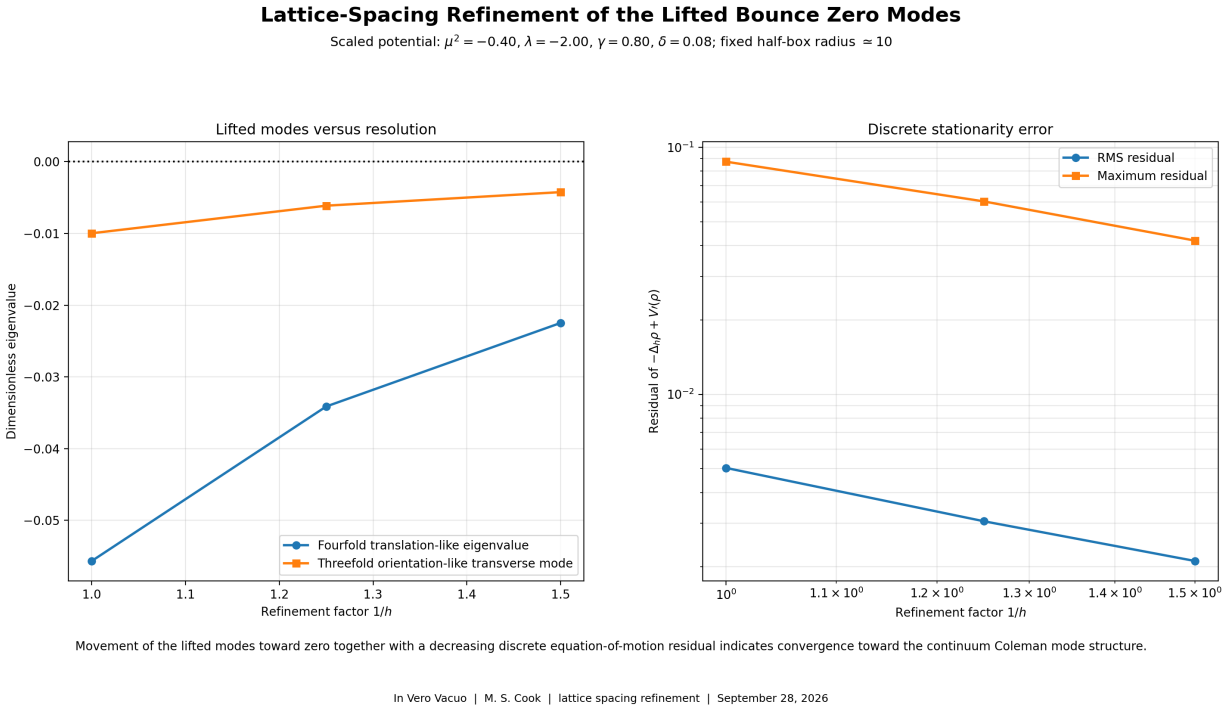



Figure 13.1: Spacing-refinement diagnostics for the sampled stationary bounce.

**Lattice-Spacing Refinement Numerical Summary**

Scaled potential: $\mu^2 = -0.40$, $\lambda = -2.00$, $\gamma = 0.80$, $\delta = 0.08$; fixed half-box radius $\simeq 10$

| L | h | Boundary | Principal negative | Translation mean | Orientation | EOM RMS |
|---|---|---|---|---|---|---|
| 20 | 1.000000 | 0.2820% | -2.357758e-01 | -5.566158e-02 | -9.976241e-03 | 5.032e-03 |
| 26 | 0.800000 | 0.2060% | -2.162747e-01 | -3.412852e-02 | -6.134877e-03 | 3.060e-03 |
| 32 | 0.666667 | 0.1673% | -2.058251e-01 | -2.248334e-02 | -4.244266e-03 | 2.114e-03 |

Eigenvalues are dimensionless in the reference coordinate system. The translation value is fourfold by hypercubic axis symmetry; the orientation value occurs in three identical internal components.

In Vero Vacuo | M. S. Cook | lattice spacing refinement | September 28, 2026

Figure 13.2: Evolution of the low collective sectors under lattice refinement. The study demonstrates why exact discrete stationarity is preferable to manually declaring shifted modes to be zero.

# Chapter 14

# The Spacing Campaign

This chapter documents a fixed-physical-extent comparison at $La = 24$, followed by a translation-mode audit and determinant-shift checks. The campaign uses $(L, h) = (24, 1)$, $(30, 0.8)$, and $(36, 2/3)$. It complements, but is distinct from, the earlier scalar spacing-refinement diagnostics. Here the central question is whether the low-mode subspace used in the longitudinal scalar SLQ trace spans all four finite-lattice translation directions. The primary audit is `SU2_Spacing_Campaign_Mode_Audit_Report_20261014.tex`; its follow-ups run through October 18 and are summarized below without merging distinct profiles, probe streams, or estimators.

## 14.1 Campaign scope and preliminary bare traces

The one-page `spacing_campaign_report.pdf`, generated on 11 October 2026, records an initial bare finite-lattice trace–log sweep at fixed $La = 24$:

| Lattice | $h$ | Bare log-determinant ratio | Probe standard error |
|---|---|---|---|
| $L = 24$ | 1 | $-1284.793$ | 6.9 |
| $L = 30$ | 0.8 | $-2053.827$ | 6.7 |
| $L = 36$ | 2/3 | $-2988.113$ | 6.9 |

Those values are regulator-dependent scalar-sector diagnostics. The PDF explicitly reports that continuum closure is not established: no continuum-renormalized reference or matching correction is included, and the probe errors omit discretization, finite-volume, matching, and bounce systematics. The later mode-audit report contains additional, separately identified reruns and follow-ups; do not combine their means with this preliminary table unless the run manifests establish a matched estimator, profile, and probe stream.

The paired mode-shift closure below concerns only the scalar longitudinal Hessian on the relaxed radial background; it does not include the coupled gauge, Goldstone, and Faddeev–Popov ghost determinants needed for a gauged physical prefactor. Separate gauge/ghost point-source inverse solves and profile-matched radial plus gauge–Goldstone–ghost trace-log pilots are also recorded here. Those pilots remain bare finite-lattice diagnostics and do not by themselves supply a counterterm-matched physical prefactor.

## 14.2 Identifying the four translation modes

In four Euclidean dimensions a localized bounce has four translation collective coordinates. On a finite lattice these are approximate modes, and the discretized eigenvalues need not be exactly zero. The baseline 12-mode deflation selected eigenvectors by sorted position. In the audited reruns, the first three selected directions tracked three translations, while the fourth selected eigenvector was instead a positive boundary mode; the positional cut consequently split the near-degenerate eigenspace and

omitted the fourth translation direction. This is why a fixed mode index is not a reliable definition of the collective-coordinate subspace.

For each direction $\mu$, the audit constructs a forward-difference tangent

$$T_\mu(x) = \frac{\rho(x + h\hat{\mu}) - \rho(x)}{h}.$$

If $Q$ is an orthonormal basis for a candidate low-eigenvalue subspace, the captured tangent fraction is

$$c_\mu = \frac{\|Q^T T_\mu\|^2}{\|T_\mu\|^2}.$$

The identification combines all four tangent overlaps with eigenvalue clustering, normalized eigenpair residuals, and tangent Rayleigh-residual checks. A targeted solve in the orthogonal complement recovered the least-captured direction where the initial solve was incomplete. The resulting four-mode basis, rather than the original positional cut, defines the corrected mode cache.

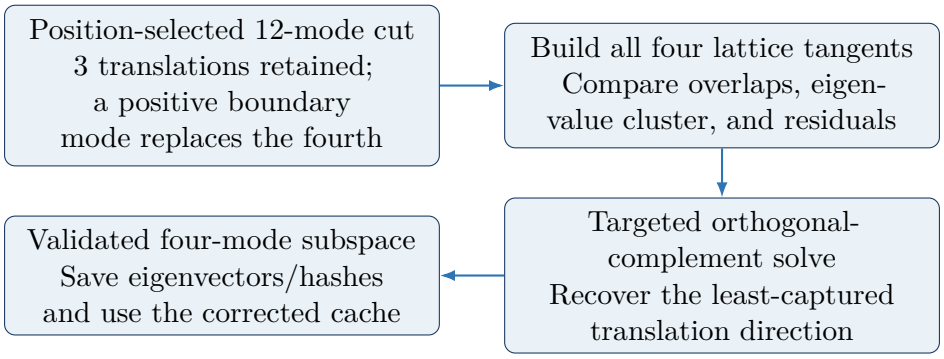



Figure 14.1: Translation-mode audit logic. The mode definition is based on the four tangent directions and eigenspace evidence, not a fixed position in the sorted spectrum.

| Lattice | $h$ | Fourfold $\lambda_T$ | Boundary $\lambda_B$ | Tangent capture | Scaled residuals |
|---|---|---|---|---|---|
| $L = 24$ | 1 | $-3.23612 \times 10^{-3}$ | 0.268131 | 90.876% | $1.0\text{–}1.6 \times 10^{-12}$ |
| $L = 30$ | 0.8 | $-2.12606 \times 10^{-3}$ | 0.273198 | 94.390% | $3.5\text{–}6.0 \times 10^{-12}$ |
| $L = 36^\dagger$ | 2/3 | $-2.46598 \times 10^{-4}$ | 0.281746 | 96.184% | $4.2\text{–}4.5 \times 10^{-11}$ |

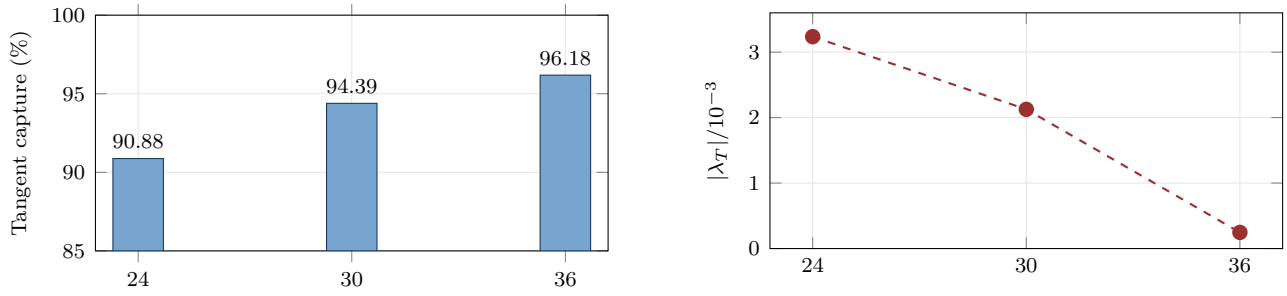



Figure 14.2: Audited four-translation cluster diagnostics at fixed $La = 24$. Capture rises and the magnitude of the common cluster eigenvalue falls, but these three points are not an extrapolation; the L36 profile is a distinct relaxation.

The listed $\lambda_T$ is the common value of a fourfold cluster; its within-cluster spread is at numerical precision. The small negative finite-$h$ values indicate residual lattice curvature, not exact zero modes.

Their magnitude decreases along these three points, and tangent capture increases, but the L36 point uses a distinct relaxation. The three-point pattern is diagnostic only: it does not establish a power law, a continuum extrapolation, or negligible finite-volume effects. The L36 capture-defect trend is likewise suggestive rather than a fitted scaling law.

## 14.3 Paired determinant-shift closure

The audited correction exchanges the omitted translation direction for the positive boundary direction in the deflation convention. With $\lambda_B > 0$ the boundary eigenvalue, $\lambda_T < 0$ the translation eigenvalue, and $v_4$ the normalized exchanged direction, the predicted per-probe log-determinant shift is

$$d_i^{\text{mode}} = \ln \lambda_B + |\ln |\lambda_T|| \, (z_i \cdot v_4)^2,$$

where $z_i$ is a shared Rademacher probe. Baseline and corrected longitudinal traces use the same vectors, so unchanged transverse and false-vacuum terms cancel in their paired difference.

| Lattice | Prediction | Paired shift, 256 probes | Difference / SE | First-16 subset shift |
|---|---|---|---|---|
| $L = 24$ | 4.4171 | $4.1665 \pm 0.5749$ | $-0.44$ | $3.9097 \pm 1.58$ |
| $L = 30$ | 4.8559 | $4.9685 \pm 0.5583$ | $+0.20$ | $2.0264 \pm 0.90$ |
| $L = 36^{\dagger}$ | 7.0410 | $7.1882 \pm 0.6691$ | $+0.22$ | $4.4479 \pm 1.34$ |

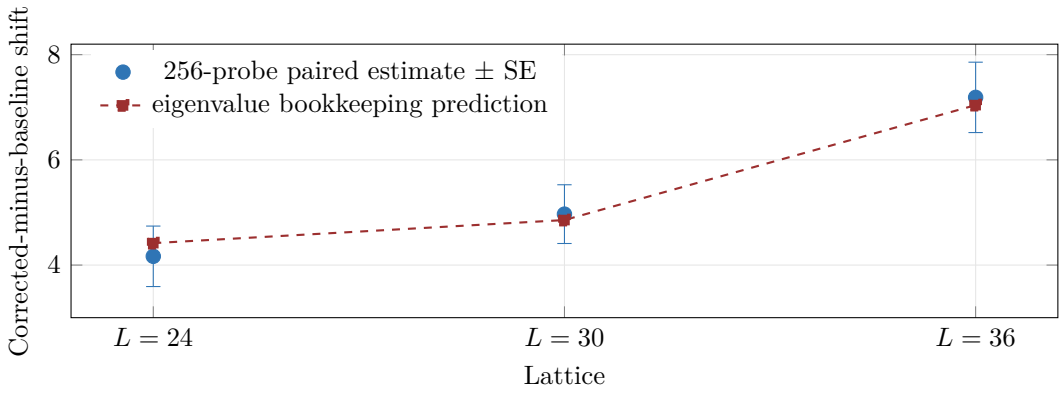



Figure 14.3: Paired scalar-longitudinal mode-exchange shifts versus their finite-mode bookkeeping predictions. All three differences are within 0.44 paired standard errors; this checks the mode exchange, not the full trace-log or a continuum limit.

These 256-probe aggregate means use 60 Lanczos steps and all lie within 0.44 paired standard errors of the eigenvalue-bookkeeping prediction. The stochastic trace-estimation background is reviewed in Refs. [5, 9]; the paired probes here are used to reduce noise in this finite-mode difference. This is stochastic consistency, not a per-probe identity or numerical-precision result. The separately reported 16-probe full-trace changes reproduce the first 16 rows of the 256-probe shift files to within $4 \times 10^{-9}$. In particular, the L30 subset mean of 2.0264 differs from the 256-probe mean because it is a finite subset, not because the estimator stream or mode exchange changed. These 16-probe shifts must not be read as a spacing trend.

A separate 64-probe L36 mode-shift order check found a low 40-step estimate ($5.180 \pm 1.052$); the shift changed by $+2.003 \pm 0.217$ from 40 to 60 steps, while the 60–120 step means clustered around the predicted 7.041. An archived-profile corrected-trace check changed by only $0.026 \pm 0.030$ between 60 and 120 steps. Neither check proves convergence of the full determinant trace: each is limited to its stated mode-shift or profile-specific comparison.

## 14.4 L36 projection regression and control variate

The completed GCS L36 follow-up is a separate 16-probe run, using matched probes at 60 and 120 Lanczos steps. Its bare full-trace results are:

| Steps | Baseline logdet | Corrected logdet | Paired shift |
|---|---|---|---|
| 60 | $-2992.071 \pm 7.164$ | $-2982.872 \pm 6.360$ | $+9.199 \pm 4.082$ |
| 120 | $-2992.106 \pm 7.176$ | $-2982.872 \pm 6.360$ | $+9.234 \pm 4.090$ |

Because this is a separate 16-vector stream, its paired means are not replacements for the 256-probe closure or for the earlier L36 traces.

Unlike the earlier paired-shift CSVs, this run stores the per-probe projection $x_i = (z_i \cdot v_4)^2$, permitting a direct check of $d_i = -1.26675 + 8.30775 x_i$. The measured slopes are 8.3901 and 8.4077 at 60 and 120 steps, respectively; the corresponding mean formula residuals are $+0.1393 \pm 0.2096$ and $+0.1743 \pm 0.2026$. Both are within one standard error of zero, while the fitted slopes are about 1.0–1.2% above the bookkeeping slope. This is a projection-level test for this sample and profile, not a precision validation of the full determinant or physical prefactor.

The same L36 probes also tested a localized-$\Delta V$ control variate. Let

$$H_0 = -\Delta_a + m_F^2, \qquad D_L = U_L(\rho) - m_F^2, \qquad D_T = U_T(\rho) - m_F^2,$$

and define

$$X_i = z_i^T H_0^{-1/2} (D_L + 3D_T) H_0^{-1/2} z_i.$$

Its exact expectation on the periodic finite lattice is

$$\mu_X = \mathrm{tr}[H_0^{-1}(D_L + 3D_T)] = G_0(0) \sum_x (D_L + 3D_T) = -2800.3036.$$

Using the prespecified coefficient $\beta = 1$ gives $Y_i^{\mathrm{CV}} = Y_i - \beta(X_i - \mu_X)$. The coefficient was fixed from the first-order log-determinant expansion, not fitted to these probes.

| Probe set | Steps | Raw mean ± SE | Adjusted mean ± SE | Variance reduction |
|---|---|---|---|---|
| 16 (seed 20261045) | 60 | $-2982.872 \pm 6.360$ | $-2997.224 \pm 4.119$ | 58.0% |
| 32 (seed 20261046) | 60 | $-3000.907 \pm 4.952$ | $-2998.303 \pm 2.986$ | 63.6% |

The 32-probe vectors are hash-verified and disjoint from the first set. For the 16-probe run, these summary values are unchanged at 120 steps to the reported precision. Across the two samples, adjusted means differ by 1.079 (0.21 combined standard errors), compared with 18.035 (2.24 combined standard errors) for the raw means. These are empirical finite-sample variance reductions, not a guarantee. Exact centering preserves the ensemble expectation but does not remove finite-Lanczos quadrature bias; report adjusted values alongside raw estimates. Applying the same control to both mode conventions cancels it in their paired difference.

## 14.5 Verified-subspace spacing comparison (16 October follow-up)

The fixed-box comparison uses $Lh = 24$ and the same verified four-translation subspace and mode-exchange convention at all three spacings. Its capture and finite-mode bookkeeping diagnostics are:

| $L$ | $h$ | Capture $C$ | $(1-C)/h^2$ | $\lambda_T$ | $\lambda_B$ | $\Delta_{\mathrm{ex}}$ |
|---|---|---|---|---|---|---|
| 24 | 1 | 90.8762% | 0.09124 | $-3.23612 \times 10^{-3}$ | 0.268131 | 4.41710 |
| 30 | 0.8 | 94.3902% | 0.08765 | $-2.12606 \times 10^{-3}$ | 0.273198 | 4.85592 |
| 36 | 2/3 | 96.1843% | 0.08585 | $-2.46598 \times 10^{-4}$ | 0.281746 | 7.04100 |

The three capture-defect values are broadly compatible with a leading $h^2$ diagnostic; the through-origin coefficient is 0.08966. This is a descriptive three-point result, not an extrapolation or an estimate with inferential uncertainty.

The matched 60-step subset comparison is:

| $L$ | $\Delta_{\mathrm{ex}}$ | Paired shift ± SE | $z$ | Corrected bare $\log\det \pm$ SE |
|---|---|---|---|---|
| 24 | 4.4171 | $3.9097 \pm 1.5796$ | $-0.32$ | $-1280.884 \pm 6.589$ |
| 30 | 4.8559 | $2.0264 \pm 0.8971$ | $-3.15$ | $-2052.087 \pm 6.213$ |
| 36 | 7.0410 | $9.1989 \pm 4.0820$ | $+0.53$ | $-2982.872 \pm 6.360$ |

L24 and L30 use the first 16 rows of their paired 256-probe streams; L36 uses all 16 probes from the completed archived-profile GCS run. The L30 subset is 3.15 SE below bookkeeping, while its 256-probe aggregate is $4.9685 \pm 0.5583$ versus 4.8559 predicted (+0.20 SE). L36's separate 256-probe closure is from the fresh-reproduction run and is not merged here. These unequal samples and run provenances preclude a spacing fit to the paired shifts. The corrected bare traces are finite-lattice scalar-sector values; their UV contributions remain unmatched.

## 14.6 Gauge and ghost point-source Green functions (16 October)

Using the campaign's verified radial profiles, a separate calculation solved the zero-background-link ($U_\mu = I$), periodic background-$R_\xi$ operators at $g = 0.65$, $q = g/2$, and $\xi = 1$. The ghost operator is $H_{\mathrm{gh}} = -\Delta_a + \xi q^2 \rho^2$; the gauge solves use the coupled link-action Hessian $H_{A\chi}$, including gauge–Goldstone mixing. Unit site-$\ell^2$ point sources are placed at the lattice center. Values below are per generator; the three identical SU(2) generator copies are not independent data.

| $L$ | $h$ | $G_{\mathrm{gh}}(x_0, x_0)$ | $\langle D_{\mu\mu}\rangle_\mu$ | $\langle D_{\mu\nu}\rangle_{\mu\neq\nu}$ | Ghost residual | Max gauge residual |
|---|---|---|---|---|---|---|
| 24 | 1 | 0.1501842 | 24.497442 | $4.16301 \times 10^{-3}$ | $9.05 \times 10^{-9}$ | $5.14 \times 10^{-8}$ |
| 30 | 0.8 | 0.0969231 | 10.025052 | $1.03276 \times 10^{-3}$ | $7.71 \times 10^{-9}$ | $6.62 \times 10^{-8}$ |
| 36 | 2/3 | 0.0676788 | 4.898023 | $3.38883 \times 10^{-4}$ | $8.81 \times 10^{-9}$ | $8.64 \times 10^{-8}$ |

The gauge columns include the four $A_\mu$ source directions and their Goldstone responses. The return matrix's diagonal spread is below $7.1 \times 10^{-8}$ and source-reciprocity error below $1.7 \times 10^{-11}$; the residuals are explicitly recomputed true residuals. L36 uses the archived-profile GCS checkpoint, not the separate fresh-reproduction profile. These are centered position-space Green functions, not an ensemble average, momentum dressing function, or gauge/ghost determinant ratio; their spacing trend is not a continuum result.

The 15 source files, checkpoint and field hashes, and solver records are retained under the campaign directory:

`spacing_campaign_audit_20261014/gauge_ghost_propagators_20261016/`

The summary file is `gauge_ghost_propagator_summary.json`. The resumable calculation is `su2_gauge_ghost_green_20261016.py`.

## 14.7 17–18 October 2026 follow-ups: volume, controls, and trace logs

The follow-up report adds a fixed-spacing volume diagnostic, channel-resolved scalar and free Green-function checks, matched trace-log pilots at L30 and L24, and an L24 gauge-parameter stability scan. These runs use distinct profiles and probe streams where noted; none should be silently merged with the fixed-extent spacing comparison above.

### 14.7.1 Fixed-spacing volume and exact small-lattice closure

This separate comparison fixes $h = 1$ and uses $L = 24, 30$, so the box lengths are 24 and 30 rather than the fixed-extent $Lh = 24$ sequence. Both profiles converged from the same continuum seed in three Newton updates. The final residual RMS/max values are $1.06 \times 10^{-15}/1.09 \times 10^{-14}$ at L24 and $8.07 \times 10^{-16}/1.83 \times 10^{-14}$ at L30.

| $L$ | $Lh$ | Sites | $J_{\mathrm{tr}}$ (forward) | $\xi q^2\overline{\rho^2}$ | Lowest ghost eigenvalue |
|---|---|---|---|---|---|
| 24 | 24 | 331,776 | 498.268778 | $5.182963 \times 10^{-4}$ | $4.016493 \times 10^{-4}$ |
| 30 | 30 | 810,000 | 498.323990 | $2.122956 \times 10^{-4}$ | $1.627839 \times 10^{-4}$ |

The forward translation measure changes by 0.011% from L24 to L30, whereas $\xi q^2\overline{\rho^2}$ and the lowest ghost eigenvalue decrease by 59.0% and 59.5%. The constant-ghost Rayleigh values agree with the reported $\xi q^2\overline{\rho^2}$ predictions. The four lowest coupled gauge modes per generator are near $7.2002 \times 10^{-8}$ (L24) and $7.8274 \times 10^{-10}$ (L30); their subspaces capture about 99.85% and 99.94% of the constant-gauge trials. The L30 scaled residuals, $7.5 \times 10^{-6}$–$1.1 \times 10^{-5}$, are larger than the L24 values, $4.2 \times 10^{-8}$–$6.5 \times 10^{-8}$, so the tiny L30 eigenvalues are not precision results without further convergence checks.

The adopted convention projects the exact four false-vacuum constant gauge modes and one constant ghost mode per generator, retains the lifted bounce modes, and treats the negative mode separately. An independent dense $L = 2$ synthetic-matrix closure agrees between slogdet and the symmetric-eigenvalue log sum to $7.816 \times 10^{-13}$. It tests projector and determinant bookkeeping, not a physical bounce determinant. The referenced low-mode archives were not included with that upload, so their eigenvectors cannot be independently rechecked from those files alone. These fixed-$h$ results are a volume/convention diagnostic, not a spacing extrapolation.

### 14.7.2 Channel-resolved scalar Born terms and free Green functions

For $H_0 = -\Delta_a + 0.4$, define $D_L = V''(\rho) - 0.4$ and $D_T = V'(\rho)/\rho - 0.4$. Since the radial Higgs and three Goldstone operators are separate blocks, the first-order trace is $\mathrm{Tr}[G_0(D_L + 3D_T)]$ and the second-order log term is

$$-\frac{1}{2}\left\{\mathrm{Tr}[(G_0 D_L)^2] + 3\,\mathrm{Tr}[(G_0 D_T)^2]\right\}, \qquad \mathrm{Tr}[(G_0 D)^2] = \sum_{x,y} G_0(x-y)^2 D(x) D(y),$$

evaluated by periodic FFT convolution. Do not square $D_L + 3D_T$ as if the channels formed one operator. The channel-resolved values are:

| $L$ | $h$ | First-order trace | Longitudinal $\mathrm{Tr}[(G_0 D_L)^2]$ | Transverse per component | Total quadratic trace | $-\frac{1}{2}$ log term |
|---|---|---|---|---|---|---|
| 24 | 1 | −1130.549 | 206.511 | 32.441 | 303.834 | −151.917 |
| 30 | 0.8 | −1882.607 | 244.317 | 37.639 | 357.235 | −178.618 |
| 36 | 2/3 | −2800.304 | 271.708 | 41.522 | 396.273 | −198.137 |

The L36 first-order trace matches its saved exact control expectation, −2800.303571579602, to $2 \times 10^{-12}$. Squaring $D_L + 3D_T$ instead overstates the quadratic trace by 351.385, 413.535, and 460.594 at L24, L30, and L36. A dense L3 periodic-matrix check agrees with the FFT result to $2.3 \times 10^{-16}$. The L24/L30 16-probe control variate was not applied because current CSV hashes disagree with historical manifests and the requisite probe vectors/projections are absent. Their first-16 paired shift cross-checks pass, but do not establish the projection regression. The separate L36 32-probe, fixed-$\beta = 1$ control variate lowers the 60-step standard error from 4.952 to 2.986; it does not remove finite-Lanczos bias and is not merged with the other samples.

For a same-code free $\rho = 0$ baseline, the four constant gauge vectors and one constant ghost vector are projected before inversion. At $\xi \neq 1$ the free gauge operator uses the exact forward-difference $R_\xi$

grad–div extension; the inhomogeneous bounce implementation is available at $\xi = 1$. The center returns are:

| $L$ | $h$ | Ghost return | Gauge diagonal $\xi = 0.5$ | Gauge diagonal $\xi = 1$ | Gauge diagonal $\xi = 2$ |
|---|---|---|---|---|---|
| 24 | 1 | 0.154689967 | 0.135353721 | 0.154689967 | 0.193362459 |
| 30 | 0.8 | 0.099057599 | 0.086675399 | 0.099057599 | 0.123821999 |
| 36 | 2/3 | 0.068811153 | 0.060209759 | 0.068811153 | 0.086013942 |

The maximum discrepancy from the exact finite-volume Fourier gauge matrix is below $7 \times 10^{-17}$ and projected true residuals are below $6.6 \times 10^{-15}$. At $\xi = 1$ the free gauge and ghost returns coincide. Compared with these free returns, the bounce ghost returns in the preceding section are lower by approximately 2.9%, 2.2%, and 1.6% at L24/L30/L36. These are normalization checks, not a spacing law or determinant result.

The independent continuum partial-wave reruns give extrapolated log determinants 40.888037, 40.932428, and 40.951217 for $\ell_{\max} = 50, 70, 90$. The 70-to-90 change is 0.0188, but the sampled 50-to-90 spread is 0.0632; the quoted 0.03 uncertainty is hard-coded, not a validated total error. This uses an independent shooting profile, so lattice-to-continuum scalar closure and the tadpole/$\delta m^2$ cancellation remain open. Action-3 artifacts are under `spacing_campaign_audit_20261014/action3_checks_20261017/`; the summary is `action3_check_summary.json`.

### 14.7.3 Profile-matched trace-log pilot at L30

The first matched determinant-sector pilot uses the archived $L = 30$, $h = 0.8$, $Lh = 24$ profile at $g = 0.65$, $\xi = 1$, with the matched false vacuum $\rho = 0$ and periodic stencil. The checkpoint SHA-256 is `e8ebeda85e400a0f59b6238482fc2e6d5cf6fea10e81453b4a01c560343aed95`; the C-order field-array hash is `3216b57de7a0bd4d5c18b80b1b7de5eb463d7830e3a5a8c17282d93735605383`. The combined bare term is $\mathcal{D}_{\text{bare}} = \frac{1}{2}\Delta_L + \mathcal{D}_{A\chi c}$; the radial bounce determinant excludes the single negative mode and four audited translations, whose factors are not included.

| Contribution | Mean | Probe SE | Convention |
|---|---|---|---|
| Radial $\Delta_L$ | −683.448 | 3.478 | Primed bounce; false radial full |
| Radial $\frac{1}{2}\Delta_L$ | −341.724 | 1.739 | Bosonic one-loop weight |
| $\mathcal{D}_{A\chi c}$ | −577.661 | 11.446 | Three generators |
| $\mathcal{D}_{\text{bare}}$ | −919.385 | 11.578 | Independent probe streams |

The radial trace uses 8 matched probes at 60 Lanczos steps; the gauge– Goldstone–ghost trace uses 8 paired probes at 120 steps. The exact false- vacuum constant gauge and ghost projectors are applied, and the displayed $+\frac{1}{2}\log\xi$ measure correction vanishes at $\xi = 1$. The combined statistical SE assumes independent sector streams. Gauge and ghost false-vacuum quadratic-form means are each within 0.75 SE of exact Fourier traces; the radial false-baseline mean is 1.24 of its across-probe SE away, consistent with eight-probe sampling noise. The combined linear versus quadratic $\epsilon \to 0^+$ fit changes by 0.0029, negligible beside the probe SE. Gauge 30-to-60 steps change the result by $-1.426 \pm 0.112$, while the 60-to-120 change is $-3.33 \times 10^{-4} \pm 8.97 \times 10^{-6}$; 30 steps are insufficient, and the 120-step check is only a finite-order comparison.

The negative eigenvalue is −0.184993 and the translation cluster is fourfold at −0.002126065; each forward tangent captures 94.390% of the cluster and $J_{\text{tr}} = 514.367$ is kept separate. The coupled gauge cluster contains four modes near $7.13183 \times 10^{-8}$, captures 99.8466% of each constant-$A_\mu$ trial, and has maximum 8-mode scaled residual $2.28 \times 10^{-10}$. The constant-ghost trial capture is 99.9663%. These are bare finite-lattice diagnostics only: this L30 pilot has no $\xi$ scan, counterterm matching, continuum/volume limit, or physical-rate normalization. The later L24 results are separate runs, not replacements for this L30 dataset. Paired outputs and run records are under `spacing_campaign_audit_20261014/action4_trace_logs_20261017/`.

### 14.7.4 Profile-matched trace-log follow-up at L24

A second $\xi = 1$ pilot uses the matched $L = 24$, $h = 1$, $Lh = 24$ relaxed profile, with 16 radial and 16 gauge–Goldstone–ghost probes. It uses 60 radial Lanczos steps and 180 gauge–ghost steps, with linear zero-shift extrapolation from $\epsilon = 10^{-9}, 10^{-8}, 10^{-7}$. Under the same normalized primed convention as the L30 pilot, the combined bare term is $\mathcal{D}_{\text{bare}} = \frac{1}{2}\Delta_L + \mathcal{D}_{A\chi c}$:

| Contribution | Mean | Probe SE |
|---|---|---|
| Radial $\Delta_L$ | $-439.395$ | 3.067 |
| Radial $\frac{1}{2}\Delta_L$ | $-219.697$ | 1.533 |
| $\mathcal{D}_{A\chi c}$, three generators | $-407.024$ | 4.906 |
| $\mathcal{D}_{\text{bare}}$ | $-626.722$ | 5.140 |

The combined SE assumes independent radial and gauge probe streams. The false-vacuum gauge and ghost constant modes are projected explicitly, the normalized $+\frac{1}{2}\log\xi$ measure term is retained (zero at $\xi = 1$), and the radial trace excludes one negative mode and four audited translations; their factors are not included. The gauge quadratic-vs-linear zero-shift fit changes by $-0.00284$, and the paired 120-to-180-step change is negligible at the reported precision. This single-$\xi$ estimate is not a gauge-stability test or a renormalized determinant ratio. Its summaries and profile hashes are recorded in `spacing_campaign_audit_20261014/action4_trace_logs_20261017/`.

### 14.7.5 L24 numerical and gauge-parameter stability (18 October)

The separate Action 5 scan evaluates only the gauge–Goldstone–ghost sector at $L = 24$, $h = 1$, $g = 0.65$, with 32 probes, 8 deflated modes, $\xi = 0.5, 1, 2$, Lanczos orders 60/120/180, and common IR shifts $10^{-9}, 10^{-8}, 10^{-7}$. It regenerates the probe stream in the recorded draw order: the first 16 coupled and ghost probe hashes match the Action 4 L24 stream, and exact false-vacuum Fourier quadratic forms reproduce the Action 4 forms to at most $9.3 \times 10^{-10}$. The four constant gauge and one constant ghost false-vacuum modes are explicitly projected, $\chi$ is left unprimed, and the normalized $+\frac{1}{2}\log\xi$ correction is retained.

| $\xi$ | Normalized bare $\mathcal{D}_{A\chi c}$ | Paired shift vs. $\xi = 1$ |
|---|---|---|
| 0.5 | $-238.719 \pm 110.405$ | $+165.509 \pm 111.699$ |
| 1 | $-404.228 \pm 5.868$ | reference |
| 2 | $-585.162 \pm 111.291$ | $-180.935 \pm 110.092$ |

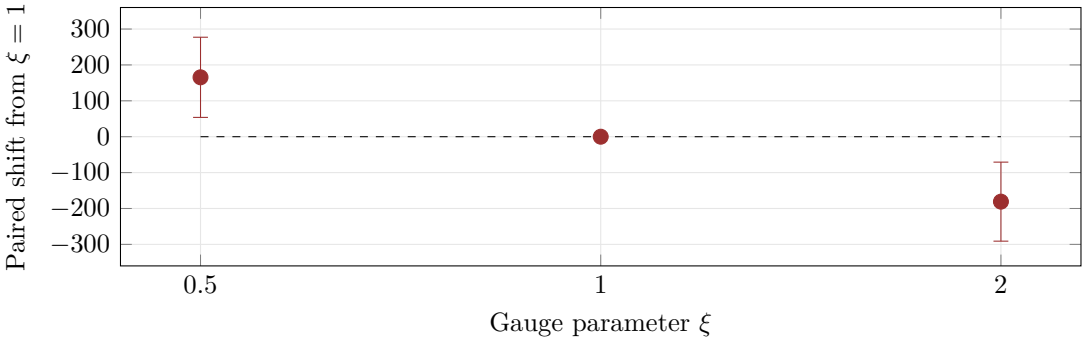



Figure 14.4: Action 5 L24, 32-probe paired gauge–Goldstone–ghost shifts from $\xi = 1$ with $\pm 1$ standard-error bars. The broad non-reference estimates are only about 1.5–1.6 SE from zero; they do not establish gauge-parameter cancellation.

The entries are 180-step linear zero-shift intercepts; the displayed uncertainties are paired-probe standard errors from 32 probes. The $\xi = 0.5$ and 2 paired differences are only about 1.5–1.6 standard

errors from zero, so this scan neither resolves a gauge dependence nor demonstrates its cancellation. The solver's automatic "PASS" flag is rejected: it accepts a spread of about 347 against three times a large unpaired standard error of about 373, which is not a valid gauge-cancellation criterion. The quadratic-vs-linear $\epsilon \to 0^+$ intercept sensitivity is below 0.0031 for all three $\xi$ values.

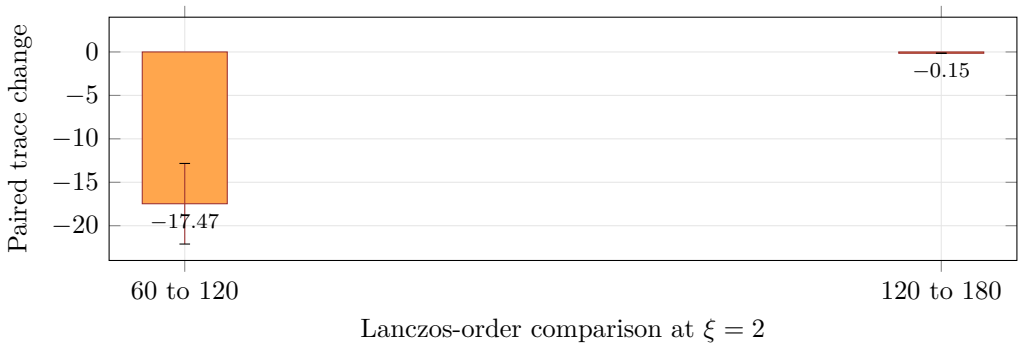



Figure 14.5: Resolved paired trace changes as Lanczos order increases in the $\xi = 2$ Action 5 run. Even the smaller 120-to-180 change is statistically resolved, so 180 steps are not a demonstrated Krylov plateau.

Lanczos changes at $\xi = 0.5$ and 1 are small in absolute size. At $\xi = 2$, the paired trace changes by $-17.471 \pm 4.638$ from 60 to 120 steps and by $-0.1516 \pm 0.0158$ from 120 to 180; both are resolved, so 180 steps are not yet stable and a longer Krylov run is needed. At $\xi = 1$, the new 32-probe 180-step intercept differs from the separate Action 4 16-probe value by $+8.707 \pm 7.446$ on their shared first 16 probes (1.17 paired SE), compatible with that comparison but not a reason to merge the datasets: the trace implementations differ. The L2 dense-matrix closure ($7.8 \times 10^{-13}$) and L3 homogeneous normalized-$\xi$ spread ($5.1 \times 10^{-13}$) are software/normalization fixtures, not bounce validations. Counterterm matching and the remaining one-loop terms are absent, so no physical gauge independence or decay prefactor follows from this scan. The paired reanalysis and source outputs are under `spacing_campaign_audit_20261014/action5_stability_20261018/L24_xi_scan_n32/`.

## 14.8 Provenance, interpretation, and next work

Keep the archived-profile trace, the earlier reproduction, the 256-probe mode-shift stream, and the GCS follow-up as distinct analysis runs. The GCS checkpoint container and recovered archived checkpoint have different hashes, but their stored field arrays are bitwise identical; the audited fourfold translation eigenvalue agrees to six significant figures. The GCS probe CSV hash matches its completed manifest. By contrast, the six earlier L24/L30/L36 16-probe baseline/corrected CSV files differ from the hashes in their historical manifests. Preserve both recorded and observed hashes rather than silently rewriting provenance. Archived-versus-rerun full-trace means also differ at L24 ($-2.72$) and L30 ($+3.98$); unrecorded selection inside the degenerate mode cluster is plausible but unconfirmed.

Established results now include the four-mode subspace identification, the 256-probe aggregate mode-shift closure, the GCS projection-level regression, the fixed control-variate checks, the verified-subspace spacing comparison, the fixed-$h$ volume and synthetic $L = 2$ closure diagnostics, the channel-resolved scalar Born terms, and the bounce/free gauge–ghost Green-function checks. Profile-matched radial plus gauge–Goldstone–ghost bare trace-log pilots are complete at L30 ($h = 0.8$) and L24 ($h = 1$), both at $\xi = 1$; they use distinct 8- and 16-probe streams. The separate 32-probe L24 $\xi$ scan is a stability diagnostic, but does not establish cancellation. Still open are production- level probe precision, a matched L36 trace-log, a longer Krylov check at $\xi = 2$, gauge-cancellation tests with adequate paired precision and matched counterterms, collective-coordinate and gauge-volume normalization in the physical rate, ultraviolet matching, lattice-to-continuum closure, and controlled continuum and volume limits. The point-source

propagators and synthetic $L = 2$ closure do not substitute for these determinant checks. All results remain finite-lattice diagnostics, not a continuum prediction or a completed SU(2)/electroweak decay rate.

The next trace-log stage is to repeat the matched radial-Higgs, coupled gauge–Goldstone, and ghost calculations at L36 using its frozen bounce and matched false vacuum; do not mix the L36 archived profile with the separate fresh reproduction, and do not add separate transverse-Goldstone determinants on top of the coupled blocks. Increase probe counts, extend the $\xi = 2$ Krylov run beyond 180 steps, and use adequately precise paired comparisons at $\xi = 0.5, 1, 2$ before drawing gauge-cancellation conclusions. Then complete counterterm matching, fixed-box spacing and larger-volume checks, and the negative-mode and collective-coordinate factors before assembling $\Gamma/V = \mathcal{A}e^{-B}$. The existing determinant-scan defaults target a different L26 checkpoint and low probe/step counts, so they must be overridden and validated.

The primary audit trail is in `SU2_Spacing_Campaign_Mode_Audit_Report_20261014.tex`, with the preliminary sweep in `spacing_campaign_report.pdf`. Machine-readable summaries, manifests, and per-probe CSVs are under `spacing_campaign_audit_20261014/`. The 16-probe control-variate summary is `spacing_campaign_audit_20261014/L36_GCS_deltaV_control_variate_summary.json`; the independent 32-probe validation artifacts are under `spacing_campaign_audit_20261014/control_variate_validation/`. The matched L24 trace-log summary and Action 5 gauge-stability reanalysis are under `spacing_campaign_audit_20261014/action4_trace_logs_20261017/L24_combined_xi1_paired_summary.json` and `spacing_campaign_audit_20261014/action5_stability_20261018/L24_xi_scan_n32/paired_stability_summary.json`, respectively.

# Chapter 15

# Output, Checkpointing, and Reproducibility

## 15.1 Checkpoint strategy

Long calculations are resumable at two levels:

- the relaxed field and Newton history are saved in a compressed NPZ file;
- in the established scalar SLQ mode, every completed stochastic probe is appended to a CSV checkpoint.

Low eigenpairs are cached separately because they are expensive and dominate memory. Random seeds are fixed through environment variables so that repeated runs can be compared probe by probe.

**SU(2) determinant restart limitation**

The present `su2_determinant` implementation writes its probe table only after all requested gauge parameters and probes have completed. Before a long cloud run, it should be extended to save low eigensystems and append each completed $(\xi, \text{probe})$ result atomically. This is a reliability requirement rather than a change in the determinant definition.

## 15.2 Cloud-migration checklist

For migration to a persistent compute environment, archive or recreate the following items before launching the production determinant:

1. the exact unified Python source and its SHA-256 digest;
2. the regenerated $L = 26$ NPZ checkpoint and a checksum stored outside the ephemeral workspace;
3. `su2_prefactor_validation_20261002/validation_summary.json` and its validation PDF;
4. the scalar relaxation and SLQ summaries used as regression references;
5. the counterterm coefficient template and all environment-variable settings; and
6. `run_higgs_parallel.py`, its source digest, and the generated campaign/per-lattice manifests; and
7. a separate persistent output directory for each lattice size, holding low modes, per-probe checkpoints, summaries, and logs.

After transfer, rerun the inexpensive Ward suite and compare the bounce action, RMS residual, field checksum, and homogeneous gauge spread before beginning the production SLQ calculation.

## 15.3 Process-parallel $L = 26$ and $L = 30$ test campaign

The portable supervisor `run_higgs_parallel.py` runs one independent solver process per requested lattice size. Within each job it executes the relaxation stage first and then the SLQ stage against that job's relaxed field. Each lattice size has its own output directory, bounce checkpoint, low-mode cache, per-probe CSV, logs, and machine-readable run manifest; the campaign directory records the combined status. The two lattice sizes do not share scratch files or mutate the existing archived $L = 26$ results.

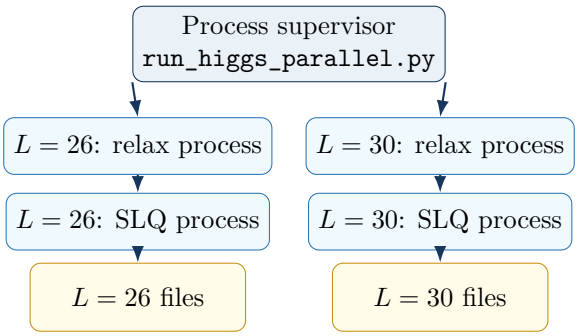



Figure 15.1: Process-level parallelism for the two requested lattice sizes. Each job runs relaxation and SLQ sequentially, while the $L = 26$ and $L = 30$ jobs may run concurrently. This supervisor does not split a single lattice operator across multiple nodes.

Preview the plan before allocating compute, then launch the two jobs in parallel on a host with sufficient memory:

```
python run_higgs_parallel.py --lattice-sizes 26 30 --workers 2 \\
  --threads-per-worker 1 --spacing 0.8 --output-root parallel_higgs_runs \\
  --dry-run

python run_higgs_parallel.py --lattice-sizes 26 30 --workers 2 \\
  --threads-per-worker 1 --spacing 0.8 --probes 16 \\
  --steps 20,40,60 --output-root parallel_higgs_runs

# Resume compatible bounce and probe checkpoints after interruption.
python run_higgs_parallel.py --lattice-sizes 26 30 --workers 2 \\
  --threads-per-worker 1 --spacing 0.8 --probes 16 \\
  --steps 20,40,60 --output-root parallel_higgs_runs --resume
```

Listing 15.1: Preview, execute, and resume the two-size process campaign.

The worker limit controls how many lattice-size processes share one host; the thread limit is applied before NumPy/SciPy import to OpenBLAS, OpenMP, MKL, NumExpr, BLIS, and Accelerate-compatible thread variables. Single-thread workers avoid nested oversubscription, but they do not reduce each process's memory requirement. Since the $L = 30$ operator and workspace are larger and the simultaneous peak memory has not yet been profiled, first measure each size separately on the target machine. Set `-workers 2` on one host only when its measured RAM headroom safely covers both processes; otherwise run one lattice per VM or scheduler task.

Google Cloud, AWS, and IBM Cloud can use the same Python runner on a compatible Linux VM or batch/HPC worker; no provider SDK is required. For separate cloud workers, invoke one lattice size per task and give each a unique output root, for example `parallel_higgs_runs/L26_worker` and `parallel_higgs_runs/L30_worker`. Use a mounted persistent filesystem or worker-local disk for the

run directory, then copy completed manifests, checkpoints, summaries, and logs to object storage. The supervisor does not submit cloud jobs, configure IAM, or mount cloud storage. Avoid two independent supervisors writing the same campaign manifest.

The campaign and per-lattice manifests record source SHA-256, lattice and SLQ settings, Python/platform/host information, thread limits, stage status, elapsed times, logs, and the relaxed-field digest. The run is process-parallel across lattice sizes, not MPI-distributed; parallel probe sharding within one SLQ job is a separate future extension.

At normal campaign finalization the supervisor automatically writes `parallel_higgs_runs/campaign_report.pdf`, exactly seven landscape pages, including completed or failed jobs. Its visuals summarize campaign status, radial bounce profiles, central field slices, Newton residual history, SLQ Lanczos-order and probe convergence, observed stage times, and the reproducibility ledger. The numerical summary is explicitly labeled as a bare finite-lattice determinant; stochastic probe errors are not presented as continuum or discretization uncertainties. The adjacent `campaign_source_bundle.zip` contains the supervisor, unified solver, nonlinear SU(2) vertex module, and a short dependency note, so the exact code can be transferred with the results. Python and its NumPy/SciPy/Pandas/ Matplotlib dependencies must still be installed on a worker. A forced process termination cannot finalize a PDF; rerun with `-resume` to complete the campaign and generate the report. The report also flags a spatially uniform stationary field as a failed localized-bounce diagnostic even when the discrete residual itself has converged.

## 15.4 Primary output files

Table 15.1: Machine-readable and publication outputs.

| Path | Purpose |
|---|---|
| `discrete_bounce_relaxation_L26_h0.8/relaxed_bounce_checkpoint.npz` | Relaxed lattice field and Newton history |
| `run_higgs_parallel.py` | Provider-neutral process supervisor for independent $L = 26$ and $L = 30$ jobs |
| `parallel_higgs_runs/campaign_manifest.json` | Combined settings, source digest, runtime summary, and job statuses |
| `parallel_higgs_runs/campaign_report.pdf` | Automatically generated seven-page visual campaign summary |
| `parallel_higgs_runs/campaign_source_bundle.zip` | Portable archive of runner, solver, local vertex module, and dependency note |
| `parallel_higgs_runs/L26_h0.8/run_manifest.json` | Per-lattice settings, stage logs/status, and relaxed-field digest; analogous for $L = 30$ |
| `discrete_bounce_relaxation_L26_h0.8/discrete_relaxation_summary.json` | Stationarity and low-spectrum summary |
| `relaxed_bounce_slq_L26_h0.8/probe_checkpoint.csv` | Resumable per-probe trace–log estimates |
| `relaxed_bounce_slq_L26_h0.8/slq_summary_by_steps.csv` | Lanczos-order convergence table |
| `relaxed_bounce_slq_L26_h0.8/relaxed_slq_summary.json` | Final bare SLQ result |
| `scaled_potential_continuum_renormalized/determinant_zero_mode_convergence.csv` | Regulated zero-mode extrapolation |

| Path | Purpose |
|---|---|
| scaled_potential_continuum_renormalized/determinant_summary.json | Continuum-renormalized determinant result |
| Higgs_Continuum_Renormalized_Determinant_Report_L26.pdf | Five-page concise production report |
| su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz | Target regenerated field; absent after the October 2 workspace reset and therefore requiring regeneration or external restoration |
| su2_prefactor_validation_20261002/validation_summary.json | October 2 PASS record for the regenerated stationary bounce |
| su2_prefactor_validation_20261002/SU2_Prefactor_Validation_Report.pdf | Gauge-identity and homogeneous-normalization plots |
| su2_determinant_scan/determinant_summary.json | Regulated gauge–Goldstone–ghost determinant status and gauge spreads |
| su2_determinant_scan/determinant_summary.csv | Determinant means, errors, IR shifts, and zero-mode policies |
| su2_determinant_scan/SU2_Determinant_Gauge_Scan.pdf | $R_\xi$ and IR-regulator diagnostic plots |
| su2_counterterm_matching/counterterm_basis.csv | Local kinetic and polynomial basis integrals on the selected background |
| su2_counterterm_matching/counterterm_summary.json | Pole-closure tests and any supplied finite matching contribution |
| su2_counterterm_coefficients_template.json | Derived pole residues and explicit placeholders for finite matching constants |
| SU2_Determinant_Counterterm_Morning_Report_20261001.tex | Standalone October 1 determinant and counterterm development report |
| SU2_Physical_Prefactor_Progress_Report.tex | Standalone report wrapper, sharing the body reproduced in this guide |
| SU2_Physical_Prefactor_Progress_Content.tex | Complete progress report and adopted EFT matching prescription |
| SU2_Physical_Prefactor_Progress_Report.pdf | Compiled five-page standalone progress report |
| SU2_L6_Single_Core_Pilot_Narrative_Report_20261005.tex | October 5 pilot and recorded increased-probe/quadrature audit |
| SU2_L6_Complete_Multiplet_Report_20261005.tex | Recorded multiplet-coverage and matched-probe variance comparison |
| SU2_Nonconstant_Prerequisite_Report_20261005.tex | Fresh $L = 3, 4$ determinant, Ward, and local-basis audit report |
| su2_nonconstant_prerequisite_audit.py | Separate single-core audit harness; production solver unchanged |
| su2_nonconstant_prerequisites_20261005/audit_summary.json | Retained aggregate audit results; full raw archive only partially retained |
| SU2_Matching_Coefficient_Progress_Report_20261006.tex | October 6 audit of matching readiness and explicit finite-coefficient status |
| SU2_Next_Steps_Finite_Matching_20261006.tex | October 19 matching status, finite-sector pilots, and remaining checks |
| SU2_Spacing_Campaign_Mode_Audit_Report_20261014.tex | Translation-mode and determinant audit; follow-ups through October 18 |

| Path | Purpose |
|---|---|
| su2_lattice_vertices.py | Nonlinear scalar-link, Wilson-gauge, ghost, and Faddeev–Popov generators |
| su2_nonlinear_vertex_audit.py | Independent numerical audit of the nonlinear vertex implementation |
| su2_nonlinear_vertex_tables_20261007_conserving.json | Momentum routings, vertex tensor dimensions, and table metadata |
| su2_full_gauged_projection_setup.py | October 8 fixed routing, projector, and common-IR setup generator |
| su2_full_gauged_projection_tables_20261008.npz | Selected full color/polarization vertex tensors and projector arrays |
| su2_full_gauged_projection_tables_20261008.json | Fixed momenta, parameters, projector definitions, and IR prescription |
| su2_full_gauged_projection_audit.py | Independent vertex, kinematic, projector, and Ward-identity audit |
| su2_full_gauged_projection_audit_20261008.json | Retained PASS status, discrepancies, and tolerances |
| su2_four_point_gauge_sector_20261008.py | Gauge-linked four-Higgs insertion evaluator at the fixed local-contact routes |
| su2_four_point_gauge_sector_20261008_ircheck.json | Four-regulator pilot values, QMC errors, extrapolation check, and diagnostics |

## 15.5 Recommended commands

```
# Preview and launch L=26/L=30 as isolated relax--SLQ processes
python run_higgs_parallel.py --lattice-sizes 26 30 --workers 2 \\
  --threads-per-worker 1 --spacing 0.8 --output-root parallel_higgs_runs --dry-run
python run_higgs_parallel.py --lattice-sizes 26 30 --workers 2 \\
  --threads-per-worker 1 --spacing 0.8 --probes 16 \\
  --steps 20,40,60 --output-root parallel_higgs_runs

# Fast analytic check
HIGGS_RUN_MODE=validation python Higgs_fluctuation_solver_stabilized.py

# Low-spectrum exploratory report
HIGGS_RUN_MODE=research HIGGS_L=20 HIGGS_NUM_MODES=132 \
python Higgs_fluctuation_solver_stabilized.py

# Individual production stages
HIGGS_RUN_MODE=relax python Higgs_fluctuation_solver_stabilized.py
HIGGS_RUN_MODE=slq python Higgs_fluctuation_solver_stabilized.py
HIGGS_RUN_MODE=continuum HIGGS_MU_SQ=-0.40 HIGGS_LAMBDA=-2.00 \
HIGGS_GAMMA=0.80 HIGGS_DELTA=0.08 HIGGS_L=20 \
python Higgs_fluctuation_solver_stabilized.py

# Complete scalar-only sequential production pipeline
HIGGS_RUN_MODE=full python Higgs_fluctuation_solver_stabilized.py

# New gauge prerequisites (not a physical prefactor)
HIGGS_RUN_MODE=su2_validation python Higgs_fluctuation_solver_stabilized.py

# Optional stationary-bounce gauge checks
HIGGS_RUN_MODE=su2_validation \
```

```
HIGGS_SU2_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_VALIDATION_OUTPUT=su2_prefactor_validation_20261002 \
python Higgs_fluctuation_solver_stabilized.py

# Regenerate the L=26 stationary field without duplicating its low spectrum
HIGGS_RUN_MODE=relax HIGGS_RELAX_L=26 HIGGS_RELAX_SPACING=0.8 \
HIGGS_RELAX_OUTPUT=su2_prefactor_bounce_L26_h0.8 \
HIGGS_RELAX_SKIP_SPECTRUM=1 \
python Higgs_fluctuation_solver_stabilized.py

# Regulated gauge--Goldstone--ghost determinant scan
HIGGS_RUN_MODE=su2_determinant \
HIGGS_SU2_DET_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_DET_OUTPUT=su2_determinant_scan_L26_h0.8 \
HIGGS_SU2_DET_XI=0.5,1,2 HIGGS_SU2_DET_METHOD=slq \
python Higgs_fluctuation_solver_stabilized.py

# Generate and independently audit nonlinear SU(2) vertex tables
HIGGS_RUN_MODE=su2_vertices python Higgs_fluctuation_solver_stabilized.py
python su2_nonlinear_vertex_audit.py

# Counterterm basis, pole closure, and supplied finite coefficients
HIGGS_RUN_MODE=su2_counterterms \
HIGGS_SU2_CT_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_CT_COEFFICIENTS=su2_counterterm_coefficients_template.json \
python Higgs_fluctuation_solver_stabilized.py

# Exploratory gauge spectrum; a Gaussian is diagnostic only
HIGGS_RUN_MODE=research HIGGS_SU2_GAUGE=1 HIGGS_SU2_G=0.65 \
HIGGS_SU2_XI=1 HIGGS_L=8 HIGGS_NUM_MODES=40 \
HIGGS_BACKGROUND_PROFILE=gaussian HIGGS_BOUNCE_WIDTH_SITES=1.2 \
HIGGS_OUTPUT_TAG=su2_diagnostic python Higgs_fluctuation_solver_stabilized.py
```

Listing 15.2: Typical commands for the unified solver.

# Chapter 16

# Limitations and Beyond-Standard-Model Extensions

## 16.1 What the present solver establishes

The completed calculation demonstrates a consistent chain from a metastable higher-order scalar potential to a stationary bounce, a matrix-free multi-component Hessian, collective-mode classification, a full bare lattice trace–log, and a finite continuum-renormalized determinant. This is already a substantial numerical test bed for scalar sectors beyond the renormalizable quartic Standard Model potential. The September 30 extension established a link-derived quadratic $SU(2)$ operator, exact homogeneous reference levels, and measured Ward-identity prerequisites on both small grids and a stationary bounce. The October 1 extension adds regulated coupled and ghost determinant backends, explicit false-vacuum constant-mode and $R_\xi$ measure policies, a local EFT counterterm basis, and derived continuum pole residues through dimension 12. On October 2 the $L = 26$, $h = 0.8$ checkpoint was reproducibly regenerated and passed the stationary Ward tests, but the ephemeral workspace subsequently lost the untracked NPZ before the production determinant scan began. It still does not extend the scalar determinant values to a physical gauged rate because the finite lattice-to-$\overline{\mathrm{MS}}$ constants and controlled regulator limits remain incomplete.

The October 5 audits in Chapter 9 add genuine four-dimensional single-core off-shell tests: $L = 6$ homogeneous stochastic consistency, independent per-probe quadrature checks, complete-multiplet variance comparisons, and $L = 3, 4$ nonconstant dense-reference, Ward, and local pole-algebra checks. Complete constants reduced homogeneous variance modestly; larger low-mode windows were not automatically better. The fresh nonconstant tests passed their deterministic gates, but sampling uncertainty, finite matching, and physical-bounce prerequisites remain. They do not supersede the earlier scalar results with a gauged rate.

## 16.2 Comparison with Euclidean-time Monte Carlo approaches

Recent Euclidean lattice Monte Carlo studies have proposed alternative estimators for false-vacuum decay in quantum-mechanical systems [12, 13]. These methods share the Euclidean functional-integral setting of the present calculation, but their observables and numerical systematics differ from those of a semiclassical one-loop prefactor. Their existing demonstrations are in low-dimensional single-particle systems; extensions to quantum field theory are discussed in the cited work, but gauge-theory implementations have not yet been demonstrated.

**Euclidean probability-density method.** Shen, Draper, and El-Khadra define a decay-rate estimator from the probability density at the classical turning point $b$, schematically

$$\Gamma \simeq \sqrt{m}\, p(b, T), \tag{16.1}$$

where $p(b, T)$ is measured with constrained Euclidean path integrals [12]. In the window $1/m \ll T \ll 1/\Gamma$, the relevant probability density becomes approximately independent of $T$, making the required continuation approximately innocuous under the assumptions of the construction. A cut on a path functional $S_V[x]$ controls the unstable direction and suppresses configurations that evolve deeply into the true-vacuum region. The method does not require an explicit numerical evaluation of the bounce determinant, although semiclassical information is used in motivating and interpreting parts of the estimator.

**Implicit decay amplitude and spectral reconstruction.** Jin and Swaim instead formulate an implicit Fermi's-Golden-Rule-type expression,

$$\Gamma = 2\pi \langle FV | (H_{\mathrm{FV}} - H)\, \delta(H_{\mathrm{TV}} - E_{\mathrm{FV}})\, (H_{\mathrm{FV}} - H) | FV \rangle, \tag{16.2}$$

and infer the required matrix element from Euclidean correlators and spectral reconstruction [13]. A multi-ensemble sampling strategy addresses the exponential suppression of false-vacuum configurations. The tested estimator does not use a semiclassical bounce expansion, but it introduces the inverse problem and associated systematics of reconstructing a spectral density from Euclidean data. Its numerical demonstrations are also currently limited to one-dimensional quantum-mechanical systems.

**Relation to the present SU(2) program.** These Monte Carlo approaches are complementary to the program developed here. The present source constructs a link-derived gauge–Goldstone Hessian and ghost operator, verifies discrete Ward identities and homogeneous normalization tests, evaluates regulated finite-lattice determinant ratios, and constructs the counterterm basis and continuum pole residues. It does not yet contain the finite-matched and extrapolated coupled determinant or a completed physical gauged prefactor. Those remaining quantities require a consistent treatment of gauge fixing, Faddeev–Popov ghosts, collective coordinates, local counterterms, and regulator limits, none of which has yet been implemented in the cited quantum-mechanical Monte Carlo demonstrations.

The semiclassical framework is therefore the presently implemented route to the weak-coupling renormalized one-loop SU(2) prefactor in this project. Euclidean-time Monte Carlo estimators remain valuable candidates for future nonperturbative cross-checks rather than being excluded in principle from a future gauge-theory generalization.

## 16.3 What remains for a gauged Higgs theory

A physical gauged-$SU(2)$ or electroweak calculation requires additional operators and consistency conditions:

1. production evaluation of the implemented regulated coupled and ghost determinants on the relaxed bounce;
2. finite lattice-to-$\overline{\mathrm{MS}}$ matching of the retained local counterterms, including the derivative term;
3. confirmation of gauge-fixing cancellation after matching and regulator limits, not only in the exact homogeneous validation;
4. continuum extrapolation at fixed physical volume and an independent infinite-volume study with the massless false-vacuum gauge sector;

5. for an electroweak extension, hypercharge and fermion sectors;
6. normalization of any remaining collective coordinates;
7. scale and scheme dependence of the complete one-loop rate.

**Scope and interpretation**

Adding a gauge coupling to the potential is not sufficient. A gauge-invariant decay rate requires the entire coupled fluctuation problem. The current global-$O(4)$ result is therefore best viewed as a controlled scalar benchmark and a foundation for the more complete calculation.

## 16.4 Natural next calculations

- repeat the relaxed SLQ calculation at several $h$ values while keeping a fixed physical box, then test counterterm-matched lattice extrapolation;
- calculate the internal $O(4)/O(3)$ collective-coordinate measure for the global model;
- factor the unified source into tested library functions while preserving the single-file production entry point;
- add automated regression tests comparing validation spectra, residuals, and determinant summaries;
- add atomic per-probe and per-$\xi$ restart files to the `su2_determinant` mode;
- regenerate or externally restore the validated $L = 26$, $h = 0.8$ stationary checkpoint, preserve its checksum on persistent storage, and run a single-$\xi$ pilot before the full deflated scan at $\xi = 0.5, 1, 2$;
- derive the finite lattice-to-$\overline{\mathrm{MS}}$ constants from homogeneous momentum integrals and a derivative expansion, then perform continuum and volume convergence checks.

# Chapter 17

# SU(2) Physical-Prefactor Foundations and EFT Matching

This chapter incorporates the foundational SU(2) physical-prefactor report first prepared on September 30, 2026. It brings its benchmark definition, link-derived Hessian, gauge-identity checks, EFT matching prescription, measured prerequisites, reproducibility record, and references into the main text. The original measurements and derivations are retained as a dated baseline. Later matching results are summarized in Section 9.7.4; later spacing and determinant follow-ups are in Chapter 14. Those updates supersede the original outstanding-work checklist where the status has changed; a physical gauged decay prefactor remains unestablished.

## 17.1 End-to-End Numerical Result

### 17.1.1 Result ledger

Table 17.1: Principal numerical outputs of the production pipeline.

| Stage | Quantity | Result |
|---|---|---|
| Continuum bounce | Center field | $\rho(0) = 1.367393286$ |
| Continuum bounce | Boundary residual | 0.281955% |
| Continuum bounce | Action | $B = 145.203947$ |
| Discrete relaxation (regenerated) | RMS EOM residual | $4.43266 \times 10^{-12}$ |
| Discrete relaxation (regenerated) | Maximum EOM residual | $6.30906 \times 10^{-11}$ |
| Hessian | Physical negative mode | $-0.1850415$ in lattice units |
| Hessian | Translation sector | four modes at $-0.00212653$ |
| Hessian | Orientation sector | three copies of a mode at $-5.33 \times 10^{-12}$ |
| SLQ | Bare trace–log | $-2079.9042 \pm 6.4504$ |
| Scalar continuum calculation | Reported trace–log | 40.9324; 0.03 hardcoded estimate |
| Continuum matching | Determinant factor | $1.2931 \times 10^{-9}$ |
| Semiclassical expression | Translation factor included | $5.9974 \times 10^{-70}$ in lattice energy units$^4$ |
| SU(2) prerequisites | Lattice Ward and homogeneous tests | PASS at $\xi = 0.5, 1, 2$ |

| Stage | Quantity | Result |
|---|---|---|
| SU(2) $L = 26$ recovery | Bounce action and RMS residual | 139.9422784742 and $4.4316323 \times 10^{-12}$ |
| SU(2) stationary Ward test | Largest on-shell residual | $5.52 \times 10^{-14}$; PASS |
| SU(2) regulated determinant | Exact homogeneous $\xi$ scan | PASS as IR shift is removed |
| SU(2) counterterms | Scalar and full Feynman-gauge pole closure | PASS to $3.11 \times 10^{-15}$ |
| October 5 $L = 6$ follow-up | 4096-probe homogeneous trace | $-20.75094 \pm 1.62888$; exact $-19.56733$ |
| October 5 complete multiplets | Best tested constants, 5/1 | 12.51% variance reduction versus fresh 8/8 |
| October 5 nonconstant audit | $L = 3, 4$ deterministic prerequisites | PASS at $10^{-10}$; all profiles off shell |
| October 5 nonconstant SLQ | Largest 80-step probe quadrature error | $1.05 \times 10^{-11}$; sampling dominates |
| SU(2) nonlinear vertices | Independent generator/table audit | PASS at tested routings; vertex APIs implemented |
| SU(2) matching interface | Full-gauged finite constants | Interface implemented; full-scope constants remain unset |
| October 7 scalar-only matching | Six radial potential coefficients | Computed at the stated benchmark; not full-$SU(2)$ values |
| October 8 full-gauged Step 1 | Fixed routings, projectors, selected vertex tensors | Independent audit PASS; first four-point gauge-linked pilot integrated |
| October 8 four-point pilot | Gauge-linked radial 4-point insertion sector | Regulated result and checks recorded; QMC/extrapolation errors remain pilot-level |
| October 18 Higgs slope | $\delta Z_H = 0.0329117974 \pm 3.60 \times 10^{-7}$ | Finite benchmark-specific estimate; QMC error only |
| October 18 six-point gauge pilot | $\delta\gamma_{\text{gauge}} = -0.01887194 \pm 0.00058491$ | Sector pilot; excludes scalar loops and matching systematics |
| October 19 higher-point pilots | 8/10/12-leg intercepts; relative QMC SE 62.4%/10.6%/16.7% | Not production precision; 10/12-leg runs use four points per scramble |
| October 18 gauge two-point | 0.4123 maximum normalized transverse residual | Lattice-only; no continuum gauge/ghost tensor or matched coefficient |
| October 14–18 spacing audit | Four-mode capture 90.88%–96.18%; 256-probe shifts within 0.44 SE | Finite-lattice mode bookkeeping; gauge cancellation and physical rate remain open |
| Full-gauged matching | Complete gauge-sector finite coefficients | Not computed; full-matching flag remains false |
| Gauged prefactor | Physical determinant/rate | Not yet calculated |

### 17.1.2 Decay-rate status

Inserting the computed action and determinant factor into the scalar part of Eq. (1.2) gives

$$\left(\frac{B}{2\pi}\right)^2 \left|\frac{\det{}' H_B}{\det H_F}\right|^{-1/2} e^{-B} \simeq 5.9974 \times 10^{-70} \quad \text{(lattice energy units)}^4. \tag{17.1}$$

This includes the four translation collective coordinates through the standard factor $(B/2\pi)^2$ [2]. It does not include the remaining three-dimensional internal-orientation group-volume Jacobian. This expression remains a global-scalar partial result. It must not be relabelled as a gauged rate or multiplied by the new diagnostic gauge spectra. The gauge-orbit measure and false-vacuum gauge/ghost zero modes require their own consistent treatment.

The source text remains shared with `SU2_Physical_Prefactor_Progress_Report.tex` through `SU2_Physical_Prefactor_Progress_Content.tex`.

**Status: prerequisites verified; physical gauged prefactor not yet calculated.** This report records newly executed lattice tests and bounce relaxation. It does not convert a diagnostic spectrum or a bare determinant into a physical rate. No gauged renormalized determinant is reported.

## 17.2 Confirmed benchmark and reproducibility

The target theory is a pure $SU(2)$ gauge theory coupled to one Higgs doublet, with vanishing background gauge field and radial amplitude $\rho$. Hypercharge and fermions are excluded. The benchmark is

$$V(\rho) = 0.2\rho^2 - 0.5\rho^4 + \frac{0.8}{6}\rho^6 + 0.01\rho^8, \qquad g = 0.65, \qquad \mu_R = \sqrt{0.4}.$$

These are the scaled production units, not an independently specified physical GeV normalization. A rate quoted in $\mathrm{GeV}^4$ requires an explicit conversion scale. The exploratory research spacing $a = 1/500\,\mathrm{GeV}^{-1}$ must not silently be identified with the production spacing $h = 0.8$.

| Quantity | New calculation | Meaning |
|---|---|---|
| Lattice | $26^4$, $h = 0.8$ | Periodic scaled box |
| Scalar sites | 456,976 | One component |
| Bosonic dimension | 7,311,616 | Radial plus three five-field blocks |
| Discrete action | 139.9422784742 | Not the continuum action |
| EOM RMS residual | $4.43266 \times 10^{-12}$ | Regenerated stationary field |
| EOM maximum residual | $6.30906 \times 10^{-11}$ | Regenerated stationary field |
| Gauge parameters tested | $0.5, 1, 2$ | Background $R_\xi$ gauge |

Table 17.2: Executed stationary-bounce prerequisites.

The regenerated field is saved at

`su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz`

The reproducible tests are embedded in `Higgs_fluctuation_solver_stabilized.py`, under the new `su2_validation` mode. The machine-readable test record is `su2_prefactor_validation/validation_summary.json`.

## 17.3 Lattice Hessian derived from links

The earlier site-centered continuum stencil has been replaced by the quadratic expansion of the covariant nearest-neighbor scalar action and the zero-background gauge-link action. This change matters because self-adjointness alone does not enforce a lattice gauge identity. Gauge components $A_{x\mu}$ occupy outgoing links; $\chi_x$ occupies sites. Define forward differences $G_\mu$ and backward divergence $D$ by

$$(G_\mu f)_x = \frac{f_{x+\hat{\mu}} - f_x}{h}, \quad (DA)_x = \sum_\mu \frac{A_{x\mu} - A_{x-\hat{\mu},\mu}}{h}, \quad G^\dagger = -D, \quad K = -DG.$$

For each generator, with $q = g/2$, the gauge-fixing functional is $F = DA - \xi q\rho\chi$. The quadratic scalar-link contribution is

$$S^{(2)}_{\text{link}} = \frac{h^4}{2}\sum_{x,\mu}\left[\frac{(\chi_{x+\hat{\mu}} - \chi_x)^2}{h^2} + q^2\rho_x\rho_{x+\hat{\mu}}A^2_{x\mu}\right] \tag{17.2}$$

$$+ h^4\sum_{x,\mu}\frac{q}{h}A_{x\mu}(\rho_{x+\hat{\mu}}\chi_x - \rho_x\chi_{x+\hat{\mu}}). \tag{17.3}$$

Adding the quadratic gauge action, scalar potential curvature, and $h^4\sum_x F_x^2/(2\xi)$ gives the implemented block

$$H_{A\chi} = \begin{pmatrix} K\mathbf{1} + (1-\xi^{-1})GD + M^2_{\text{link}} & C \\ C^\dagger & K + U_T + \xi q^2\rho^2 \end{pmatrix}, \qquad H_c = K + \xi q^2\rho^2,$$

where $U_T = V\prime(\rho)/\rho$ is understood by its smooth polynomial extension at $\rho = 0$,

$$(M^2_{\text{link}})_{x\mu} = q^2\rho_x\rho_{x+\hat{\mu}}, \quad (C\chi)_{x\mu} = \frac{q}{h}(\rho_{x+\hat{\mu}} - \rho_x)(\chi_x + \chi_{x+\hat{\mu}}).$$

The transpose $C^\dagger$ is implemented explicitly. In the continuum limit $C\chi \to 2q(\partial_\mu\rho)\chi$, recovering the gradient mixing of the gauge–Goldstone system [11]. The radial block remains $K + V\prime\prime(\rho)$.

## 17.4 Gauge identities and measured checks

For a site gauge parameter $\alpha$, the gauge direction is $R\alpha = (G\alpha, q\rho\alpha)$. The discrete off-shell identity is

$$H_{A\chi}R\alpha = \begin{pmatrix} \xi^{-1}G(H_c\alpha) \\ q\rho H_c\alpha \end{pmatrix} + \begin{pmatrix} 0 \\ q\alpha\,[K\rho + V\prime(\rho)] \end{pmatrix}.$$

The residual term must be retained on an arbitrary background. It vanishes on a stationary bounce; it must not be discarded on the old Gaussian diagnostic profile. The implemented tests check both forms.

| $\xi$ | Small-grid off-shell error | Bounce off-shell error | Bounce on-shell error |
|---|---|---|---|
| 0.5 | $1.43 \times 10^{-16}$ | $1.41 \times 10^{-16}$ | $1.38 \times 10^{-14}$ |
| 1.0 | $1.83 \times 10^{-16}$ | $1.62 \times 10^{-16}$ | $2.76 \times 10^{-14}$ |
| 2.0 | $3.20 \times 10^{-16}$ | $2.78 \times 10^{-16}$ | $5.52 \times 10^{-14}$ |

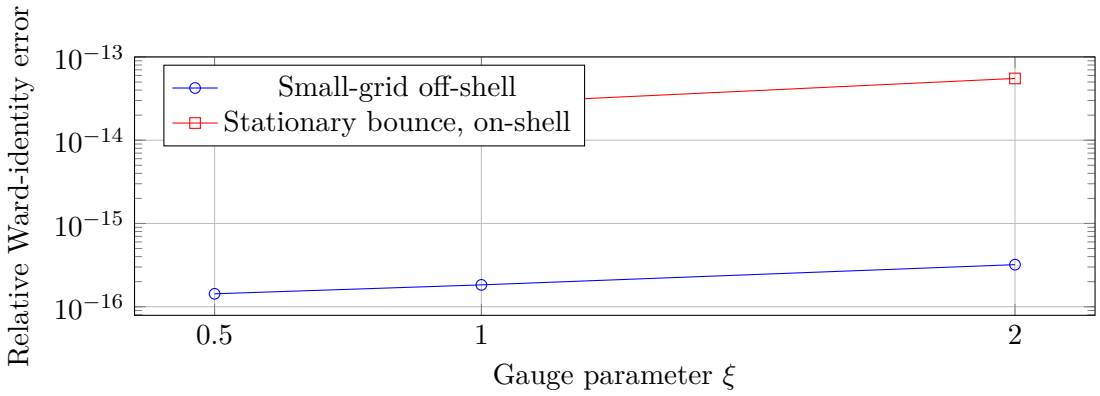
Table 17.3: Relative vector-norm errors. The on-shell error reflects the nonzero, tightly converged lattice EOM residual.

The $g \to 0$, $\xi = 1$ decoupling test gives zero measured relative error. Homogeneous coupled and ghost spectra agree with complete Fourier levels to relative errors below $9.3 \times 10^{-16}$. False-vacuum research spectra now use these exact levels, preserving momentum degeneracies that the old single-vector ARPACK calculations did not always recover.

For a constant stationary broken background, the normalized one-generator one-loop expression

$$\frac{1}{2}\log\det H_{A\chi} - \log\det H_c + \frac{N}{2}\log\xi$$

is 296.100090843863 at all three tested gauges, with a spread of $5.12 \times 10^{-13}$. This tests the gauge-fixing normalization in a positive, finite-volume example. It is not a false-vacuum/bounce ratio and is not a gauged-prefactor gauge-independence test.




## 17.5 New renormalization issue: the potential is not closed

The previously proposed prescription specified the sixth- and eighth-order Wilson coefficients, but it did not specify all local operators generated when those terms are included nonperturbatively in a one-loop determinant. Already the scalar part of the local divergence is proportional to $U_L^2 + 3U_T^2$, where

$$U_L = m_F^2 + 3\lambda\rho^2 + 5\gamma\rho^4 + 7\delta\rho^6, \quad U_T = m_F^2 + \lambda\rho^2 + \gamma\rho^4 + \delta\rho^6.$$

Direct symbolic expansion gives

$$U_L^2 + 3U_T^2 = 4m_F^4 + 12\lambda m_F^2\rho^2 + (12\lambda^2 + 16\gamma m_F^2)\rho^4 \tag{17.4}$$
$$+ (36\lambda\gamma + 20\delta m_F^2)\rho^6 + (28\gamma^2 + 48\lambda\delta)\rho^8 \tag{17.5}$$
$$+ 76\gamma\delta\rho^{10} + 52\delta^2\rho^{12}. \tag{17.6}$$

The last two terms require higher potential counterterms under this nonperturbative treatment. Their finite renormalized coefficients are not fixed just by declaring $\gamma$ and $\delta$ to be $\overline{\text{MS}}$ coefficients. Gauge fluctuations additionally require the appropriate derivative/field-normalization counterterm [11]. This algebra is a counterterm-closure diagnostic, not a complete $SU(2)$ matching computation.

### 17.5.1 EFT matching prescription and counterterm closure

The scalar contribution to the one-loop local divergences contains $U_L^2+3U_T^2$. Its expansion above generates $\rho^{10}$ and $\rho^{12}$ terms absent from the original production potential. This establishes the need for additional local potential counterterms when the sixth- and eighth-order interactions are retained nonperturbatively in the leading fluctuation operators. It does not identify the complete gauge–Goldstone–ghost divergence: the gauged calculation also requires its potential and derivative contributions [11].

**Gauge-invariant definition and radial restriction.** Let $\mathcal{H}$ denote the Higgs doublet and define $\rho = \sqrt{2\mathcal{H}^\dagger\mathcal{H}}$. The invariant Euclidean Lagrangian relevant to the benchmark is

$$\mathcal{L}_{\text{inv}} = Z_{\mathcal{H}}(\mu_R)(D_\nu\mathcal{H})^\dagger D_\nu\mathcal{H} + \frac{1}{4}F^a_{\nu\sigma}F^a_{\nu\sigma} + V_{\text{EFT}}(\rho;\mu_R), \tag{17.7}$$

where $D_\nu$ is the $SU(2)$ covariant derivative. Gauge-fixing and Faddeev–Popov ghost terms are added to the quantized action. On the radial background with fixed internal orientation and vanishing background gauge field, the scalar part reduces to

$$\mathcal{L}_{\text{radial}} = \frac{1}{2}Z_{\mathcal{H}}(\mu_R)(\partial_\nu\rho)^2 + V_{\text{EFT}}(\rho;\mu_R).$$

The radial expression alone is therefore a background restriction, not the full gauged EFT.

Use $m^2 = -\mu^2$ to distinguish the positive false-vacuum mass coefficient from the source-code convention. The renormalized potential is written as

$$V_{\rm EFT}(\rho;\mu_R) = \frac{1}{2}m^2(\mu_R)\rho^2 + \frac{1}{4}\lambda(\mu_R)\rho^4 + \frac{1}{6}\gamma(\mu_R)\rho^6 + \frac{1}{8}\delta(\mu_R)\rho^8 + \frac{1}{10}c_{10}(\mu_R)\rho^{10} + \frac{1}{12}c_{12}(\mu_R)\rho^{12}. \tag{17.8}$$

**1. Closure at the specified loop order.** For this benchmark $c_{10}$ and $c_{12}$ are counted as *one-loop-order* coefficients. They enter the local one-loop action, but are not inserted into the leading bounce equation or leading Hessian used inside the one-loop determinant. Inserting them there would contribute beyond the retained loop order. If they were instead treated as independent nonperturbative leading interactions, the enlarged potential would not be closed at dimension 12: for example, the curvature of a $\rho^{12}$ interaction is proportional to $\rho^{10}$, whose square generates $\rho^{20}$.

At the retained order, the local potential counterterms must include those for $m^2, \lambda, \gamma, \delta, c_{10}, c_{12}$, together with the covariant kinetic counterterm $\delta Z_{\mathcal{H}}$ needed for the bounce-background effective action. The field-independent vacuum term cancels in the bounce-minus-false-vacuum action. This is a closure prescription for the specified background sector and loop order, not a claim that these terms alone renormalize every amplitude of the full gauge theory. Gauge-field and gauge-coupling renormalization, and any further operators required by the chosen EFT expansion, must be treated consistently where they contribute.

**2. Finite matching at the reference scale.** We adopt $M_R = \sqrt{0.4}$ in the scaled production units and impose the renormalized benchmark conditions

$$Z_{\mathcal{H}}(M_R) = 1, \qquad c_{10}(M_R) = c_{12}(M_R) = 0, \tag{17.9}$$

$$m^2(M_R) = 0.4, \qquad \lambda(M_R) = -2.00, \qquad \gamma(M_R) = 0.80, \qquad \delta(M_R) = 0.08, \qquad g(M_R) = 0.65. \tag{17.10}$$

The required poles are to be subtracted in the $\overline{\rm MS}$ prescription. Canonical normalization here defines the renormalized local kinetic coefficient at the reference scale; it does not assert that the divergent bare counterterm vanishes. The vanishing higher-operator coefficients fix additional finite matching conditions rather than following from the original four potential parameters.

These numerical conditions reproduce the *tree-level* production potential, its barrier, and its vacuum depths. They do not impose that the loop-corrected barrier height, false-vacuum curvature, and true-vacuum depth remain unchanged. Such requirements would define additional finite renormalization conditions and would need to be distinguished from the benchmark chosen here.

**Wilson-coefficient dimensions and scale dependence.** In four dimensions, for a dimensionful canonically normalized scalar, one may parameterize

$$c_{10}(\mu_R) = \frac{C_{10}(\mu_R)}{\Lambda^6}, \qquad c_{12}(\mu_R) = \frac{C_{12}(\mu_R)}{\Lambda^8},$$

with dimensionless $C_{10}, C_{12}$. These dimensional relations do not specify a UV completion or establish a numerical suppression in this scaled benchmark. No value of $\Lambda$ or physical energy conversion has yet been supplied.

Running away from $M_R$ is part of the retained-order renormalization group description, not itself an EFT truncation error. The running coefficients must compensate the explicit scale dependence of the loop effective action to the calculated order. A controlled estimate of omitted operators additionally requires power counting, the dimensionless Wilson coefficients, and checks of the bounce field amplitudes and gradient scales against $\Lambda$; a suppression by $(\rho/\Lambda)^2$ is not assumed without those checks.

This prescription defines a reproducible one-loop benchmark for the specified bounce-background sector. It fixes the additional finite matching conditions but does not, by itself, establish the EFT validity range or complete the renormalized gauged prefactor calculation. These conditions are adopted here as the calculation definition; their counterterm and determinant implementation remains part of the numerical work described below.

## 17.6 Remaining steps and safeguards

The complete one-loop gauge contribution uses the radial block, three coupled generator blocks, and three complex ghost determinants:

$$\Delta\Gamma_1 = \frac{1}{2}\log\left|\frac{\det{}' H_{r,B}}{\det H_{r,F}}\right| + \frac{3}{2}\log\left|\frac{\det H_{A\chi,B}}{\det H_{A\chi,F}}\right| - 3\log\left|\frac{\det H_{c,B}}{\det H_{c,F}}\right| + \Delta S_{\rm ct}.$$

This equation is schematic until gauge-volume and zero-mode conventions are fixed. In particular the symmetric false vacuum has constant gauge and ghost zero modes on the periodic box. Gauge-orbit normalization must be handled consistently; the old global-$O(4)$ orientation Jacobian cannot simply be multiplied into the gauged result. A consistent gauge-invariant treatment of the decay determinant and its renormalization is essential [10].

With the matching conditions defined above, work remains on coupled determinants, the false-vacuum zero modes, ultraviolet matching, the $h \to 0$ and volume limits, translation normalization, and uncertainty estimation. No Gaussian spectral window supplies these steps. The program now rejects `HIGGS_SU2_GAUGE=1` in scalar-only `full`, `slq`, `continuum`, and `relax` modes, so a scalar pipeline cannot silently masquerade as a gauge calculation. The old scalar continuum uncertainty 0.03 is hardcoded in the source; it is not a measured gauged error and is not reused here.

## 17.7 Commands for the completed prerequisites

```
HIGGS_RUN_MODE=relax HIGGS_RELAX_SKIP_SPECTRUM=1 \
HIGGS_RELAX_OUTPUT=su2_prefactor_bounce_L26_h0.8 \
python Higgs_fluctuation_solver_stabilized.py

HIGGS_RUN_MODE=su2_validation \
HIGGS_SU2_BOUNCE_FILE=su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
python Higgs_fluctuation_solver_stabilized.py
```

# Bibliography


[1] S. Coleman, *The Fate of the False Vacuum: Semiclassical Theory*, Phys. Rev. D 15 (1977) 2929–2936; Erratum: Phys. Rev. D 16 (1977) 1248.

[2] C. G. Callan, Jr. and S. Coleman, *The Fate of the False Vacuum. II. First Quantum Corrections*, Phys. Rev. D 16 (1977) 1762–1768.

[3] S. Coleman, V. Glaser and A. Martin, *Action Minima among Solutions to a Class of Euclidean Scalar Field Equations*, Commun. Math. Phys. 58 (1978) 211–221.

[4] I. M. Gel'fand and A. M. Yaglom, *Integration in Functional Spaces and Its Applications in Quantum Physics*, J. Math. Phys. 1 (1960) 48–69.

[5] M. F. Hutchinson, *A Stochastic Estimator of the Trace of the Influence Matrix for Laplacian Smoothing Splines*, Commun. Stat. Simul. Comput. 19 (1990) 433–450.

[6] A. J. McKane and M. B. Tarlie, *Regularisation of Functional Determinants Using Boundary Perturbations*, J. Phys. A 28 (1995) 6931–6942, arXiv:cond-mat/9509126.

[7] J. Baacke and G. Lavrelashvili, *One-Loop Corrections to the Metastable Vacuum Decay*, Phys. Rev. D 69 (2004) 025009, arXiv:hep-th/0307202.

[8] G. V. Dunne and K. Kirsten, *Functional Determinants for Radial Operators*, J. Phys. A 39 (2006) 11915–11928, arXiv:hep-th/0607066.

[9] S. Ubaru, J. Chen and Y. Saad, *Fast Estimation of* $\mathrm{tr}(f(A))$ *via Stochastic Lanczos Quadrature*, SIAM J. Matrix Anal. Appl. 38 (2017) 1075–1099.

[10] M. Endo, T. Moroi, M. M. Nojiri and Y. Shoji, *False Vacuum Decay in Gauge Theory*, JHEP 11 (2017) 074, arXiv:1704.03492.

[11] W.-Y. Ai, J. S. Cruz, B. Garbrecht and C. Tamarit, *Gradient effects on false vacuum decay in gauge theory*, arXiv:2006.04886.

[12] J. Shen, P. Draper and A. X. El-Khadra, *Vacuum Decay and Euclidean Lattice Monte Carlo*, Phys. Rev. D 107 (2023) 094506, arXiv:2210.05925.

[13] L. Jin and J. Swaim, *Non-perturbative False Vacuum Decay Using Lattice Monte Carlo in Imaginary Time*, arXiv:2512.00542 (2025), revised 2026.

[14] H. Kawai, R. Nakayama, and K. Seo, *Comparison of the lattice* $\Lambda$ *parameter with the continuum* $\Lambda$ *parameter in massless QCD*, Nucl. Phys. B 189 (1981) 40–62, doi:10.1016/0550-3213(81)90080-8.

[15] T. Reisz, *A power counting theorem for Feynman integrals on the lattice*, Commun. Math. Phys. 116 (1988) 81–126, doi:10.1007/BF01239027.

[16] J. C. Taylor, *Ward identities and charge renormalization of the Yang-Mills field*, Nucl. Phys. B 33 (1971) 436–444, doi:10.1016/0550-3213(71)90297-5.

[17] A. B. Owen, *Monte Carlo variance of scrambled net quadrature*, SIAM J. Numer. Anal. 34 (1997) 1884–1910, doi:10.1137/S0036142994277468.

[18] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D 10 (1974) 2445–2459, doi:10.1103/PhysRevD.10.2445.

[19] M. Lüscher and P. Weisz, *Efficient numerical techniques for perturbative lattice gauge theory computations*, Nucl. Phys. B 266 (1986) 309–356, doi:10.1016/0550-3213(86)90094-5.

[20] L. F. Abbott, *The background field method beyond one loop*, Nucl. Phys. B 185 (1981) 189–203, doi:10.1016/0550-3213(81)90371-0.

[21] S. Coleman and E. Weinberg, *Radiative corrections as the origin of spontaneous symmetry breaking*, Phys. Rev. D 7 (1973) 1888–1910, doi:10.1103/PhysRevD.7.1888.

# Chapter 18

# SU(2) Classical-to-Quantum Modal Bridge

The following chapter material incorporates the substantive sections of `SU2_Quantum_Modal_Bridge_Implementation_Guide_20261001.tex`. Its standalone cover, table of contents, and footer are omitted here; the parent guide supplies those document-level elements. The source guide remains independently compilable.

**Scope and interpretation**

The modal-bridge guide is dated October 1, 2026. Its statements about implementation status describe that date and should be read together with later updates elsewhere in this technical guide. In particular, the production $L = 26$ checkpoint was subsequently recovered and regenerated, then lost from the ephemeral workspace before the production determinant scan.

## 18.1 Purpose and present status

This guide defines the first executable bridge between the classical four-dimensional lattice calculation in `Higgs_fluctuation_solver_stabilized.py` and the quantum spectral program described in `Project2_proposal.tex`. The bridge exports low eigenpairs from the same gauge–Goldstone, ghost, and radial-scalar operators used in the vacuum-decay work. A separate preparation utility verifies the exported data, pads its state vectors to a power-of-two Hilbert-space dimension, and records an affine spectral transformation for later quantum algorithms.

**Implemented result**

The modal exporter is implemented in both `su2_validation` and `su2_determinant`. The homogeneous $3^4$ validation run exports seven datasets: coupled and ghost sectors at $\xi = 0.5, 1, 2$, plus the radial scalar sector. The test run passed all pre-existing Ward, Fourier, and gauge normalization checks. The coupled $\xi = 1$ modal dataset was independently reloaded with a maximum recomputed relative eigenpair residual of $9.77 \times 10^{-15}$.

The production $L = 26$ hook is also implemented, but it cannot be exercised until the relaxed bounce NPZ checkpoint is restored or regenerated.

## 18.2 Dark-energy relevance and investment decision

### 18.2.1 What the present calculation does and does not show

The present $SU(2)$–Higgs benchmark does *not* contain evidence for a physical dark-energy-scale excitation. This conclusion is useful because it establishes the scale separation that any successful extension must overcome. The available $L = 20$ stationary-bounce spectrum contains positive scalar curvatures beginning at approximately

$$\lambda_{\min}^{(+)} \simeq 6.85 \times 10^{3}\ \mathrm{GeV}^2, \qquad m_{\min}^{(+)} \simeq 83\ \mathrm{GeV}, \tag{18.1}$$

while the available coupled $SU(2)$ modes are also at particle-physics mass scales. These results are recorded in the existing spectral CSV files and `run_summary_scaled_stationary_bounce_L20.json`.

By contrast, a canonical slowly rolling dark-energy field normally requires a potential-dominated energy density and a curvature no larger than roughly the present Hubble scale,

$$\rho_{\mathrm{DE}} \sim 10^{-47}\ \mathrm{GeV}^4, \qquad m_Q^2 \lesssim H_0^2 \sim 10^{-84}\ \mathrm{GeV}^2. \tag{18.2}$$

The qualitative scalar-field conditions and their cosmological interpretation are reviewed by the Particle Data Group and originate in classic rolling-field models [1, 2].

The current model's false-to-true vacuum energy difference is approximately

$$|\Delta V| \simeq \frac{0.3803}{(0.002\ \mathrm{GeV}^{-1})^4} \simeq 2.4 \times 10^{10}\ \mathrm{GeV}^4. \tag{18.3}$$

It is therefore about $10^{57}$ times the observed dark-energy density. Its smallest available physical positive curvature is about $10^{87}$ times $H_0^2$. Neither discrepancy can be repaired by simply interpreting an existing electroweak-scale mode differently.

The negative bounce eigenmode is also not a dark-energy particle. It is the single unstable direction associated with tunneling. Translation modes, Goldstone modes, constant gauge modes, and ghost modes arise from collective coordinates, symmetries, gauge redundancy, or determinant bookkeeping. A dark-energy candidate must remain after all of these nonphysical directions are projected out.

**Present scientific conclusion**

The current solver supplies no positive dark-energy candidate. A very small vacuum-decay rate, if ultimately obtained through the factor $e^{-B}$, would not by itself imply a dark-energy-scale vacuum splitting or an ultralight physical scalar. Decay probability, vacuum energy density, and fluctuation mass are distinct observables.

### 18.2.2 What can transfer successfully to the quantum project

Although the current spectrum is too heavy, the modal bridge transfers the right *diagnostic machinery* for a serious dark-energy search:

- explicit removal and classification of gauge, Goldstone, translation, ghost, and tunneling directions;
- normalized eigenvectors that define candidate collective field coordinates;
- lattice-spacing and volume tests that reject numerical near-zero modes;
- determinant and counterterm machinery that tests whether light curvatures survive quantum corrections;
- bounce methods that test whether a candidate direction is metastable or decays too rapidly; and

- versioned classical answers for validating quantum eigensolvers.

For a normalized physical mode $v_i$, define the collective path

$$\mathbf{\Phi}(x;Q) = \mathbf{\Phi}_{\text{background}}(x) + Qv_i(x). \tag{18.4}$$

The full action can then be evaluated along this path to reconstruct

$$V_{\text{eff}}(Q) = V_0 + \frac{1}{2}m_i^2Q^2 + \frac{g_3}{3!}Q^3 + \frac{g_4}{4!}Q^4 + \cdots . \tag{18.5}$$

This calculation distinguishes a genuinely flat potential from an isolated small Hessian eigenvalue. It also supplies the cubic and quartic couplings needed for radiative-stability and cosmological-evolution tests.

### 18.2.3 The model extension most likely to be informative

The realistic route is not to force the present Higgs mode down by nearly ninety orders of magnitude. It is to introduce an additional scalar with a protecting principle, such as an approximate shift symmetry, approximate Goldstone symmetry, or another technically natural suppression mechanism. A minimal exploratory system is a Higgs-portal singlet with a block Hessian

$$H_{\rho Q} = \begin{pmatrix} H_{\rho\rho} & H_{\rho Q} \\ H_{Q\rho} & H_{QQ} \end{pmatrix}. \tag{18.6}$$

Mixing may then produce a light eigenvector dominated by $Q$ with only a small Higgs component. Higgs-portal quintessence constructions exist in the literature and demonstrate that the structural idea is meaningful, while also illustrating the severe naturalness and phenomenological constraints [3].

The search should be classical first. Only parameter regions that produce a stable, non-gauge, radiatively credible light direction should be promoted to expensive quantum simulations. Quantum algorithms can then test spectral extraction, large coupled systems, and eventually model families that are classically difficult; they do not remove the need for a protecting physical mechanism.

### 18.2.4 Is the project worth undertaking?

**Go/no-go assessment**

Yes, as a staged research program with explicit stopping criteria. It is worth undertaking to build and validate a reusable quantum-spectral bridge, to search protected extended scalar sectors, and to obtain meaningful constraints if no viable light direction is found. It is not justified to promise that the present $SU(2)$–Higgs model contains dark energy, or to move directly to costly hardware calculations before a classical extended model passes the physical screening tests.

Table 18.1: Scientific value and recommended commitment.

| Objective | Assessment | Recommendation |
|---|---|---|
| Validate the classical-to-quantum modal interface | High value, low scientific risk | Proceed now |
| Reinterpret a present Higgs or gauge zero mode as dark energy | Not physically supported | Do not pursue |
| Add a symmetry-protected singlet or related scalar sector | High potential, high model risk | Proceed through classical screening gates |
| Use quantum algorithms to benchmark low-mode extraction | Valuable methods research | Proceed on small validated operators |
| Claim quantum advantage or a dark-energy discovery | Premature | Require a classical candidate, resource analysis, and independent cosmology checks |
| Use a null result to constrain extended Higgs models | Scientifically useful | Retain as an explicit project outcome |

## 18.3 Files and responsibilities

Table 18.2: Components of the modal bridge.

| File | Responsibility |
|---|---|
| `Higgs_fluctuation_solver_stabilized.py` | Constructs the physical lattice operators, solves their low modes, and writes versioned modal datasets |
| `quantum_modal_bridge.py` | Reloads a modal dataset, verifies it, pads state vectors, and records the physical-to-encoded spectral map |
| `*_eigenpairs.npz` | Numerical eigenvalues, eigenvectors, residuals, and physical lattice norms |
| `*_metadata.json` | Operator, lattice, background, indexing, normalization, and provenance contract |
| `*_operator_csr.npz` | Explicit sparse operator for small benchmark systems |
| `*_quantum_ready.npz` | Padded vectors and physical/encoded eigenvalues |
| `*_quantum_ready.json` | Qubit count, padding policy, spectral map, and reload diagnostics |

The exporter is deliberately embedded in the established solver rather than reimplementing its operators in a second program. This ensures that the classical determinant calculation and the quantum benchmark use identical discretizations and conventions.

## 18.4 Modal-export controls

The same controls are accepted by `su2_validation` and `su2_determinant`.

Table 18.3: Modal-export environment variables.

| Variable | Default | Meaning |
|---|---|---|
| `HIGGS_SU2_MODAL_EXPORT` | 0 | Enable the exporter with 1, true, yes, or on |
| `HIGGS_SU2_MODAL_MODES` | 12 | Number of lowest eigenpairs exported per sector |
| `HIGGS_SU2_MODAL_OUTPUT` | mode output subdirectory | Override the modal output directory |
| `HIGGS_SU2_MODAL_MAX_OPERATOR_DIM` | 4096 | Largest dimension for which an explicit CSR operator is written |

The mode count controls the exported benchmark window. In the stochastic determinant path, the solver computes enough eigenvectors to satisfy both the SLQ deflation request and the modal-export request. Only the requested deflated subset is removed from the stochastic trace, so enabling export does not silently change the determinant definition.

## 18.5 Step 1: generate the homogeneous benchmark

The homogeneous benchmark is the correct first quantum target because it is small, exactly checkable in Fourier space, and already part of the gauge-cancellation validation suite. Run:

```
HIGGS_RUN_MODE=su2_validation \
HIGGS_SU2_VALIDATION_OUTPUT=su2_modal_validation \
HIGGS_SU2_MODAL_EXPORT=1 \
HIGGS_SU2_MODAL_MODES=12 \
python Higgs_fluctuation_solver_stabilized.py
```

The principal results appear below `su2_modal_validation/modal_exports`. For every value of $\xi$, the run exports one coupled $(A_\mu, \chi)$ dataset and one ghost dataset. It also exports a gauge-independent radial-scalar dataset.

For the $3^4$ lattice, the dimensions and minimum state-register sizes are

$$N_{A\chi} = 5(3^4) = 405, \qquad n_{A\chi} = \lceil \log_2 405 \rceil = 9, \tag{18.7}$$

$$N_c = N_{\text{radial}} = 3^4 = 81, \qquad n_c = \lceil \log_2 81 \rceil = 7. \tag{18.8}$$

The amplitude vectors are consequently padded to dimensions 512 and 128.

## 18.6 Step 2: understand the exported contract

### 18.6.1 Numerical NPZ payload

Each eigenpair archive contains:

**`eigenvalues`** The unshifted physical lattice eigenvalues.

**`eigenvectors`** Unit-normalized real eigenvectors stored as columns.

**`residual_norms`** Values of $\|Hv_i - \lambda_i v_i\|_2$.

**`relative_residuals`** Residuals divided by a stable operator scale.

`physical_lattice_norms`
Values of $\sqrt{a^4 \sum_j |v_{ji}|^2}$.

The quantum amplitude convention is the ordinary discrete normalization

$$\sum_j |v_{ji}|^2 = 1. \tag{18.9}$$

The $a^4$-weighted norm is recorded separately and is not used as a quantum state normalization.

### 18.6.2 JSON metadata

The sidecar uses schema name `ivv.su2.modal_dataset` and version `1.0.0`. Its principal fields are:

- source filename and SHA-256 digest;
- background type, numerical field digest, and optional checkpoint path;
- lattice shape, spacing, boundary conditions, and C-order flattening;
- sector name, dimension, component order, and eigenvalue units;
- gauge coupling, gauge parameter, IR shift, and scalar couplings;
- normalization and deterministic sign conventions;
- residual, orthogonality, and degeneracy diagnostics; and
- minimum system-qubit count and padded dimension.

The coupled component order is

$$(A_x, A_y, A_z, A_t, \chi), \tag{18.10}$$

with each component flattened over $(x, y, z, t)$ in C order.

### 18.6.3 Degenerate subspaces

Homogeneous momentum spectra contain exact degeneracies. A numerical eigensolver may return any orthonormal basis spanning a degenerate eigenspace. Therefore, individual columns should not be required to agree between two implementations. The correct invariant comparison is between eigenvalues and subspace projectors,

$$P_{\mathcal{D}} = V_{\mathcal{D}} V_{\mathcal{D}}^{\mathsf{T}}. \tag{18.11}$$

The JSON file records the detected degenerate clusters for this reason.

## 18.7 Step 3: prepare a dataset for quantum work

Run the preparation utility on any exported metadata file. For example:

```
python quantum_modal_bridge.py \
su2_modal_validation/modal_exports/\
homogeneous_L3_xi_1p0_coupled_A_chi_metadata.json \
--output su2_quantum_ready
```

The utility performs the following checks and transformations:

1. verify the schema and array dimensions;

2. recompute normalization and orthogonality errors;
3. reload the CSR operator, when present, and recompute residuals;
4. compute the minimum number of system qubits;
5. append zero amplitudes to each vector up to dimension $2^n$;
6. calculate Gershgorin bounds for the exported sparse operator; and
7. write physical and encoded eigenvalues with a reversible affine map.

The encoded operator convention is

$$H_{\mathrm{enc}} = \frac{H - cI}{s}, \qquad c = \frac{\lambda_{\mathrm{upper}} + \lambda_{\mathrm{lower}}}{2}, \qquad s = \frac{\lambda_{\mathrm{upper}} - \lambda_{\mathrm{lower}}}{2}. \tag{18.12}$$

The physical eigenvalue is reconstructed using

$$\lambda_{\mathrm{physical}} = c + s\lambda_{\mathrm{encoded}}. \tag{18.13}$$

When a CSR operator is unavailable, the preparation utility can only use the exported modal window as a provisional scale. Such a bound is explicitly marked as uncertified and must not be used as a final block-encoding bound.

## 18.8 Step 4: reproduce the classical benchmark

Before adding a quantum SDK, a benchmark notebook or test program should:

1. load the CSR matrix and quantum-ready NPZ file;
2. reconstruct $Hv_i$ for every exported state;
3. confirm the physical and encoded eigenvalue maps;
4. compare eigenspaces using projectors in degenerate windows;
5. verify the zero padding has not changed any state norm; and
6. save a machine-readable PASS/FAIL result.

Recommended acceptance thresholds for the homogeneous double-precision test are

$$\max_i r_i < 10^{-10}, \qquad \|V^{\mathsf{T}}V - I\|_2 < 10^{-10}. \tag{18.14}$$

The first tested exports are substantially more accurate than these limits.

## 18.9 Dark-energy screening protocol

A low eigenvalue should enter the dark-energy study only after passing five successive gates. This ordering prevents numerical artifacts or symmetry directions from consuming quantum-computing resources.

### 18.9.1 Gate A: numerical reality

The eigenvalue and its eigenspace must converge under changes in lattice spacing, physical volume, eigensolver tolerance, mode count, and random starting vector. Near-degenerate modes must be compared as subspaces. A mode that scales to zero only because the box is enlarged, or moves strongly when the ultraviolet cutoff changes, is not yet a particle candidate.

### 18.9.2 Gate B: physical classification

The candidate must be orthogonal to known translation, gauge, Goldstone, ghost, and negative-mode directions. Its norm must be positive in the physical fluctuation space, and its eigenvalue must be gauge independent in the consistently matched calculation. The sector weights and projectors in the modal metadata provide the starting point for this classification.

### 18.9.3 Gate C: effective-potential and radiative stability

Equation (18.5) must be reconstructed beyond quadratic order. The one-loop corrected mass should satisfy

$$\left|\delta m_Q^2\right| \lesssim \mathcal{O}(m_Q^2) \tag{18.15}$$

without repeated fine-tuning. Portal couplings that regenerate an electroweak-scale mass for $Q$ fail this gate unless a demonstrated symmetry enforces the cancellation.

### 18.9.4 Gate D: cosmological viability

For a canonically normalized homogeneous coordinate, calculate

$$R_m = \frac{m_{\rm eff}^2}{H_0^2}, \qquad R_V = \frac{V_{\rm eff}}{\rho_{\rm DE}}, \tag{18.16}$$

and the potential slow-roll diagnostics

$$\epsilon_V = \frac{M_{\rm Pl}^2}{2}\left(\frac{V'}{V}\right)^2, \qquad \eta_V = M_{\rm Pl}^2\frac{V''}{V}. \tag{18.17}$$

A serious candidate should have the correct energy density, sufficiently slow evolution, an acceptable equation of state, and no excluded fifth-force or time-varying-coupling signal. Its cosmological lifetime must also be much longer than the present age of the Universe.

### 18.9.5 Gate E: justification for quantum execution

The classical model must first produce a scientifically interesting candidate or a spectral problem whose scaling makes classical treatment inadequate. The quantum stage should then state which resource is improved: memory, spectral resolution, parameter throughput, or access to a larger coupled field space. Qubit count alone is not evidence of practical quantum advantage because state preparation, sparse-oracle construction, arithmetic, and measurement costs may dominate.

### 18.9.6 Worked example: promotion or rejection of a hypothetical mode

Suppose a protected two-field scan returns a lowest positive eigenvalue corresponding provisionally to $m_Q^2 = 3 \times 10^{-85}\,\mathrm{GeV}^2$, with an eigenvector that is 99.99% singlet by norm. The numerical value alone is not enough to promote it. The candidate would proceed as follows:

**Gate A:** Recompute the eigenspace at two or more lattice spacings and volumes, with independent eigensolver starts. If the eigenvalue changes by an order of magnitude or tracks the inverse box size, reject it as a discretization or finite-volume mode. Otherwise retain it provisionally.

**Gate B:** Project out translation, gauge, Goldstone, ghost, and tunneling directions and repeat the calculation across the adopted $R_\xi$ gauges. If the mode disappears under projection or its matched eigenvalue is gauge dependent, classify and reject it. If it remains a positive physical mode, continue.

**Gate C:** Reconstruct $V_{\text{eff}}(Q)$ and calculate the one-loop mass shift. If, for example, $|\delta m_Q^2| = 10^{-20}\,\text{GeV}^2$, the candidate fails by roughly sixty-five orders of magnitude despite its attractive tree-level curvature. It proceeds only if a demonstrated symmetry keeps the corrected mass naturally near the target scale.

**Gate D:** Evolve the canonically normalized coordinate cosmologically and test $R_m$, $R_V$, $w_Q(z)$, lifetime, fifth-force, and varying-coupling bounds. A mode with the right mass but the wrong vacuum energy or excessive visible sector coupling is rejected as dark energy, although it may remain useful as a spectral benchmark.

**Gate E:** Promote the surviving operator to quantum execution only if the quantum calculation answers a defined question—for example, whether a larger coupled-sector scan preserves the protected low eigenspace—and only after the required oracle, precision, depth, sampling, and state-preparation costs are estimated.

Thus the same hypothetical mode can have three legitimate outcomes: early rejection as a numerical or gauge artifact, retention as a physically real but noncosmological light scalar, or promotion as a dark-energy candidate and quantum benchmark. The gate record should be stored with the modal metadata so that the decision is reproducible.

Table 18.4: Recommended decision rule for a candidate light mode.

| Decision | Required evidence | Action |
|---|---|---|
| Reject | Gauge/collective identity, poor lattice convergence, or large loop mass | Archive as a classified artifact or excluded parameter point |
| Retain for study | Stable physical mode but incomplete effective potential or cosmology | Complete classical reconstruction before quantum execution |
| Promote to quantum benchmark | Stable physical mode and a clear algorithmic test, even if not yet cosmologically viable | Run state-vector and controlled hardware studies |
| Dark-energy candidate | Passes numerical, physical, radiative, and cosmological gates | Seek independent theoretical and observational validation |

## 18.10 Step 5: introduce quantum algorithms in controlled stages

### 18.10.1 Stage A: exact state-vector experiments

Use the exported low modes as exact initial states and verify expectation values of the padded operator. This stage tests qubit ordering, padding, spectral shifts, and inverse transformations without state-preparation error.

### 18.10.2 Stage B: approximate state preparation

Replace exact initialization with a documented circuit or variational ansatz. Measure fidelity with the exported classical modes. For degenerate levels, measure total overlap with the target subspace rather than a selected vector.

### 18.10.3 Stage C: VQE and subspace methods

Begin with the radial scalar and ghost sectors because their dimensions and component structures are simpler. Then move to the 405-dimensional coupled sector. Ground-state VQE alone is not sufficient for a 50–200 mode program; excited-state or subspace methods must be added with explicit orthogonality constraints.

### 18.10.4 Stage D: phase estimation

Use the encoded operator and exported eigenstates to separate Hamiltonian simulation error, phase resolution, and state-overlap error. Convert every reported phase back to solver mass-squared units through the JSON spectral map.

### 18.10.5 Stage E: sparse block encoding and qubitization

The small CSR benchmark can validate an initial sparse-access construction. The production implementation should use row/value or stencil oracles rather than decompose a multimillion-dimensional matrix into a stored Pauli sum. Resource estimates must include system, index, value, and signal ancillas, as well as arithmetic precision and state preparation.

## 18.11 Step 6: export the production bounce modes

After restoring or regenerating the checkpoint, execute:

```
HIGGS_RUN_MODE=su2_determinant \
HIGGS_SU2_DET_BOUNCE_FILE=\
su2_prefactor_bounce_L26_h0.8/relaxed_bounce_checkpoint.npz \
HIGGS_SU2_DET_OUTPUT=su2_determinant_scan \
HIGGS_SU2_MODAL_EXPORT=1 \
HIGGS_SU2_MODAL_MODES=12 \
python Higgs_fluctuation_solver_stabilized.py
```

This writes the radial scalar modes once and the coupled and ghost modes at each requested value of $\xi$. For SLQ, the exporter reuses the ARPACK eigenvectors computed for low-mode deflation. If the export window is larger than the deflation window, the additional vectors are computed but are not removed from the stochastic trace.

For $L = 26$, one coupled generator block has dimension

$$5(26^4) = 2{,}284{,}880, \tag{18.18}$$

which requires 22 system qubits after padding. A twelve-vector uncompressed double-precision coupled export is approximately 209 MiB before NPZ compression. Storage, transfer, and checksum verification should therefore be treated as part of the production workflow.

**Important limitation**

The production export contains low modes, not a compact block encoding of the full operator. The complete quantum Hamiltonian must reproduce the periodic nearest-neighbor stencil, background-dependent link masses, gauge-fixing term, and gauge–Goldstone mixing. The JSON metadata fixes the conventions, but a sparse oracle implementation is still required.

## 18.12 Production quality-control checklist

Every production dataset should satisfy the following checklist before it is used in a quantum calculation:

1. the source and background SHA-256 digests are archived;
2. the checkpoint passes the stationary-bounce residual test;
3. the Ward identities pass at the same lattice parameters;
4. all exported eigenpair residuals meet the declared tolerance;
5. low negative and near-zero modes are physically classified;
6. results are repeated for more than one ARPACK starting condition;
7. degenerate or nearly degenerate subspaces are compared projectively;
8. the quantum spectral shift and scale are independently reconstructed;
9. physical units are restored after every encoded calculation; and
10. quantum results are compared with the archived classical eigenvalues.

## 18.13 Separation from the decay-rate calculation

Modal export is read-only with respect to the physical determinant. It does not add the exported modes to a determinant, change counterterm coefficients, or claim a gauged decay rate. In the SLQ route, determinant deflation remains controlled by `HIGGS_SU2_DET_DEFLATED_MODES`; modal output is controlled independently by `HIGGS_SU2_MODAL_MODES`.

This separation permits the two projects to advance together:

- the decay-rate program continues finite matching, gauge cancellation, and regulator extrapolations; while
- the quantum program develops encodings and eigensolvers against frozen, versioned outputs from the same operators.

## 18.14 Recommended next milestones

The effort bands below are planning estimates rather than measured performance promises. “Core” means one conventional CPU core; elapsed times assume the existing software remains functional and that personnel can work continuously. Quantum columns give the system-register size of the small benchmark, not the total physical-qubit requirement after ancillas, error correction, routing, and fault tolerance.

Table 18.5: Near-term execution sequence.

| Stage | Deliverable | Status |
|---|---|---|
| 1 | Homogeneous coupled, ghost, and radial modal exporter | Implemented |
| 2 | JSON schema, checksums, residuals, and CSR benchmark operators | Implemented |
| 3 | Power-of-two padding and reversible spectral map utility | Implemented |
| 4 | Automated regression test committed with fixed acceptance thresholds | Next |
| 5 | Dark-energy screening report for the present benchmark | Next |
| 6 | Initial exact state-vector or circuit-level benchmark | Next |
| 7 | Classical two-field protected-scalar prototype | Conditional next phase |
| 8 | Restored $L = 26$ checkpoint and production modal export | Blocked by checkpoint |
| 9 | Sparse stencil-oracle specification and resource estimate | Planned |
| 10 | VQE/subspace and QPE comparison against classical modes | Planned |
| 11 | Extended Higgs-sector parameter scan with screening gates | Planned |

Table 18.6: Indicative time and resource bands for the next decision points.

| Deliverable | Personnel time | Classical resource band | Quantum resource band |
|---|---|---|---|
| Regression suite and frozen homogeneous datasets | 1–2 weeks | Workstation; 4–16 cores; below 32 GB RAM | 7–9 system qubits in exact simulation; hardware optional |
| Present-control dark-energy screening report | 1–2 weeks | Existing outputs plus selected reruns; 8–32 cores | None required |
| First state-vector and circuit benchmark | 2–4 weeks | Workstation or one small node; 16–64 GB RAM | 7–9 system qubits, plus method-dependent ancillas |
| Protected two-field classical prototype | 4–8 weeks | 32–128 cores for parameter scans; memory determined by lattice size | 8–12 system-qubit reduced benchmarks |
| Restored $L = 26$ modal export | 1–3 weeks after checkpoint recovery | One high-memory node; preliminary target 16–64 cores and 128–512 GB RAM | 22 system qubits mathematically; not a near-term full-state hardware target |
| Stencil-oracle specification and resource audit | 3–6 weeks | Classical oracle tests on small lattices | Must report all index, value, signal, and arithmetic ancillas, depth, and precision |
| Extended-sector gated scan | 2–6 months | Multi-node campaign after prototype profiling | Quantum execution only for Gate-E-qualified cases |

These bands should be replaced by measured wall times, peak memory, circuit depths, shot counts, and queue times after the regression and state-vector milestones. In particular, the statement that the production operator needs 22 system qubits describes amplitude-index capacity only; it is not a claim that a useful fault-tolerant calculation fits in 22 physical qubits.

## 18.15 Conclusion

The implemented modal bridge creates a reproducible boundary between the lattice vacuum-decay code and the quantum-simulation project. It already supports exact homogeneous benchmarks, carries

sufficient metadata to recover the operator conventions, and preserves the classical low modes needed for state preparation and validation. The immediate next technical task is an automated regression suite followed by a quantitative dark-energy screening report and a first quantum state-vector benchmark. The present $SU(2)$–Higgs model is a control system rather than a dark-energy candidate. The project is worth continuing if it remains staged: validate the quantum methods on this control, introduce a technically protected scalar extension, require it to pass the classical screening gates, and only then commit substantial quantum hardware resources. A well-controlled exclusion of candidate sectors is a valid scientific result. The production bounce path can use the same modal schema as soon as its checkpoint is available.

# Bibliography

[1] D. H. Weinberg and M. White, "Dark Energy," in the 2025 update of the *Review of Particle Physics*, Particle Data Group.

[2] B. Ratra and P. J. E. Peebles, "Cosmological Consequences of a Rolling Homogeneous Scalar Field," *Phys. Rev. D* **37**, 3406 (1988).

[3] O. Bertolami and R. Rosenfeld, "The Higgs Portal and an Unified Model for Dark Energy and Dark Matter," arXiv:0708.1784.

# Appendix A

# Environment Variable Reference

Table A.1: Principal environment variables accepted by the unified program.

| Variable | Typical value | Meaning |
|---|---|---|
| HIGGS_RUN_MODE | full | Select execution mode |
| HIGGS_SU2_GAUGE | 1 | Enable research-mode gauge diagnostics; rejected in scalar production modes |
| HIGGS_SU2_G | 0.65 | SU(2) gauge coupling for research diagnostics |
| HIGGS_SU2_XI | 0.5, 1, 2 | Positive background gauge parameter |
| HIGGS_SU2_NUM_MODES | 12 | Requested modes in each diagnostic generator/ghost window |
| HIGGS_SU2_VALIDATION_OUTPUT | directory | Output directory for SU(2) prerequisite tests |
| HIGGS_SU2_BOUNCE_FILE | NPZ path | Optional scaled L=26, h=0.8 stationary-bounce test input |
| HIGGS_SU2_DET_BOUNCE_FILE | NPZ path | Relaxed bounce used by the regulated gauge determinant |
| HIGGS_SU2_DET_OUTPUT | directory | Determinant CSV, JSON, low-mode, and PDF output directory |
| HIGGS_SU2_DET_METHOD | slq or exact | Select stochastic or small-matrix determinant backend |
| HIGGS_SU2_DET_XI | 0.5,1,2 | Gauge parameters included in the determinant scan |
| HIGGS_SU2_DET_IR_SHIFTS | 0.001,0.003,0.01 | Positive auxiliary mass-squared regulators |
| HIGGS_SU2_DET_STEPS | 30,60 | Lanczos orders for the determinant scan |
| HIGGS_SU2_DET_PROBES | 8 | Common Rademacher probes at each gauge parameter |
| HIGGS_SU2_DET_DEFLATED_MODES | 12 | Exact low modes removed before SLQ |
| HIGGS_SU2_CT_BOUNCE_FILE | NPZ path | Background used to evaluate local counterterm invariants |
| HIGGS_SU2_CT_OUTPUT | directory | Counterterm basis and matching output directory |
| HIGGS_SU2_CT_COEFFICIENTS | JSON path | Pole and optional finite matching coefficients |
| HIGGS_SU2_CT_SCALE | $\sqrt{0.4}$ | $\overline{\mathrm{MS}}$ matching scale in solver units |
| HIGGS_SU2_VERTEX_SPACING | 0.8 | Lattice spacing for vertex tables |
| HIGGS_SU2_VERTEX_G0 | 0.65 | Bare gauge coupling for vertex tables |
| HIGGS_SU2_VERTEX_OUTPUT | workspace-relative NPZ path | Vertex arrays; JSON sidecar uses the same basename |
| HIGGS_MU_SQ | $-0.40$ | Quadratic coefficient convention |
| HIGGS_LAMBDA | $-2.00$ | Quartic coupling |
| HIGGS_GAMMA | 0.80 | Sixth-order coupling |
| HIGGS_DELTA | 0.08 | Eighth-order coupling |

| Variable | Typical value | Meaning |
|---|---|---|
| HIGGS_L | 20 | Research/continuum reference size |
| HIGGS_NUM_MODES | 120–132 | Requested low spectral modes |
| HIGGS_BACKGROUND_PROFILE | stationary | Stationary or Gaussian profile |
| HIGGS_RELAX_L | 26 | Relaxation lattice length |
| HIGGS_RELAX_SPACING | 0.8 | Relaxation grid spacing |
| HIGGS_RELAX_MAX_NEWTON | 14 | Maximum Newton steps |
| HIGGS_RELAX_SKIP_SPECTRUM | 1 | Skip duplicate spectrum in full mode |
| HIGGS_SLQ_L | 26 | SLQ lattice length |
| HIGGS_SLQ_SPACING | 0.8 | SLQ lattice spacing |
| HIGGS_SLQ_PROBES | 16 | Rademacher probes |
| HIGGS_SLQ_DEFLATED_MODES | 12 | Exact low eigenpairs per channel |
| HIGGS_SLQ_STEPS | 20,40,60 | Lanczos orders |
| HIGGS_PARTIAL_WAVE_LMAX | 70 | Maximum continuum partial wave |
| HIGGS_RADIAL_CUTOFF | 20 | Continuum radial cutoff |
| HIGGS_DETERMINANT_OUTPUT | directory | Continuum output directory |

# Appendix B

# Section Map of the Unified Source

Table B.1: Guide from source-code regions to numerical responsibilities. Line numbers refer to the October 8, 2026 workspace versions.

| Approximate lines/file | Section | Responsibility |
|---|---|---|
| 1–227 | Imports and SU(2) shared operators | Imports nonlinear vertex APIs; modal export and quadratic link-Hessian tools |
| 228–329 | SU(2) reference operators | Homogeneous spectra and exact momentum multiplicities |
| 330–582 | SU(2) validation | Ward identities, decoupling, normalization, and optional bounce checks |
| 583–1091 | SU(2) determinant | Exact/SLQ trace–logs, zero-mode policies, and gauge scan |
| 1092–1427 | SU(2) counterterms | Local basis, pole residues, coefficient interface, and report |
| 1428–1561 | Vertex-table mode | Configured nonlinear SU(2) momentum/color table generation |
| 1562–1619 | Dispatcher/guard | Ten-mode dispatch and sequential scalar pipeline |
| 1620–2800 | Scalar production stages | Relaxation, SLQ, and continuum calculation modes |
| 2801–2982 | Physics/configuration | Units, scalar potential, and run settings |
| 2983–3187 | Operators and validation | Scalar Hessians, SU(2) wrappers, Fourier spectrum, exact checks |
| 3188–3462 | Background | Stationary/Gaussian profiles and bounce setup |
| 3463–3754 | Hessian and spectral extraction | Operator assembly, low modes, and regularized spectral ratio |
| 3755–4204 | Diagnostics/reporting | Spectrum visualization and scalar/gauge report outputs |
| 4205–end | Console summary | Scalar results and optional gauge diagnostic status |
| `su2_lattice_vertices.py` | Nonlinear vertex library | Scalar-link, Wilson-gauge, gauge-ghost, scalar-ghost, and FP-matrix APIs |
| `su2_nonlinear_vertex_audit.py` | Independent audit | Separate action expansions, finite transformations, and vertex-table comparisons |

# Appendix C

# Compact Algorithm Summary

**Step 1:** Read the selected execution mode and numerical parameters.

**Step 2:** Validate the periodic matrix-free operator on an exactly solvable test.

**Step 3:** Locate the false vacuum, barrier, and deeper nonzero minimum.

**Step 4:** Shoot the continuum $O(4)$ bounce using overshoot/undershoot bisection.

**Step 5:** Sample the bounce on a periodic $L^4$ lattice.

**Step 6:** Relax the sample until the discrete equation of motion is satisfied.

**Step 7:** Construct longitudinal, transverse, and false-vacuum Hessian operators.

**Step 8:** Compute and classify the low negative and collective-coordinate modes.

**Step 9:** Deflate these modes and estimate the full bare trace–log with SLQ.

**Step 10:** Evaluate the continuum partial-wave determinant with asymptotic subtraction and a regulated primed zero-mode limit.

**Step 11:** Combine the action, translation factor, and renormalized determinant, while clearly retaining the missing internal-orientation normalization.

**Step 12:** Write checkpoints, data tables, figures, JSON summaries, and PDF reports.

The steps above describe the scalar pipeline. The separate nonlinear-vertex mode generates and audits link, plaquette, and ghost rules; it does not feed the scalar pipeline or gauged determinant. The quadratic gauge branch builds the link Hessian and ghost operator, runs Ward, Fourier, and normalization checks, and can report exploratory spectra. The October 1 modes further evaluate regulated determinant ratios, separate false-vacuum constant-mode policies, construct the local counterterm basis, and verify continuum pole closure. Finite lattice-to-$\overline{\mathrm{MS}}$ matching and controlled continuum and volume limits now have selected sector results, including the provisional Higgs slope and higher-point pilots in Section 9.7.4. Production-precision projections, the continuum gauge/ghost two-point tensor, full Ward and regulator checks, and the complete validated full-gauged coefficient set remain open. The foundational EFT prescription is in Chapter 17; the October 5 test evidence and its provenance are incorporated in Chapter 9. Scalar-only radial coefficients are available, but they do not supply the missing full-gauged matching constants.

# Appendix D

# Annotated Solver Source Overview

**Three-page scope and source dependency**

This is an annotated, selective source overview, not a complete printout of the 4,000-plus-line program. The full implementation is `Higgs_fluctuation_solver_stabilized.py`. It imports the nonlinear SU(2) vertex library `su2_lattice_vertices.py`, which must remain alongside it for the corresponding modes. The excerpts below emphasize the program contract, matrix-free operators, bounce construction, and spectral determinant bookkeeping; comments in the source explain the conventions and numerical choices at those transitions.

## Program contract

The header documents the available run modes and separates scalar production from gauge diagnostics. The default validation path is deterministic and exits before the more expensive bounce and spectrum stages.

```
"""Unified Higgs fluctuation, bounce, SLQ, and continuum determinant
      solver.

Execution modes selected with HIGGS_RUN_MODE:

    validation  analytic periodic-lattice validation (default)
    su2_validation  lattice Ward identities and gauge-normalization tests
    su2_determinant  regulated gauge--Goldstone--ghost determinant scan
    su2_counterterms  local EFT basis and lattice-to-MSbar matching stage
    su2_vertices  generate nonlinear SU(2) momentum/color vertex tables
    research    Hessian spectrum and finite IR determinant
    relax       discrete-action bounce relaxation
    slq         deflated stochastic-Lanczos full lattice trace-log
    continuum   partial-wave/Gel'fand--Yaglom continuum renormalization
    full        run relax, SLQ, and continuum stages sequentially

The research mode can additionally construct the coupled SU(2)-Higgs
quadratic operator by setting HIGGS_SU2_GAUGE=1.  HIGGS_SU2_G sets the
      gauge
coupling, HIGGS_SU2_XI selects a positive background R_xi gauge parameter,

and HIGGS_SU2_NUM_MODES controls the diagnostic spectral window.  This
option reports gauge-Goldstone and ghost spectra but does not silently
      combine
them with the scalar-only determinant.

Set HIGGS_SU2_MODAL_EXPORT=1 in su2_validation or su2_determinant to
      write
versioned low-mode eigenpairs and JSON metadata for quantum-simulation
benchmarks. HIGGS_SU2_MODAL_MODES selects the exported window and
HIGGS_SU2_MODAL_OUTPUT selects its directory.

The specialized calculation implementations are included directly below
      so
this file is self-contained.  The older standalone scripts may still be
      used
for reproducibility, but they are no longer required by this program.
"""
```

## Matrix-free operator interface

The scalar kernel applies the periodic nearest-neighbor stencil directly to an array. A `LinearOperator` exposes products to sparse eigensolvers without constructing a dense or sparse global Hessian matrix.

```
def make_periodic_scalar_operator(curvature, spacing):
    """Return -Delta plus a site curvature on a periodic four-dimensional
       grid."""
    import numpy as np
    import scipy.sparse.linalg as spla

    curvature_grid = np.asarray(curvature, dtype=float)
    if curvature_grid.ndim != 4 or not np.all(np.isfinite(curvature_grid))
      :
        raise ValueError("The scalar curvature must be a finite four-
      dimensional grid.")
    if not np.isfinite(spacing) or spacing <= 0.0:
        raise ValueError("The scalar-operator spacing must be finite and
      positive.")

    def matvec(vector):
        scalar = np.asarray(vector, dtype=float).reshape(curvature_grid.
      shape)
        laplacian = sum(
            (2.0 * scalar - np.roll(scalar, 1, axis=axis)
             - np.roll(scalar, -1, axis=axis)) / spacing**2
            for axis in range(4)
        )
        return (laplacian + curvature_grid * scalar).ravel()

    site_count = curvature_grid.size
    return spla.LinearOperator(
        (site_count, site_count), matvec=matvec, rmatvec=matvec,
        dtype=np.float64
    )
```

The same matrix-free convention is used by the component Hessians in the scalar pipeline; the SU(2) vertex rules themselves are supplied by the separate module named above.

## Potential and extrema

The potential is expressed in $s = \rho^2$. Solving its cubic stationarity equation identifies positive extrema, which are classified by the radial curvature before bounce shooting begins.

```
def potential_derivative(rho):
    """Return dV/drho for the O(4)-symmetric bounce equation."""
    return (
        -mu_sq * rho + lambda_val * rho**3
        + gamma * rho**5 + delta * rho**7
    )


def stationary_radial_curvature(rho_sq):
    """Return d^2V/drho^2 at a nonzero radial stationary point."""
    return 2.0 * rho_sq * (
        lambda_val + 2.0 * gamma * rho_sq
        + 3.0 * delta * rho_sq**2
    )


# The false vacuum is at rho=0. Positive real roots in s=rho^2 locate the
# nonzero extrema; the curvature distinguishes the barrier and true well.
# Solve for nonzero extrema in s=rho^2 rather than by a nonlinear search
      in rho.
# Positive real roots are classified by their potential and radial
      curvature.
# Nonzero stationary points obey
# delta*s^3 + gamma*s^2 + lambda*s - mu_sq = 0, with s=rho^2.
stationary_roots = np.roots([delta, gamma, lambda_val, -mu_sq])
stationary_points = []
for root in stationary_roots:
    if abs(root.imag) < 1.0e-10 and root.real > 0.0:
        rho_sq_stationary = root.real
        stationary_points.append({
            'rho_sq': rho_sq_stationary,
            'rho': np.sqrt(rho_sq_stationary),
            'potential': potential_from_rho_sq(rho_sq_stationary),
            'curvature': stationary_radial_curvature(rho_sq_stationary)
        })
stationary_points.sort(key=lambda point: point['rho'])

if -mu_sq <= 0.0:
    raise ValueError('A stable rho=0 false vacuum requires mu_sq < 0.')
barrier_points = [p for p in stationary_points if p['curvature'] < 0.0]
nonzero_minima = [p for p in stationary_points if p['curvature'] > 0.0]
if not barrier_points or not nonzero_minima:
    raise ValueError('Potential requires a barrier and a nonzero minimum.'
      )
true_vacuum = min(nonzero_minima, key=lambda p: p['potential'])
if true_vacuum['potential'] >= potential_from_rho_sq(0.0):
    raise ValueError('The nonzero true vacuum must lie below rho=0.')
barrier_point = min(barrier_points, key=lambda p: p['rho'])
```

## O(4)-symmetric bounce

The radial equation includes the $3\rho'/r$ friction term. Regular initial data are started at a small positive radius using the origin expansion; overshoot/undershoot bisection then locates the separatrix. The action uses the four-dimensional sphere measure $2\pi^2 r^3$.

```
def bounce_rhs(radius, state):
    """First-order O(4) bounce equation away from the regular origin."""
    rho, rho_prime = state
    return [rho_prime, potential_derivative(rho) - 3.0 * rho_prime /
      radius]


def trial_overshoots(rho_center, radius_max):
    """Return True when a trial solution crosses below the false vacuum."
      ""
    # Regularity gives rho'(0)=0; a local series supplies finite start
      values
    # at epsilon without evaluating the singular 3/r term at the origin.
    epsilon = 1.0e-6
    force = potential_derivative(rho_center)

    def crosses_false_vacuum(radius, state):
        return state[0]

    crosses_false_vacuum.terminal = True
    crosses_false_vacuum.direction = -1
    trial = solve_ivp(
        bounce_rhs, (epsilon, radius_max),
        [rho_center + force * epsilon**2 / 8.0,
         force * epsilon / 4.0],
        events=crosses_false_vacuum,
        rtol=3.0e-10, atol=1.0e-12, max_step=0.03
    )
    return bool(trial.t_events[0].size)
```

```
if background_profile == 'stationary':
    # Bracket the separatrix between undershooting and overshooting
      centers.
    # Bisection is robust because the target is a saddle trajectory.
    shooting_radius = max(80.0, 4.0 * L)
    lower = barrier_point['rho'] * (1.0 + 1.0e-8)
    upper = true_vacuum['rho'] * (1.0 - 1.0e-12)
    for _ in range(70):
        midpoint = 0.5 * (lower + upper)
        if trial_overshoots(midpoint, shooting_radius):
            upper = midpoint
        else:
            lower = midpoint
    rho_center_value = upper

    # Integrate only until the physical tail is negligible.  Continuing
      an
    # approximate separatrix far beyond this point amplifies roundoff
      along
    # the exponentially growing false-vacuum solution.
    epsilon = 1.0e-7
    tail_threshold = 1.0e-9 * max(1.0, rho_center_value)
    force = potential_derivative(rho_center_value)

    def reaches_tail(radius, state):
```

```
    radial_r = bounce_solution.t
    radial_rho = bounce_solution.y[0]
    radial_prime = bounce_solution.y[1]
    # The O(4) angular measure is 2*pi^2*r^3; subtract V(false) so the
      action
    # is integrated relative to the homogeneous false-vacuum
      configuration.
    action_density = 2.0 * np.pi**2 * radial_r**3 * (
        0.5 * radial_prime**2
        + potential_from_rho(radial_rho) - potential_from_rho(0.0)
    )
    bounce_action = float(np.trapezoid(action_density, radial_r))
    leading_suppression = float(np.exp(-bounce_action))
```

## Hessian channels and low modes

Because the potential depends only on the radial magnitude, its second variation separates into one longitudinal channel and three degenerate transverse channels. The bounce channels are solved separately; their eigenvalues are merged only after each solve.

```
# The O(4)-symmetric four-component potential depends only on rho^2.
      These
# derivatives generate the radial and transverse fluctuation curvatures.
# A. First Derivative of the Potential with respect to (\rho^2): V'(\rho
      ^2)
V_prime_B = (
    -0.5 * mu_sq
    + 0.5 * lambda_val * rho_sq_bounce
    + 0.5 * gamma * (rho_sq_bounce**2)
    + 0.5 * delta * (rho_sq_bounce**3)
)

# B. Second Derivative of the Potential with respect to (\rho^2): V''(\
      rho^2)
V_pprime_B = (
    0.5 * lambda_val
    + gamma * rho_sq_bounce
    + 1.5 * delta * (rho_sq_bounce**2)
)

# C. Longitudinal Fluctuation Channel (\phi_1 Direction):
V_diag_long_B = (
    2 * V_prime_B + 4 * V_pprime_B * rho_sq_bounce
) / (a_spacing**2)

# D. Transverse Fluctuation Channels (\phi_2, \phi_3, \phi_4 Directions):
V_diag_trans_B = (2 * V_prime_B) / (a_spacing**2)

# E. Matrix-Free Bounce Hessian Channels:
# The two local curvature arrays are retained. Separate scalar
      LinearOperators
# are created immediately before the eigensolve, avoiding a four-
      component
# sparse matrix and making the exact threefold transverse degeneracy
      explicit.
```

```
# %% 5. HIGH-PERFORMANCE LANCZOS SPECTRAL EXTRACTION
print('Extracting physical eigenvalues via Sparse ARPACK Solvers...')
start_time = time.time()

# The requested low end includes any negative and near-zero modes, so use
# algebraically smallest eigenvalues (SA), not eigenvalues nearest zero (
      SM).
# Solve the bounce longitudinal and transverse blocks separately. The
      three
# transverse field components share exactly the same scalar operator, so
      one
# transverse solve is replicated three times. This prevents ARPACK from
      losing
# members of the exact threefold component degeneracy.
research_rng = np.random.default_rng(RANDOM_SEED)
channel_modes = min(V_sites - 1, num_modes)
channel_ncv = min(V_sites - 1, max(2 * channel_modes + 1, 20))

longitudinal_operator = make_scalar_hessian_operator(V_diag_long_B)
transverse_operator = make_scalar_hessian_operator(V_diag_trans_B)
ev_longitudinal, vectors_longitudinal = spla.eigsh(
    longitudinal_operator, k=channel_modes, which='SA', ncv=channel_ncv,
    tol=1.0e-10, maxiter=20000,
    v0=research_rng.standard_normal(V_sites)
)
ev_transverse, vectors_transverse = spla.eigsh(
    transverse_operator, k=channel_modes, which='SA', ncv=channel_ncv,
    tol=1.0e-10, maxiter=20000,
    v0=research_rng.standard_normal(V_sites)
)
```

## Primed determinant and interpretation

Near-zero collective-coordinate modes are excluded from the primed trace-log; the negative mode is retained as a separately classified instability. The absolute-value logarithm defines the real determinant magnitude and does not turn the scalar-only result into a gauged decay rate.

```
# %% 6. FLUCTUATION SPECTRUM REGULARIZATION & LOG-DETERMINATION RATIO
print('Regularizing eigenvalue arrays and extracting the log-determinant
      ratio...')
# Collective-coordinate modes are omitted from the primed determinant,
      while
# the physical negative mode is classified separately for the rate
      formula.
zero_tolerance = 1.0e-8 * max(
    1.0, np.max(np.abs(ev_bounce)), np.max(np.abs(ev_false))
)
bounce_negative_modes = ev_bounce[ev_bounce < -zero_tolerance]
bounce_zero_modes = ev_bounce[np.abs(ev_bounce) <= zero_tolerance]
false_negative_modes = ev_false[ev_false < -zero_tolerance]
false_zero_modes = ev_false[np.abs(ev_false) <= zero_tolerance]
if false_negative_modes.size:
    raise RuntimeError('Negative modes remain in the false-vacuum
      spectrum.')

neg_mode = bounce_negative_modes[0] if bounce_negative_modes.size else np.
      nan
zero_modes_removed = bounce_zero_modes
filtered_ev_bounce = ev_bounce[np.abs(ev_bounce) > zero_tolerance]
filtered_ev_false = ev_false[np.abs(ev_false) > zero_tolerance]
determinant_window_size = min(
    filtered_ev_bounce.size, filtered_ev_false.size
)
if determinant_window_size == 0:
    raise RuntimeError('No nonzero determinant modes remain.')
filtered_ev_bounce = filtered_ev_bounce[:determinant_window_size]
filtered_ev_false = filtered_ev_false[:determinant_window_size]

# The absolute value keeps a real trace-log after the negative direction
      has
# been separated; its sign remains available in neg_mode above.
log_det_bounce = np.sum(np.log(np.abs(filtered_ev_bounce)))
log_det_false = np.sum(np.log(np.abs(filtered_ev_false)))
log_determinant_ratio = log_det_bounce - log_det_false
functional_determinant_factor = np.exp(-0.5 * log_determinant_ratio)
negative_mode_scale = (
    np.sqrt(abs(neg_mode)) if np.isfinite(neg_mode) else np.nan
)
max_zero_mode_residual = (
    float(np.max(np.abs(zero_modes_removed)))
    if zero_modes_removed.size else 0.0
)
```

**Reading the excerpts**

The listings are illustrative excerpts, not a standalone executable. Use the full source and its imported vertex library for execution; matching and gauge diagnostic outputs retain the scope limits described in the main text.